UCT MECHATRONICS

University of Cape Town · iYunivesithi yaseKapa · Universiteit van Kaapstad

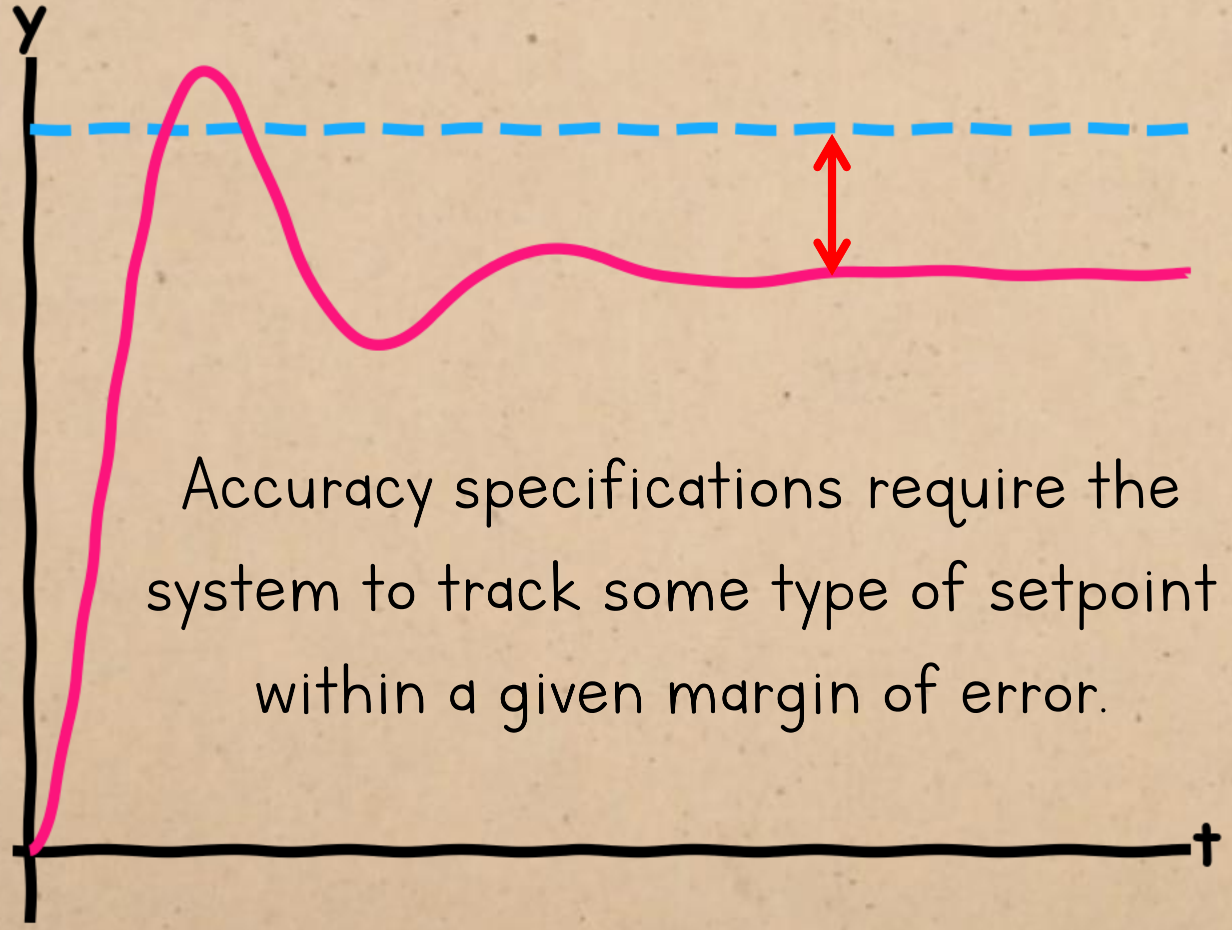
SPES BONA

CHAPTER 8:

COMPENSATION

TIME RESPONSE DESIGN IN THE FREQUENCY DOMAIN

DESIGN SPECIFICATIONS

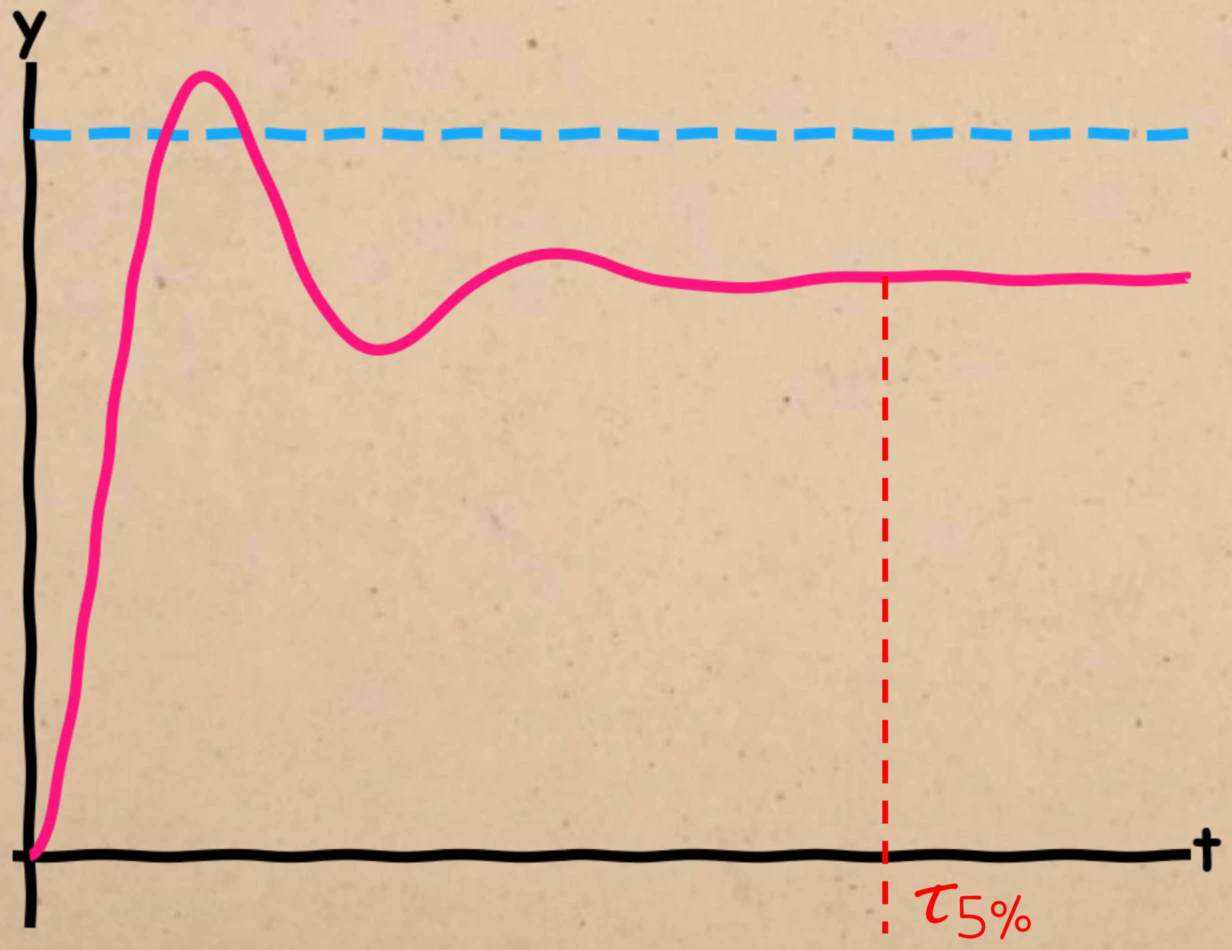


Accuracy specifications require the system to track some type of setpoint within a given margin of error.

Time-domain performance requirements typically fall into three categories:

1. **Accuracy (steady-state error)**

DESIGN SPECIFICATIONS

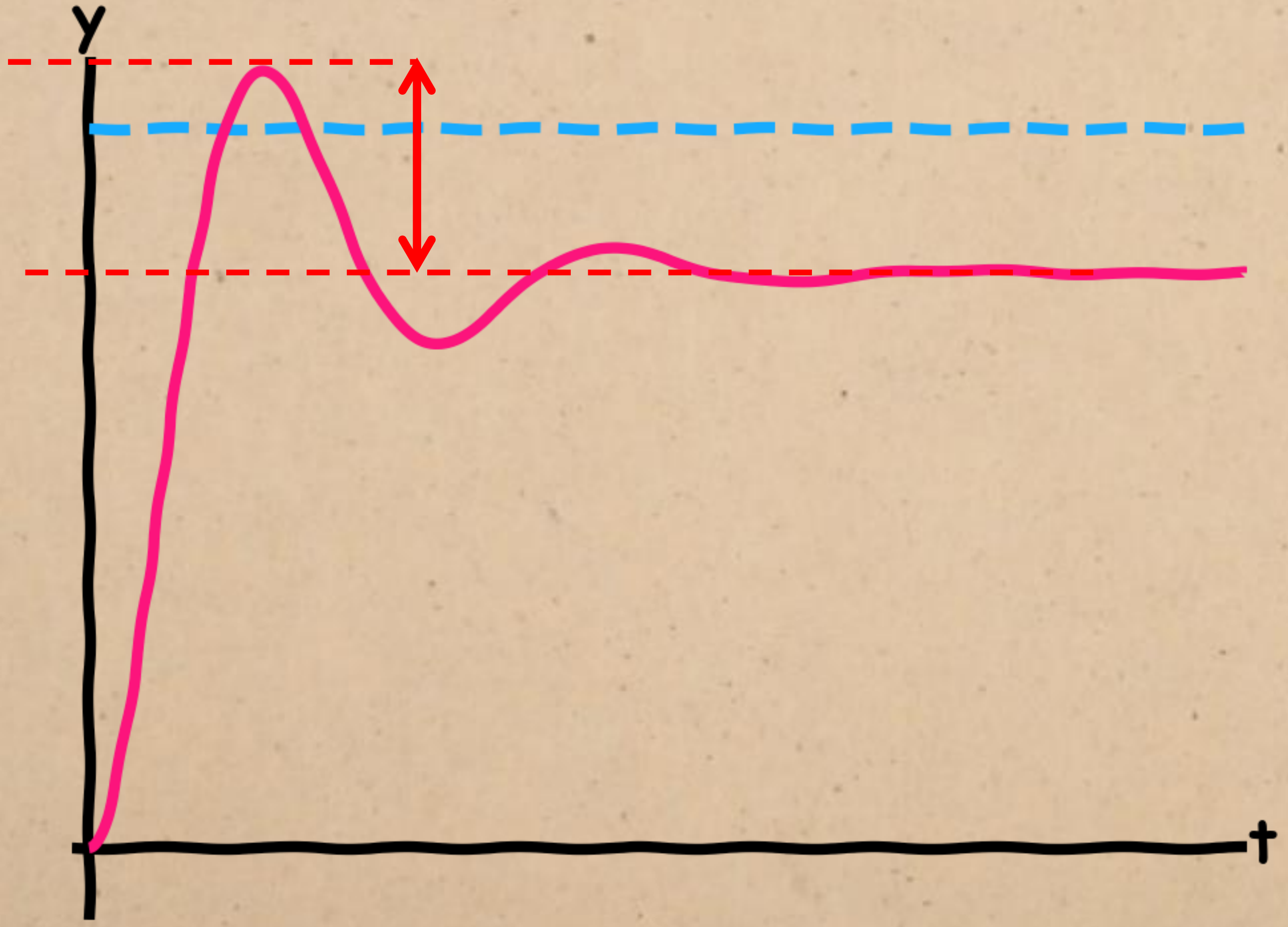


Time-domain performance requirements typically fall into three categories:

1. **Accuracy (steady-state error)**

2. **Speed (time constant)** - settling time, rise time, time to peak, etc.

DESIGN SPECIFICATIONS



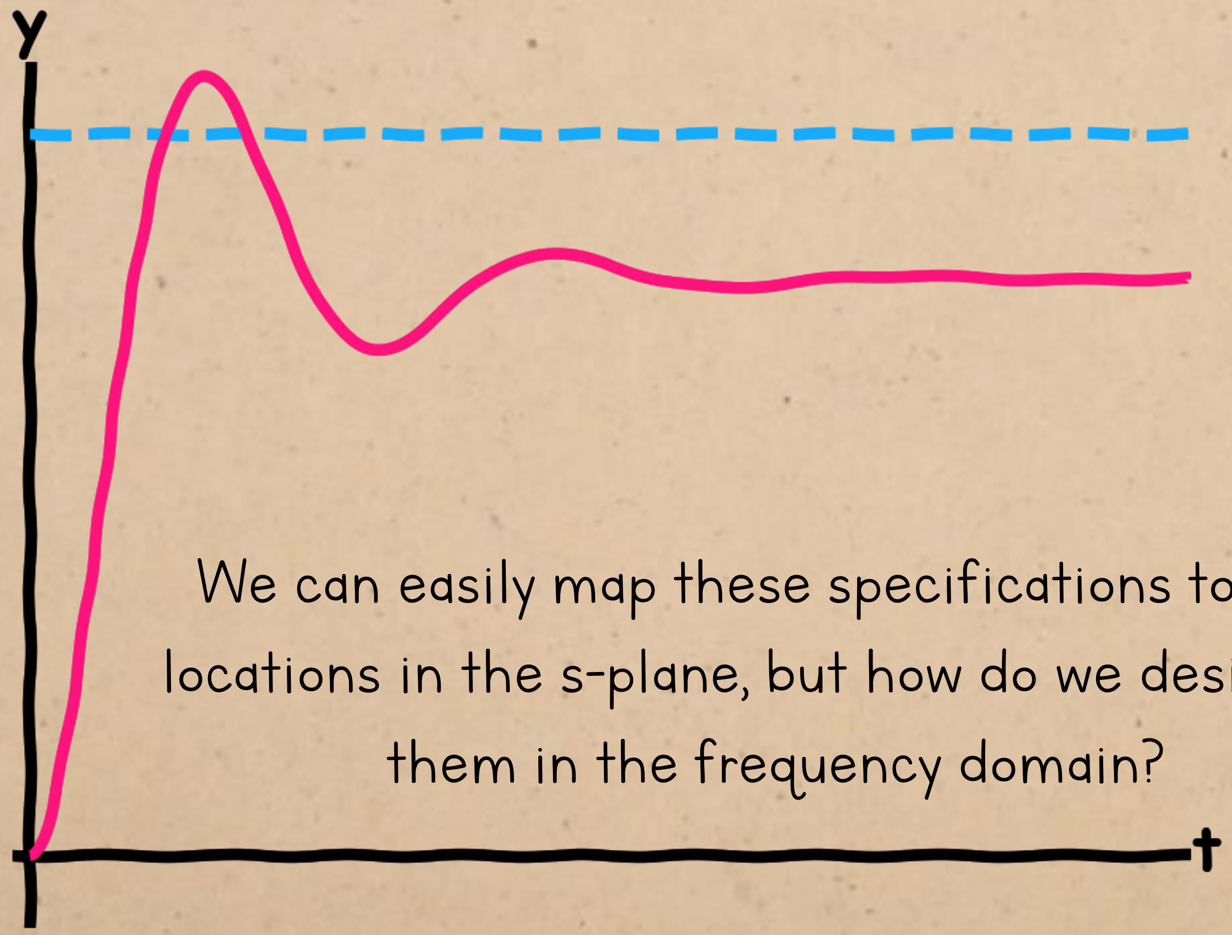
Time-domain performance requirements typically fall into three categories:

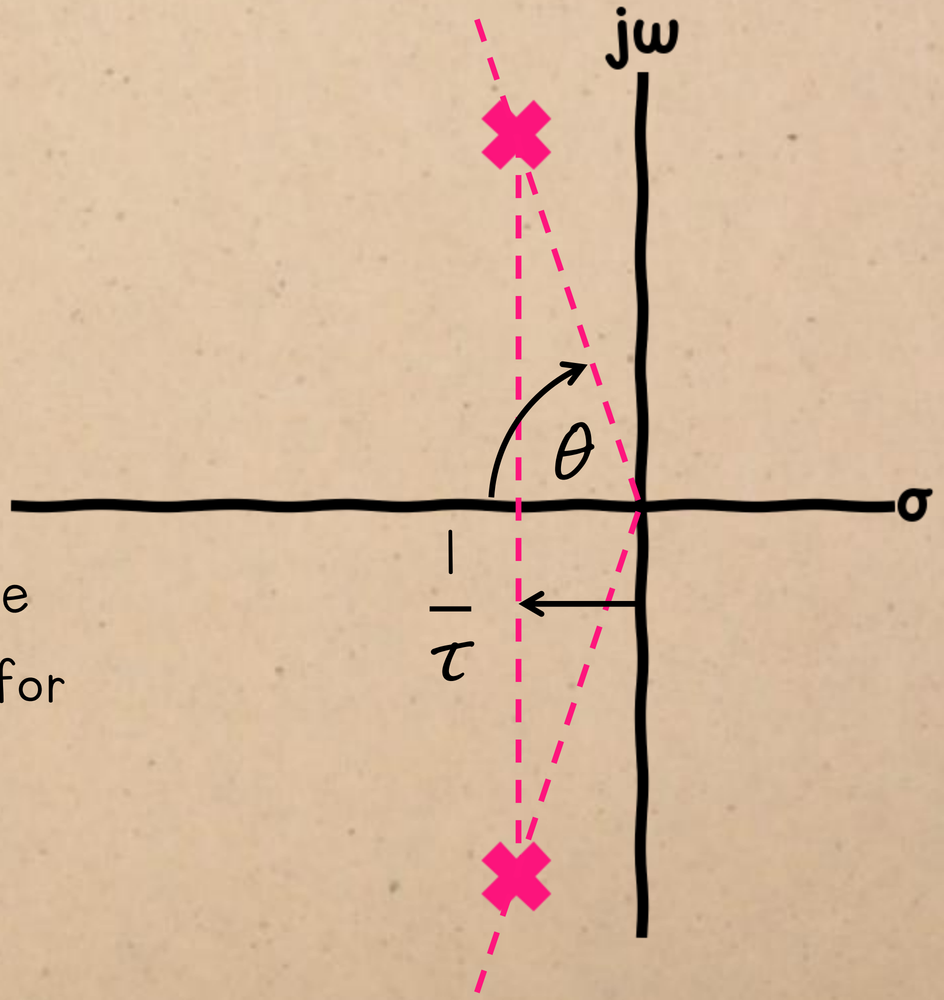
1. **Accuracy (steady-state error)**

2. **Speed (time constant)** - settling time, rise time, time to peak, etc.

3. **Damping (damping ratio)** – damping type, % overshoot, “must not oscillate”, etc.

DESIGN SPECIFICATIONS

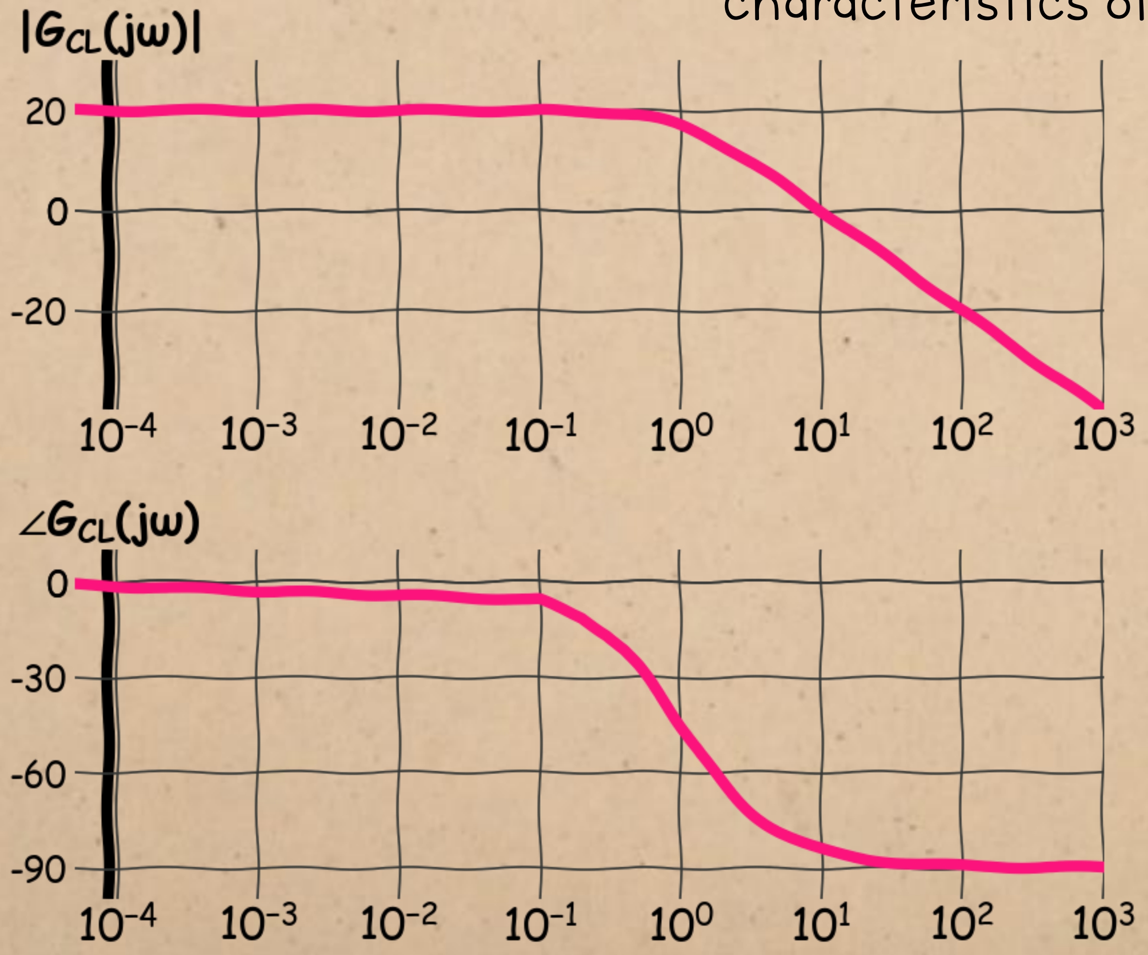




We can easily map these specifications to pole locations in the s-plane, but how do we design for them in the frequency domain?

DESIGN SPECIFICATIONS

We can relate each of the time-domain parameters to characteristics of the frequency response:

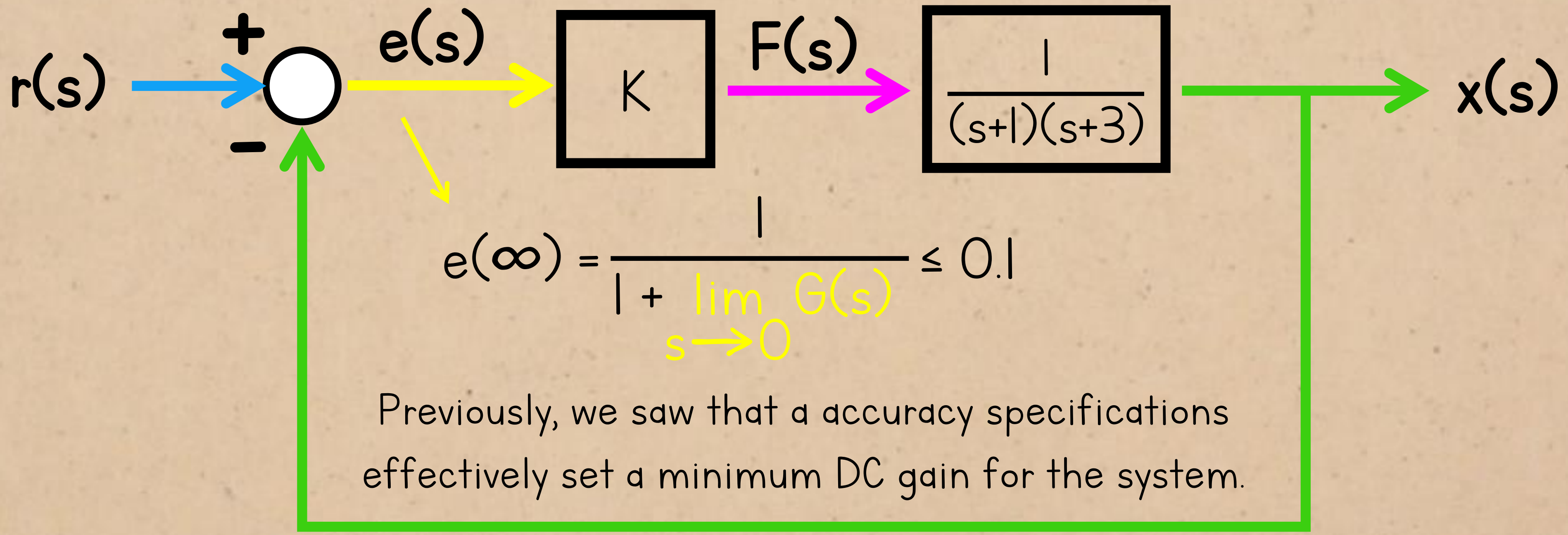


1. Accuracy (steady-state error)

2. Speed (time constant).

3. Damping (damping ratio)

DESIGN SPECIFICATIONS

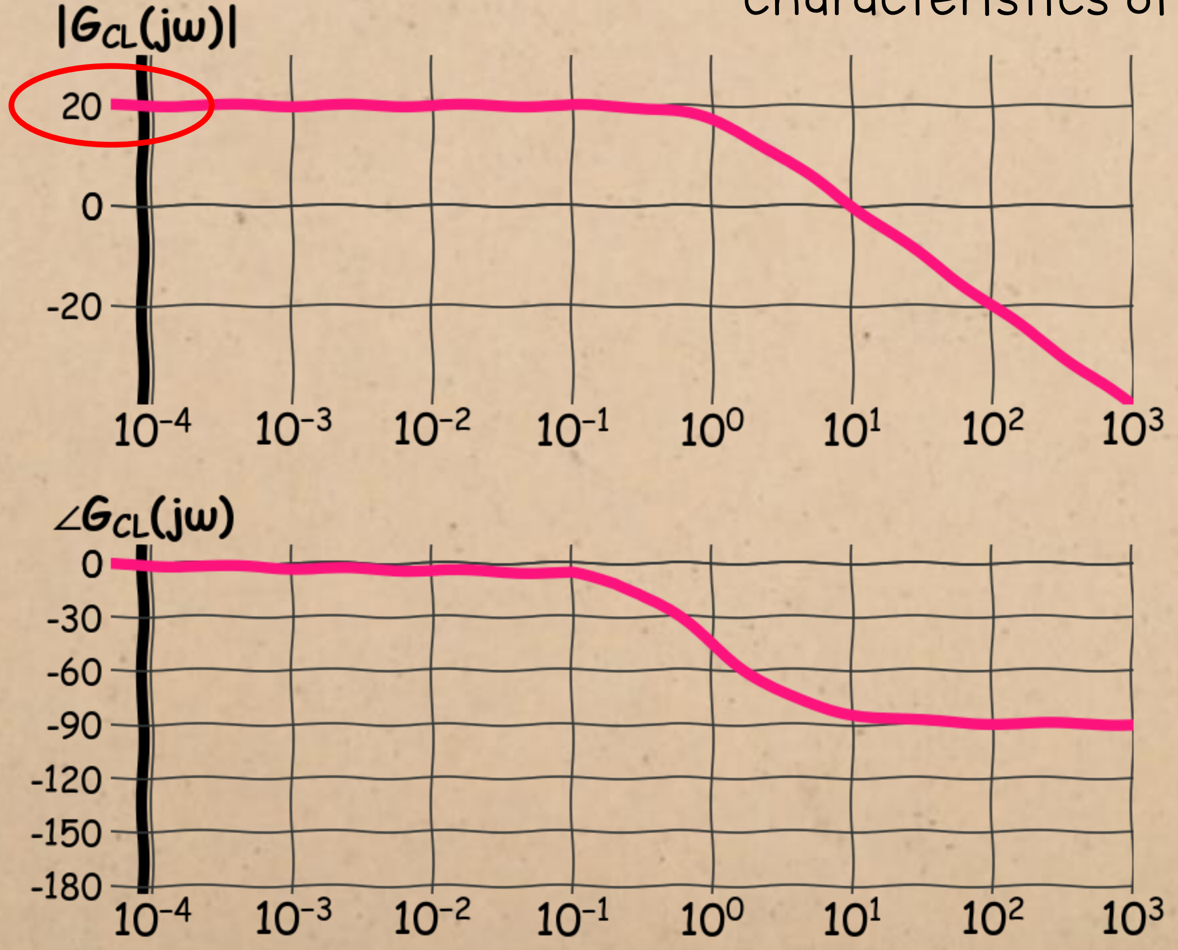


$$e(\infty) = \frac{1}{1 + \lim_{s \to 0} G(s)} \leq 0.1$$

Previously, we saw that a accuracy specifications effectively set a minimum DC gain for the system.

DESIGN SPECIFICATIONS

We can relate each of the time-domain parameters to characteristics of the frequency response:



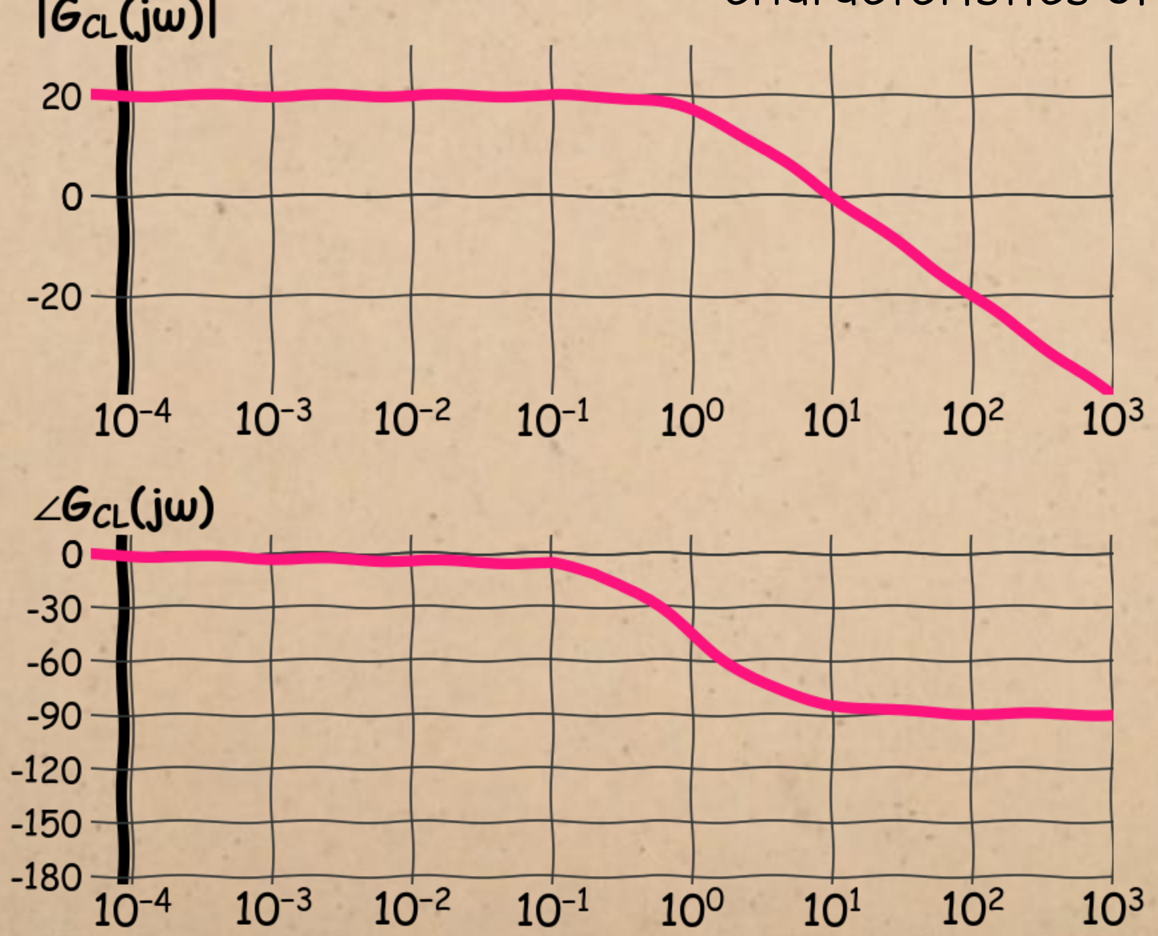
1. **Accuracy (steady-state error)**
   depends on the **DC gain**

2. **Speed (time constant)**.

3. **Damping (damping ratio)**

DESIGN SPECIFICATIONS

We can relate each of the time-domain parameters to characteristics of the frequency response:



1. **Accuracy (steady-state error)** depends on the **DC gain**

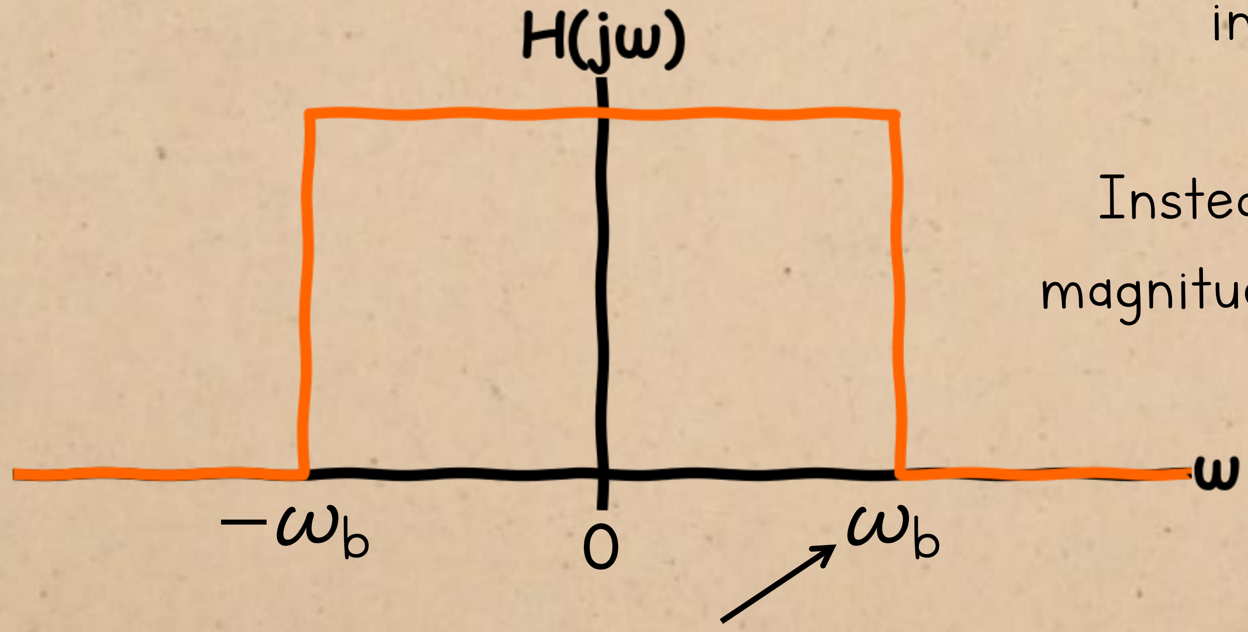
2. **Speed (time constant)** depends on the **bandwidth**

3. **Damping (damping ratio)**

BANDWIDTH

This is how we defined bandwidth in signals and systems last year, but the bandwidth we're interested in in control systems is slightly different.

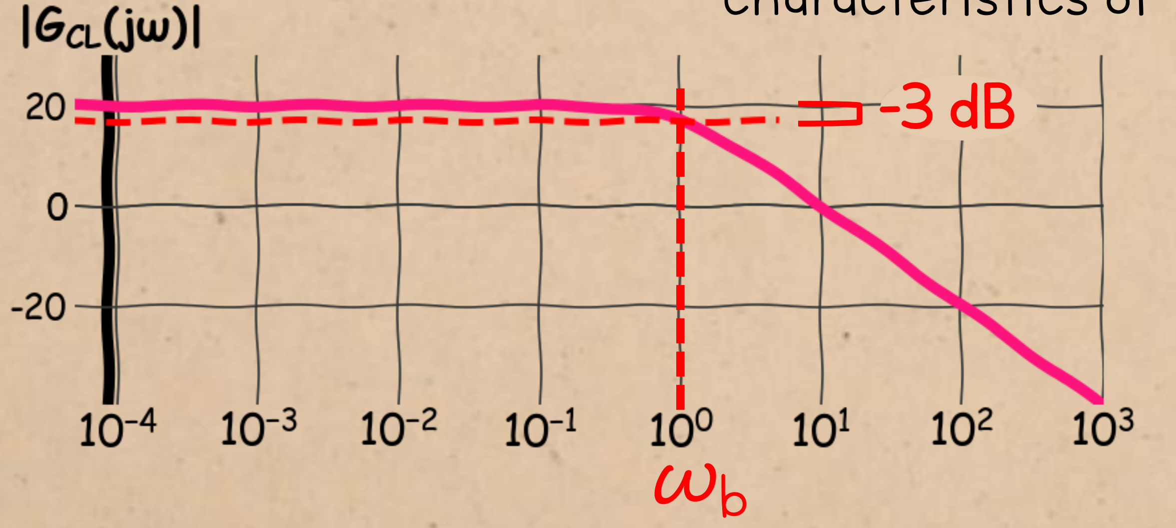
Instead of looking for the frequency where the magnitude drops all the way to zero, we want to find where it's dropped by 3 decibels.



bandwidth = the highest frequency for which a signal has a nonzero magnitude.

BANDWIDTH

We can relate each of the time-domain parameters to characteristics of the frequency response:



-3 dB bandwidth = the frequency where the magnitude drops to 3 decibels below the peak value.

1. **Accuracy (steady-state error)**
   depends on the **DC gain**

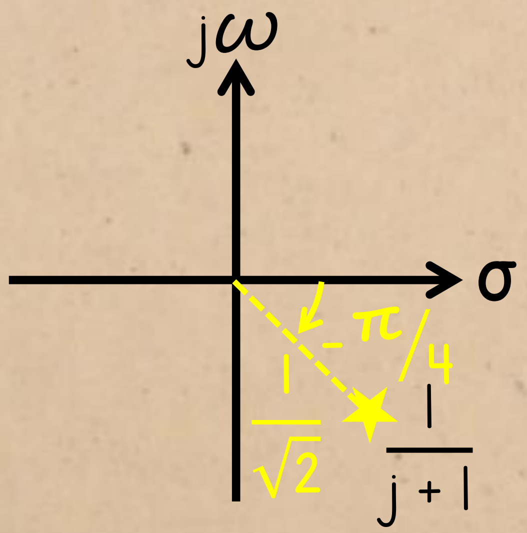
2. **Speed (time constant)**
   depends on the **bandwidth**

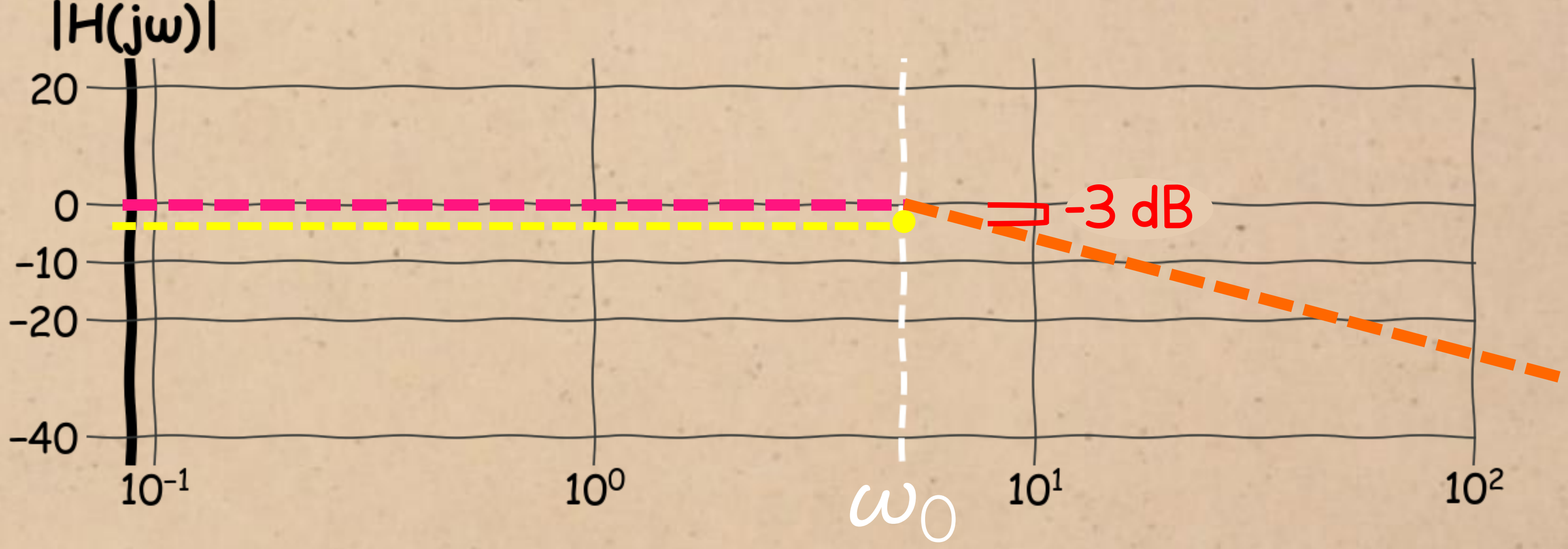
3. **Damping (damping ratio)**

BANDWIDTH

$$\left|\frac{1}{\frac{1}{\omega_0} j\omega_0 + 1}\right| = \left|\frac{1}{j+1}\right| = \frac{1}{\sqrt{2}}$$

$= -3\text{dB}$



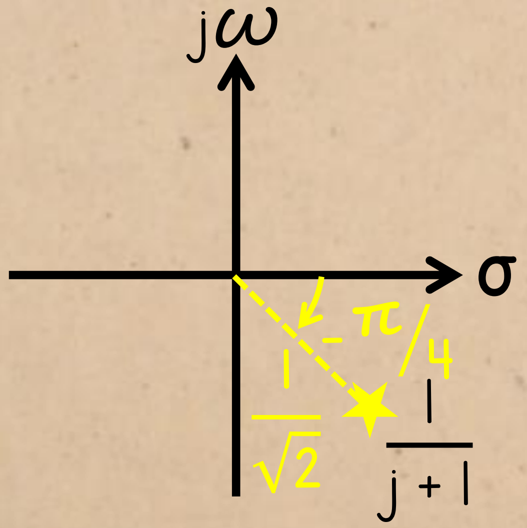


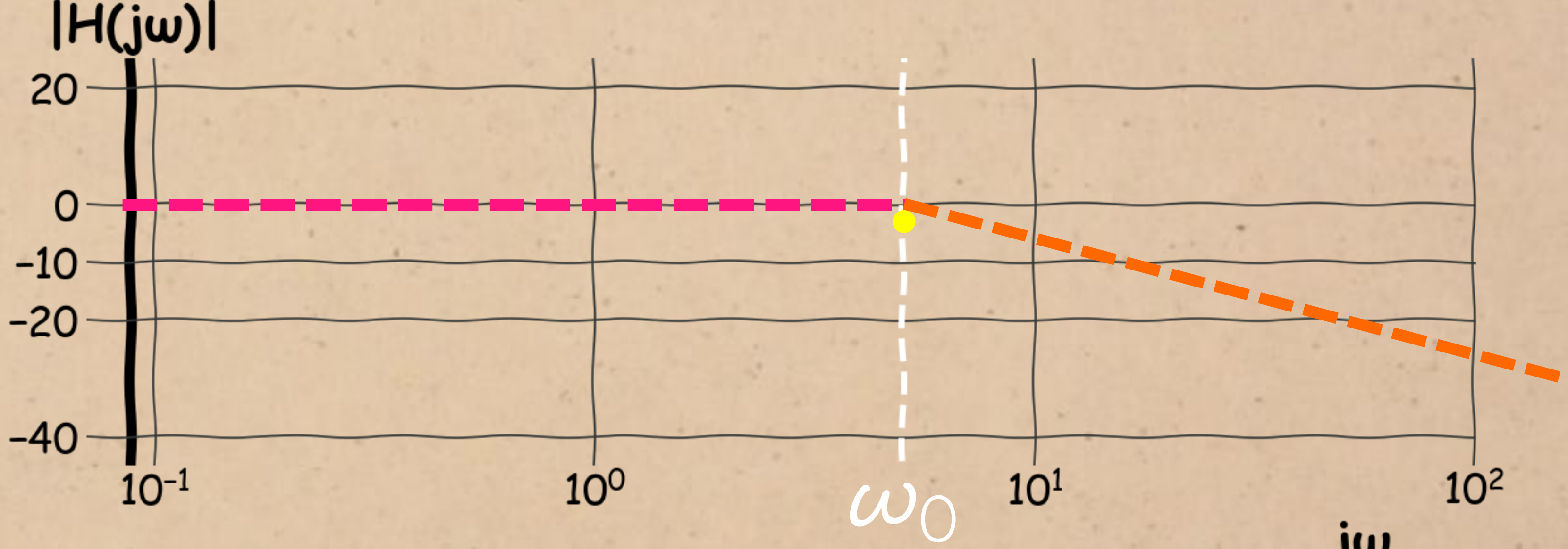
-3 dB from the peak is a significant value, because this is the magnitude at the **break frequency**.

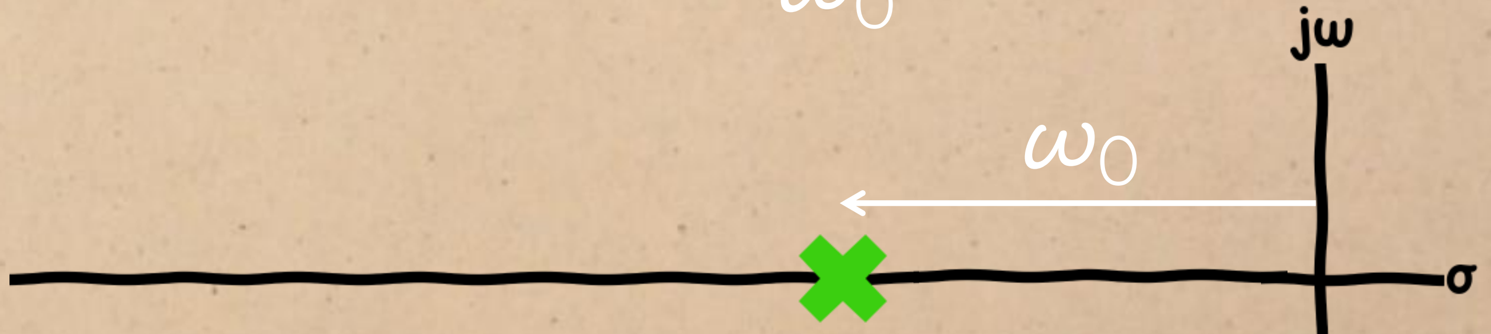
BANDWIDTH

$$\left|\frac{1}{\frac{1}{\omega_0}j\omega_0+1}\right| = \left|\frac{1}{j+1}\right| = \frac{1}{\sqrt{2}}$$

$$= -3\text{dB}$$



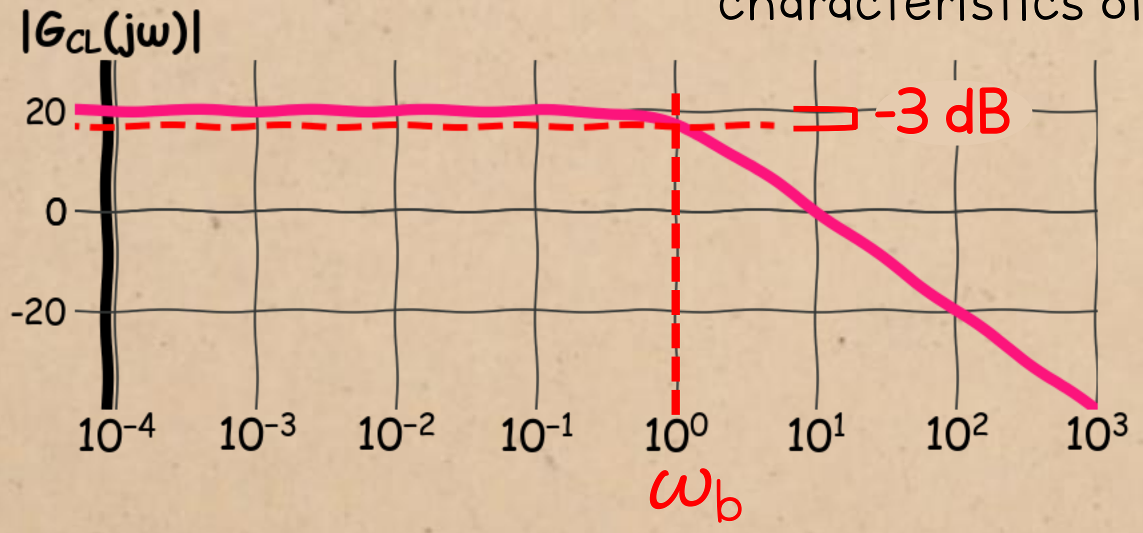




The break frequency is related to the real part of the pole.

BANDWIDTH

We can relate each of the time-domain parameters to characteristics of the frequency response:



You know the -3 dB bandwidth

∴ you know the break frequency

∴ you know the distance from the pole to the origin

∴ you know the time constant.

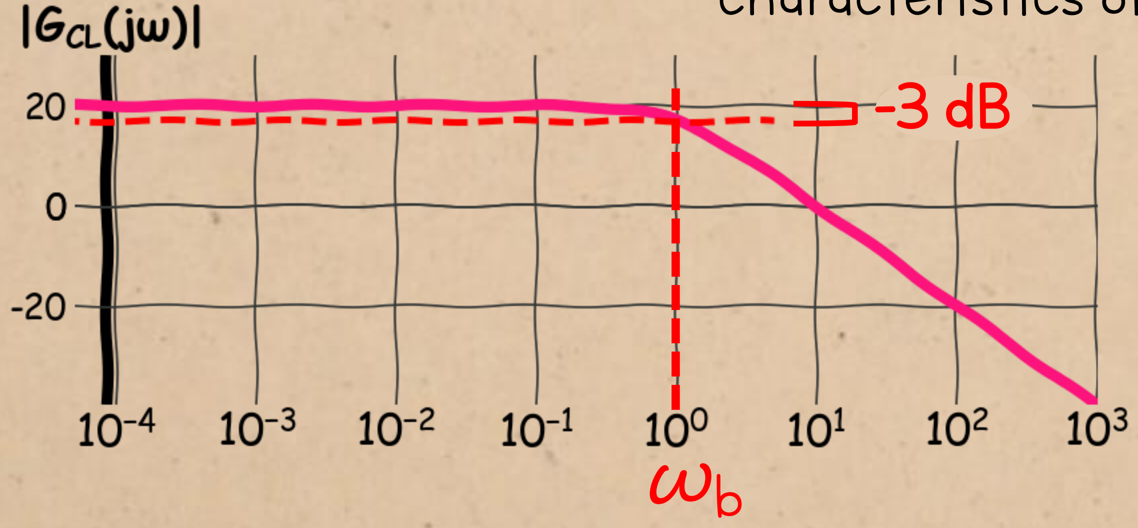
1. **Accuracy (steady-state error)**
   depends on the **DC gain**

2. **Speed (time constant)**.
   depends on the **bandwidth**

3. **Damping (damping ratio)**

BANDWIDTH

We can relate each of the time-domain parameters to characteristics of the frequency response:



It's easy to find the bandwidth for the **open-loop** system: you just read it off the Bode plot.

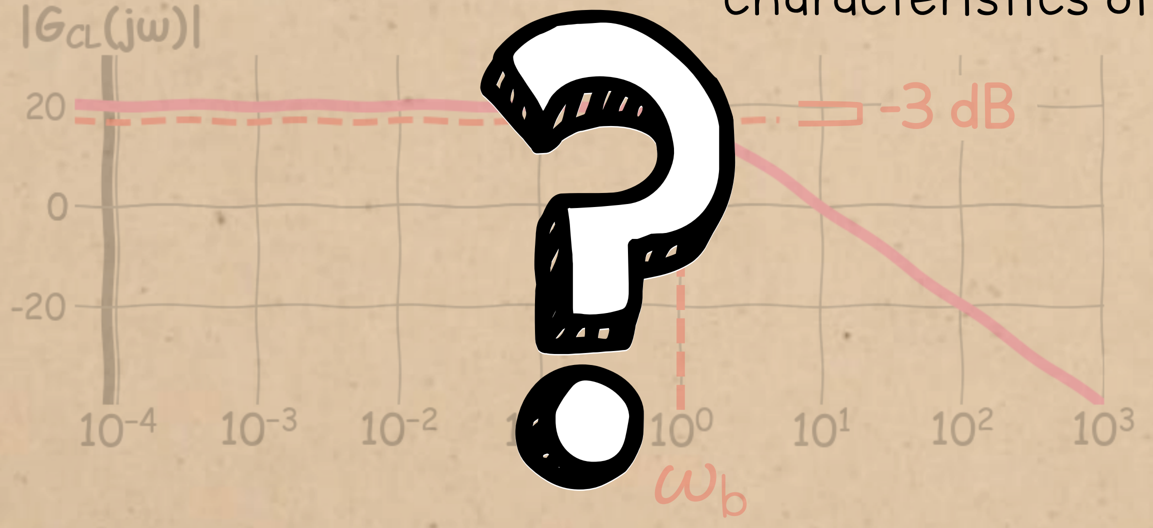
1. **Accuracy (steady-state error)** depends on the **DC gain**

2. **Speed (time constant)** depends on the **bandwidth**

3. **Damping (damping ratio)**

BANDWIDTH

We can relate each of the time-domain parameters to characteristics of the frequency response:



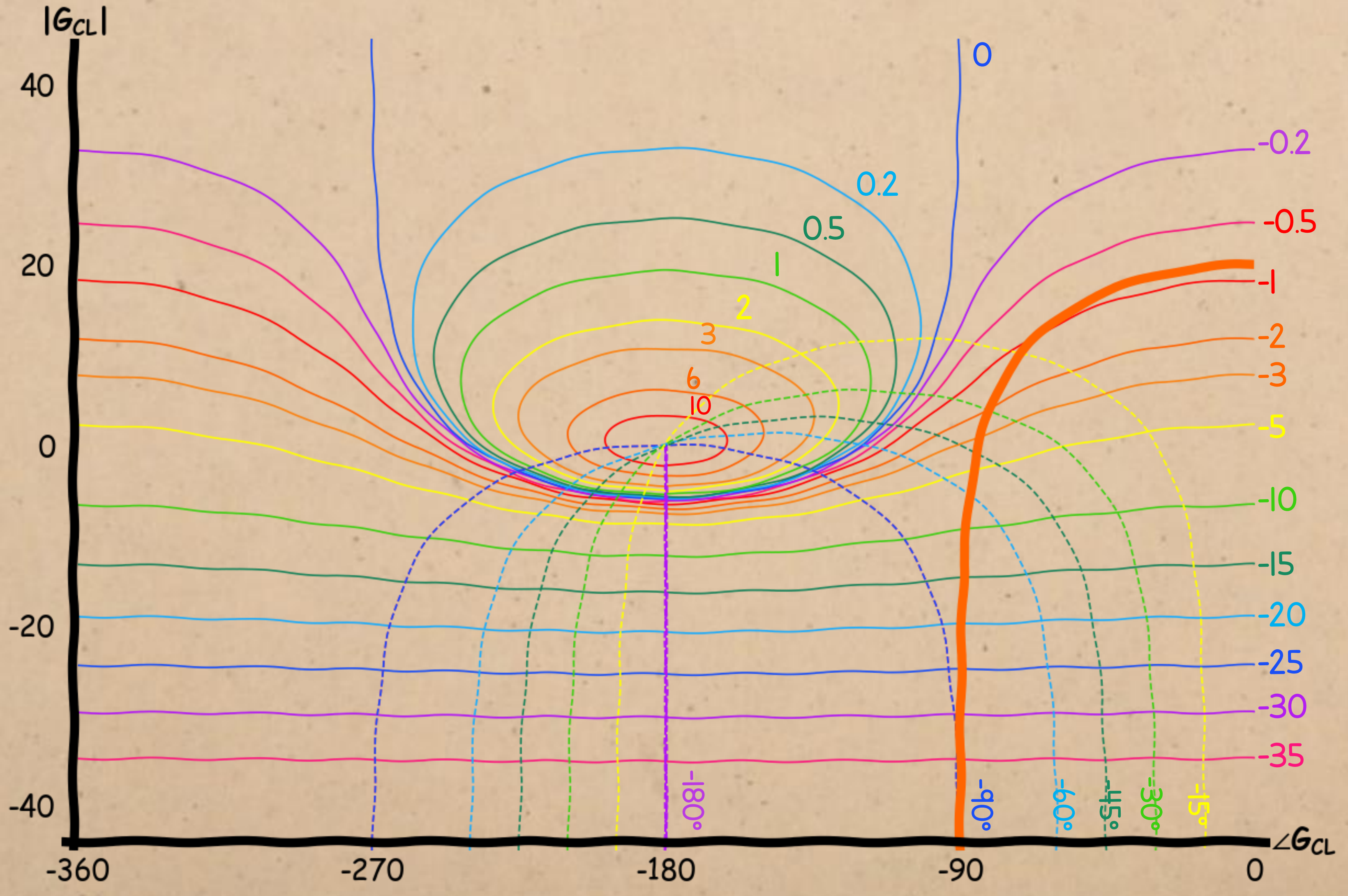
But for the **closed-loop** system, we have to explicitly calculate the closed-loop frequency response before we can create a Bode plot that will give the bandwidth.

1. **Accuracy (steady-state error)** depends on the **DC gain**

2. **Speed (time constant)** depends on the **bandwidth**

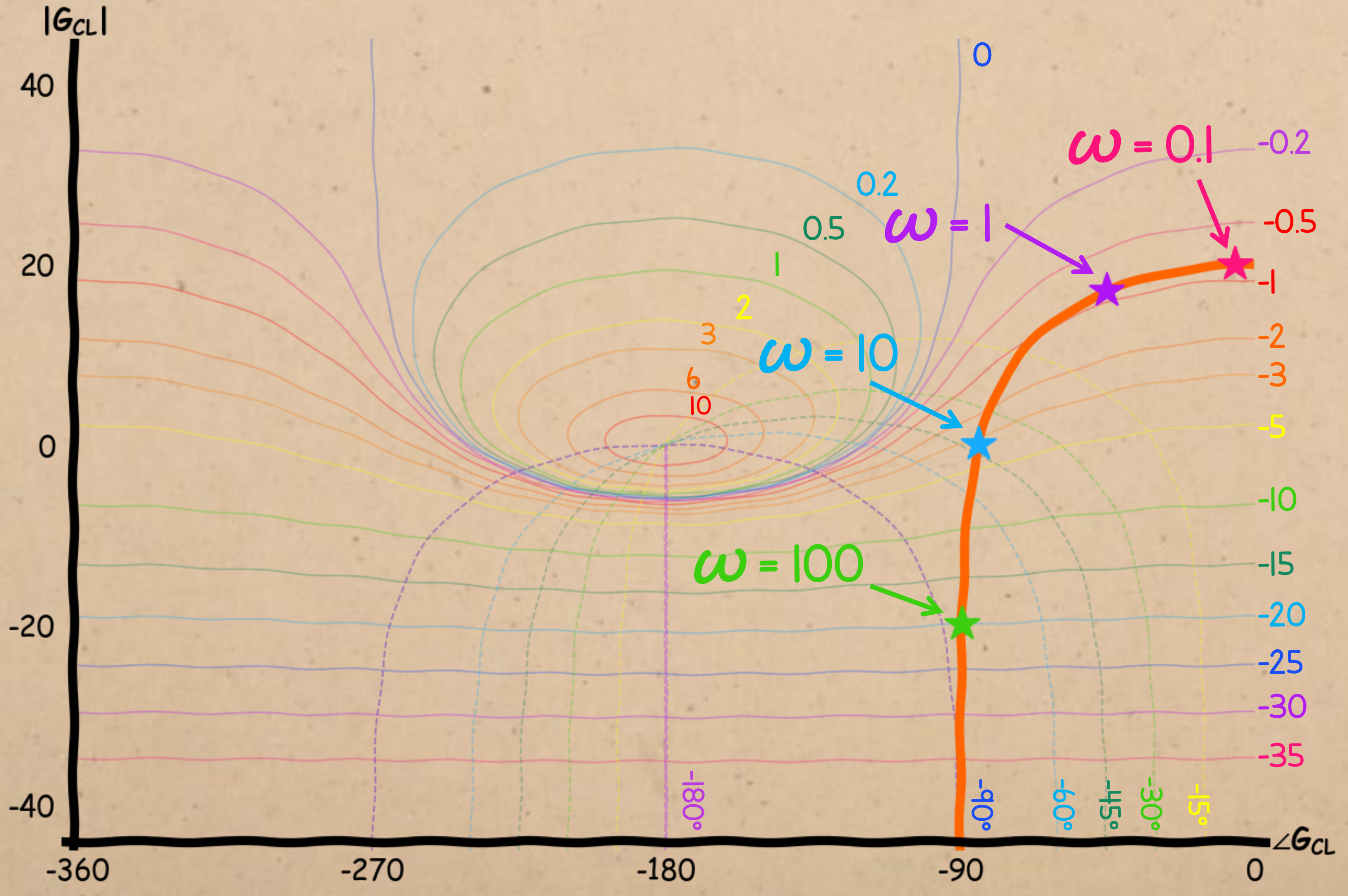
3. **Damping (damping ratio)**

BANDWIDTH



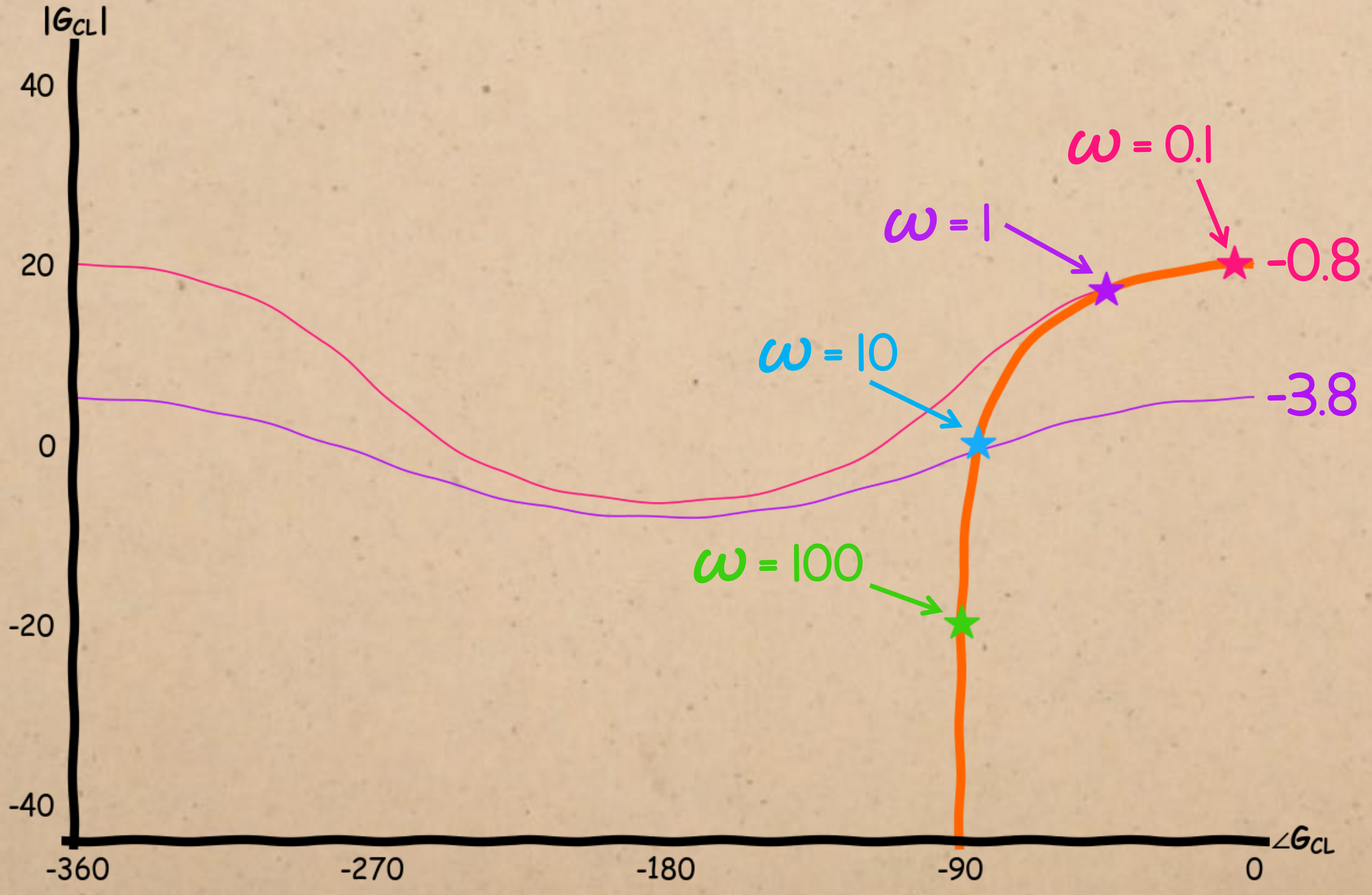
We can use the **M circles** on the Nyquist or Nichols chart to find the closed-loop bandwidth from the open-loop response.

BANDWIDTH



First, we need to annotate the response with the frequencies corresponding to each point.

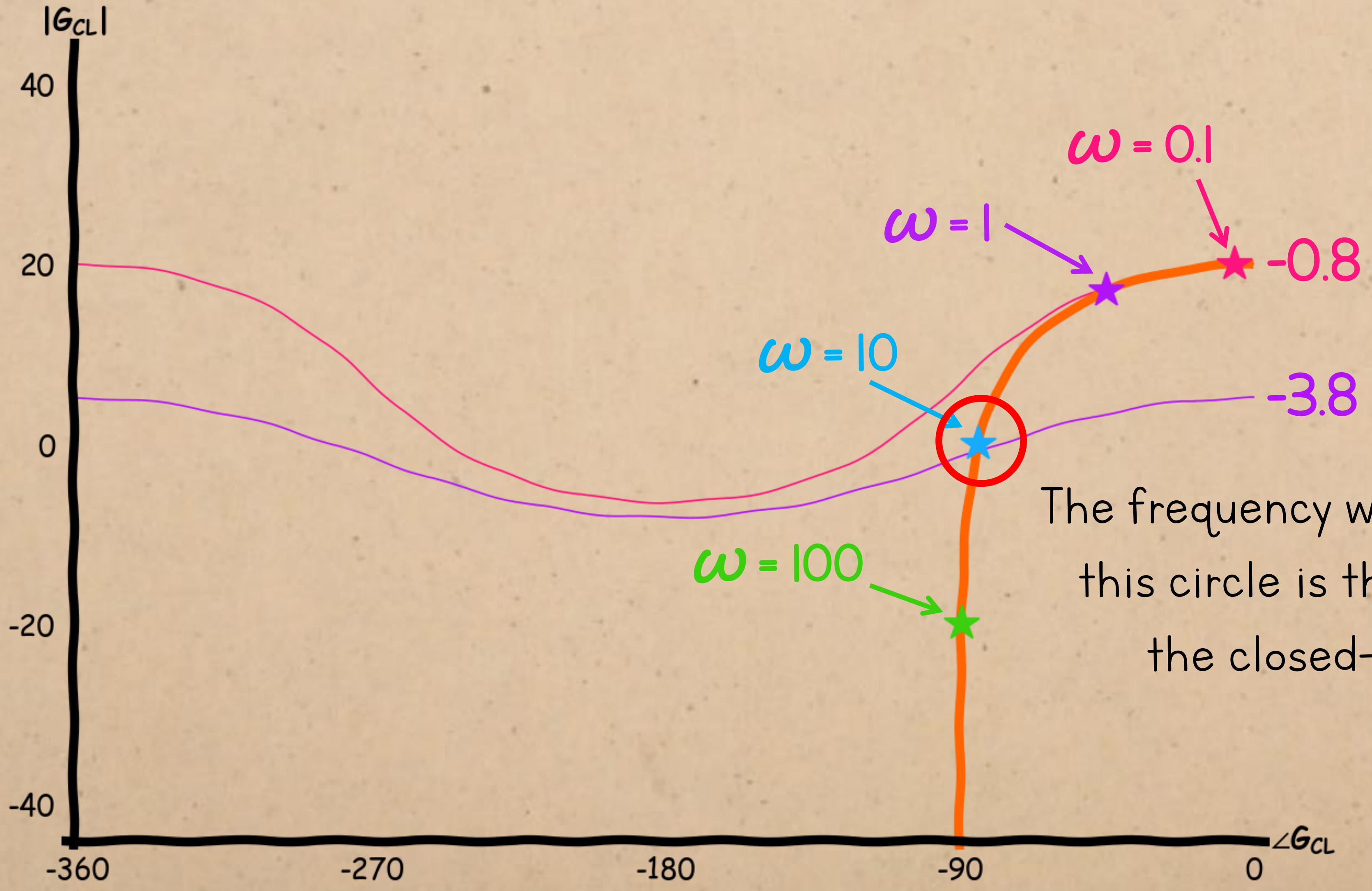
BANDWIDTH



Once you find the closed-loop DC gain

you look for the circle representing the gain 3 dB below that.

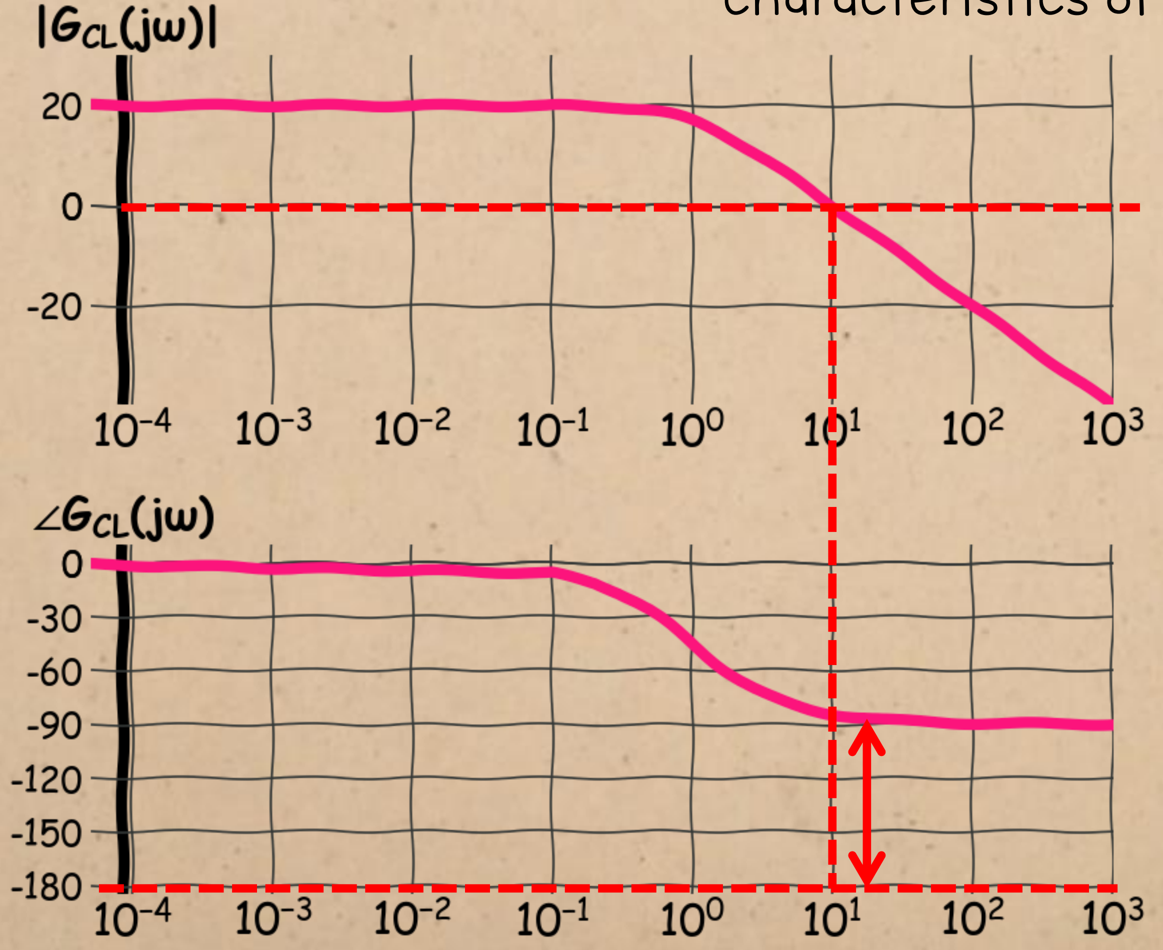
BANDWIDTH



The frequency where it intersects this circle is the bandwidth of the closed-loop system.

DESIGN SPECIFICATIONS

We can relate each of the time-domain parameters to characteristics of the frequency response:



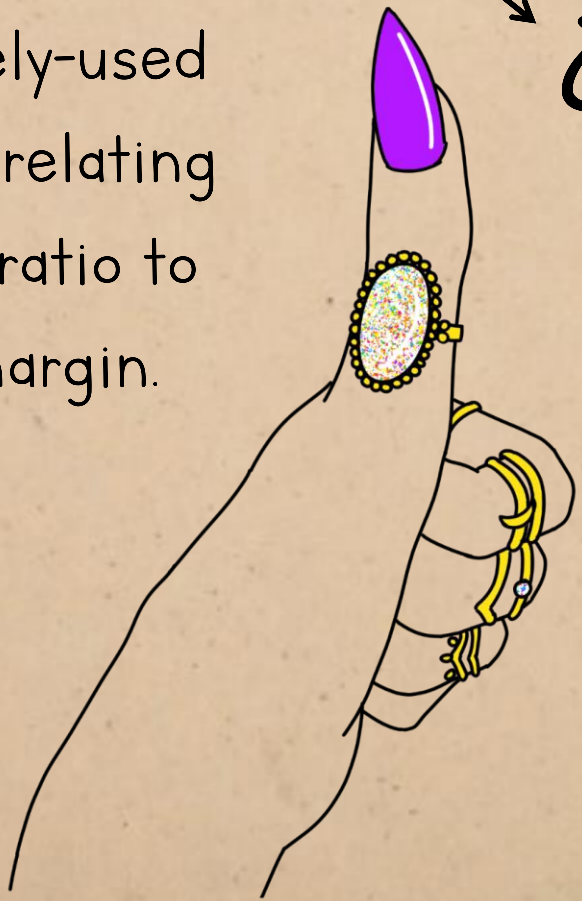
1. **Accuracy (steady-state error)**
   depends on the **DC gain**

2. **Speed (time constant)**
   depends on the **bandwidth**

3. **Damping (damping ratio)**
   depends on the **phase margin**

DAMPING vs. PHASE MARGIN

There's a widely-used **rule of thumb** relating the damping ratio to the phase margin.



damping ratio

closed-loop gain

$$\zeta = \frac{A}{100}\phi_m$$

phase margin (in degrees)

It's useful, but not perfect, so it's worth understanding the true relationship between these parameters so you know its limitations.

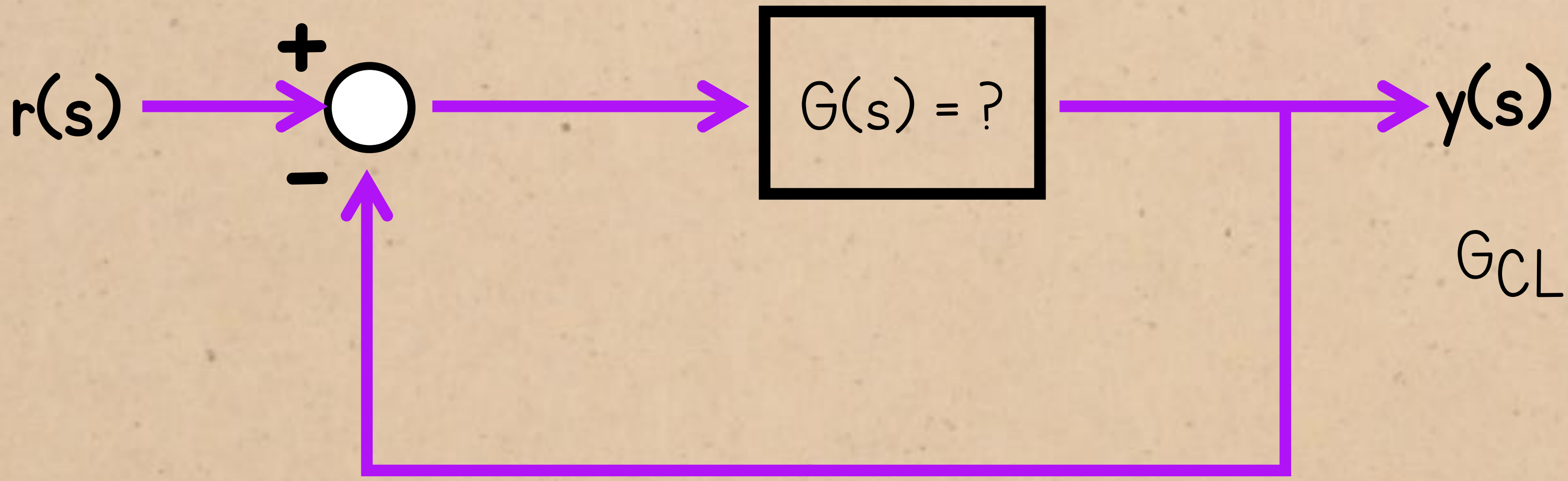
DAMPING vs. PHASE MARGIN

If the system is second order, we can write the closed-loop transfer function in this general form.

$$G_{CL}(s) = \frac{A\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2}$$

(We're assuming the system is second-order just by considering the damping ratio, since this parameter is only defined for second-order systems.)

DAMPING vs. PHASE MARGIN

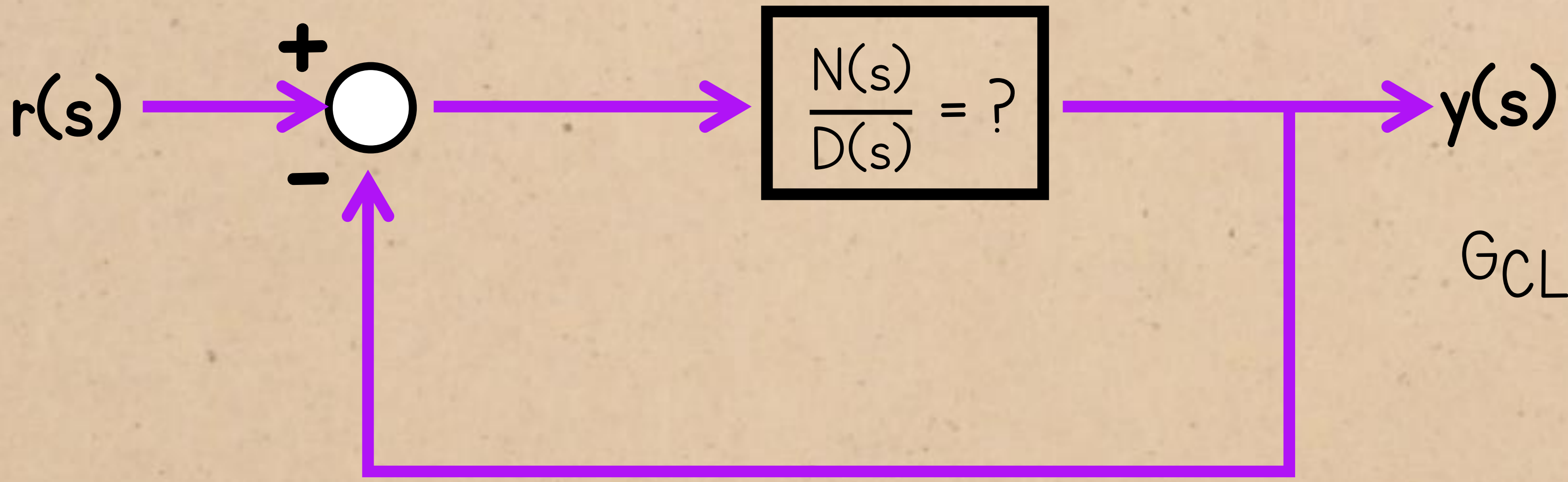


$$G_{CL}(s) = \frac{A\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2}$$

$$= \frac{G(s)}{1 + G(s)}$$

If this is the closed-loop system, what is the open-loop transfer function that produced it?

DAMPING vs. PHASE MARGIN

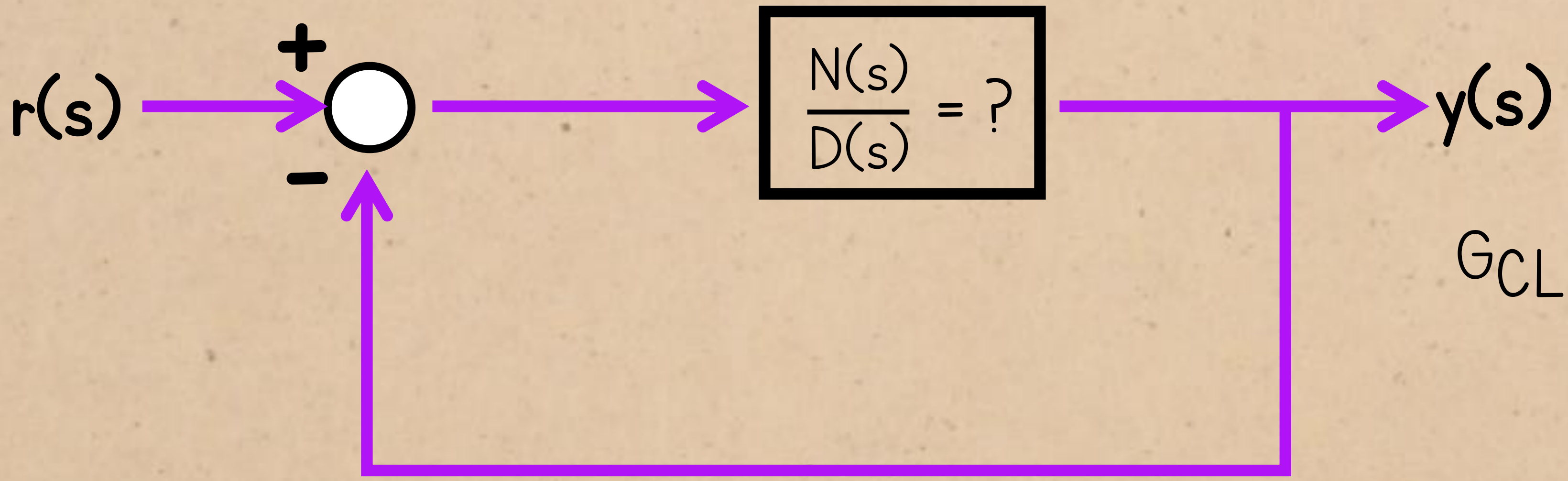


$$G_{CL}(s) = \frac{A\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2}$$

$$= \frac{\frac{N(s)}{D(s)}}{1 + \frac{N(s)}{D(s)}}$$

Write the open-loop transfer function in terms of its numerator N and denominator D:

DAMPING vs. PHASE MARGIN

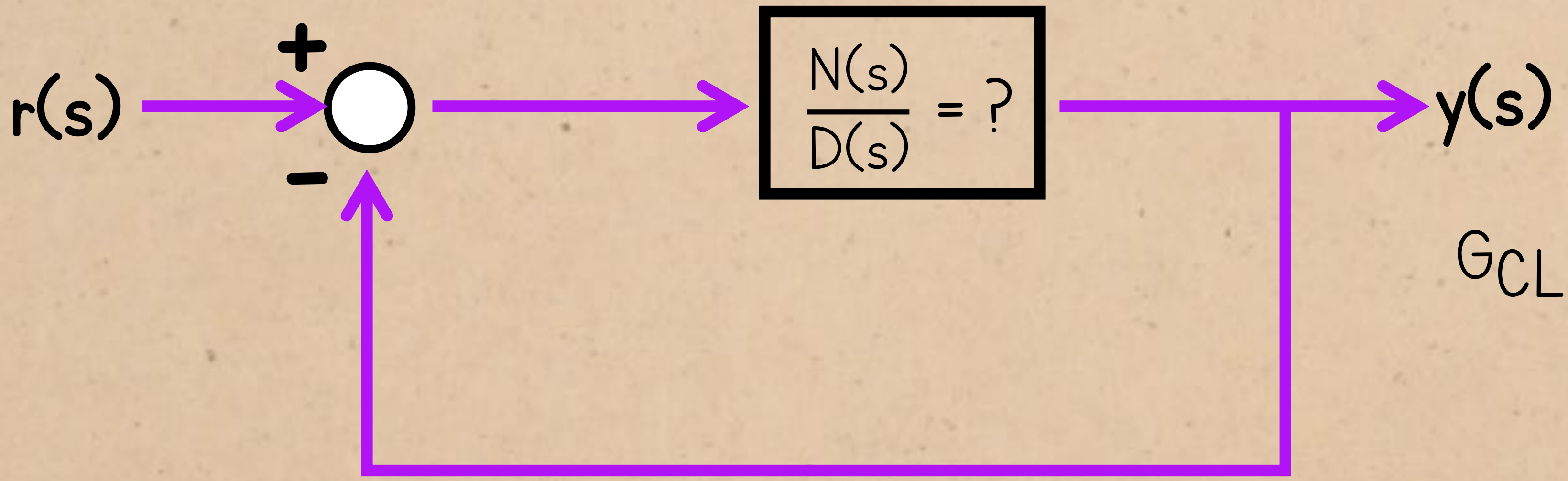


$$G_{CL}(s) = \frac{A\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2}$$

$$= \frac{N(s)}{N(s) + D(s)}$$

Write the open-loop transfer function in terms of its numerator N and denominator D:

DAMPING vs. PHASE MARGIN

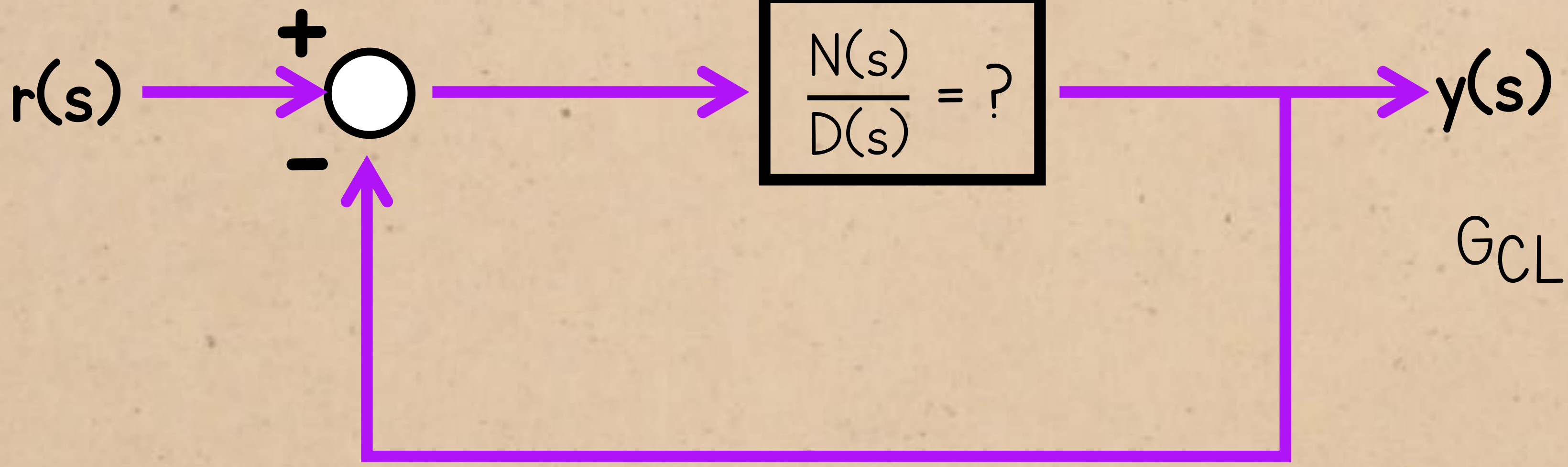


$$G_{CL}(s) = \frac{A\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2}$$

$$= \frac{A\omega_n^2}{A\omega_n^2 + D(s)}$$

By inspection, $N(s) = A\omega_n^2$

DAMPING vs. PHASE MARGIN



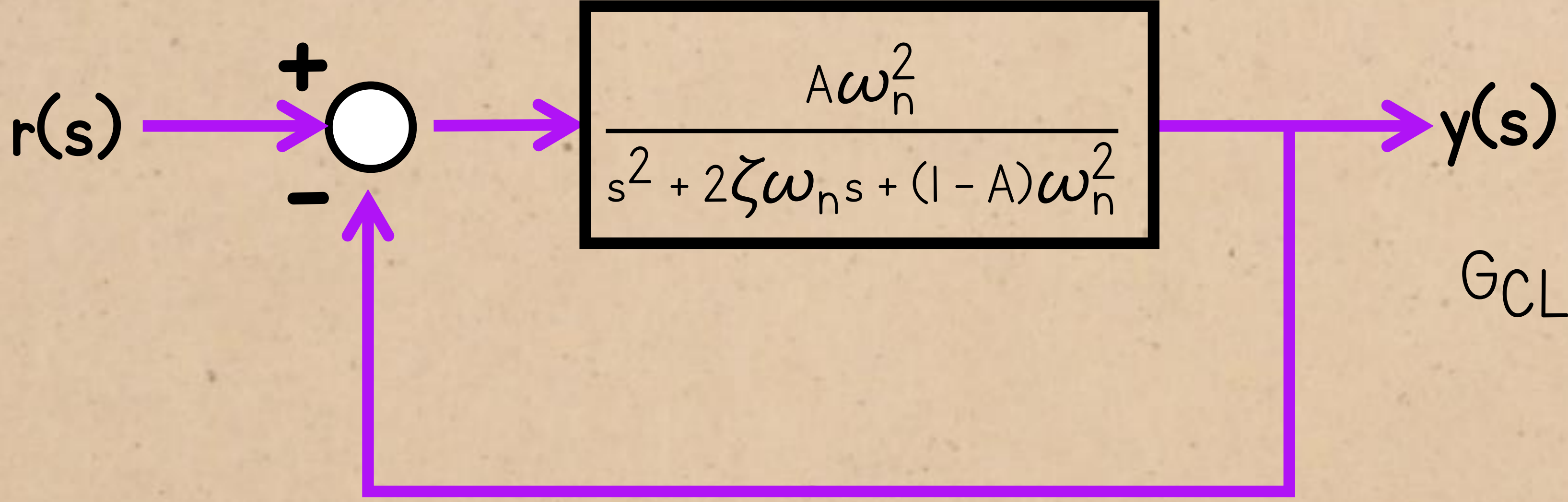
$$G_{CL}(s) = \frac{A\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2}$$

$$= \frac{A\omega_n^2}{A\omega_n^2 + s^2 + 2\zeta\omega_n s + (1 - A)\omega_n^2}$$

So, $D(s) = s^2 + 2\zeta\omega_n s + \omega_n^2 - A\omega_n^2$

$$= s^2 + 2\zeta\omega_n s + (1 - A)\omega_n^2$$

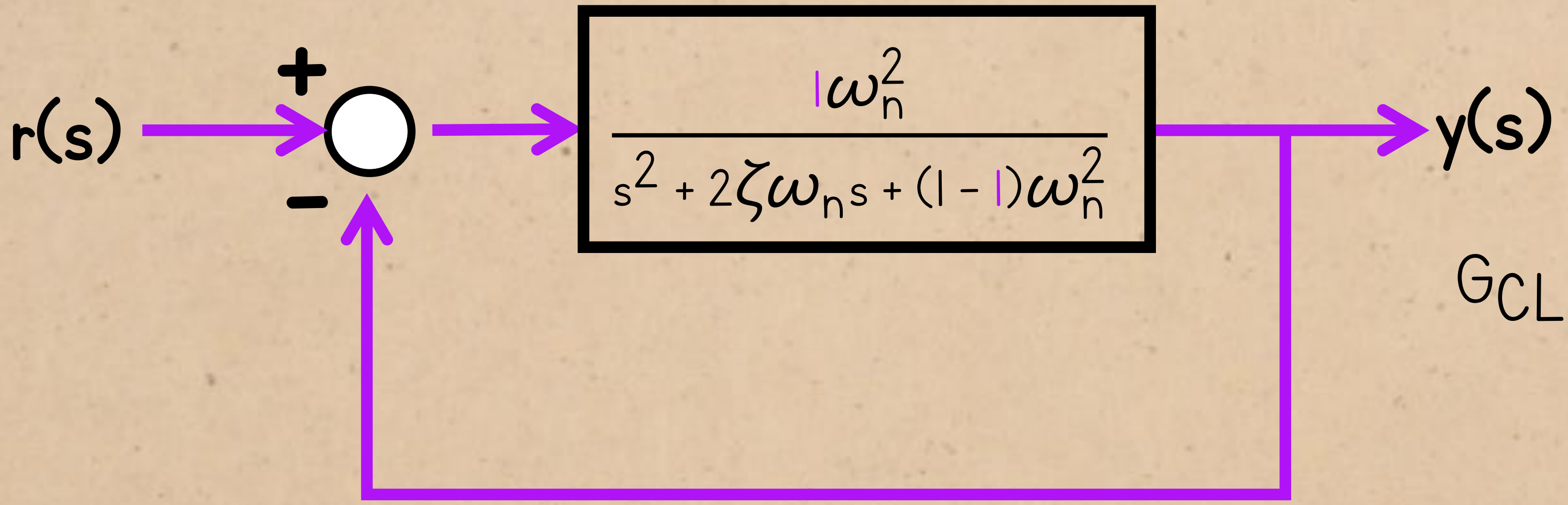
DAMPING vs. PHASE MARGIN



$$G_{CL}(s) = \frac{A\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2}$$

So this is the open-loop transfer function that gives the standard closed-loop form.

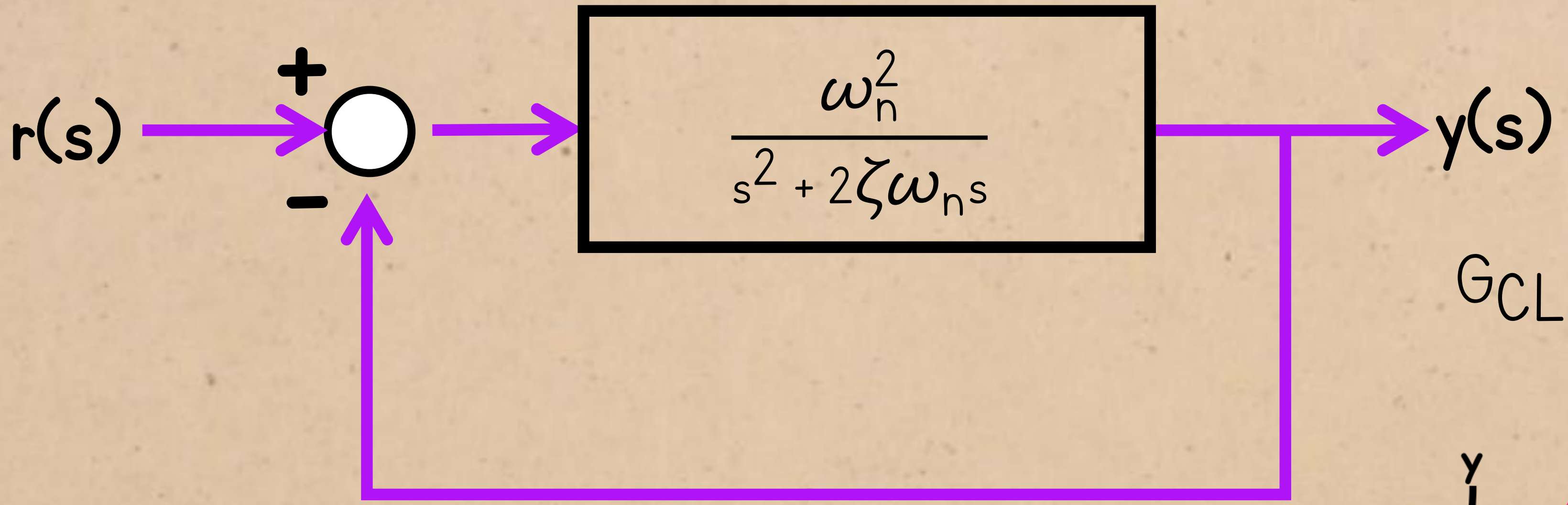
DAMPING vs. PHASE MARGIN



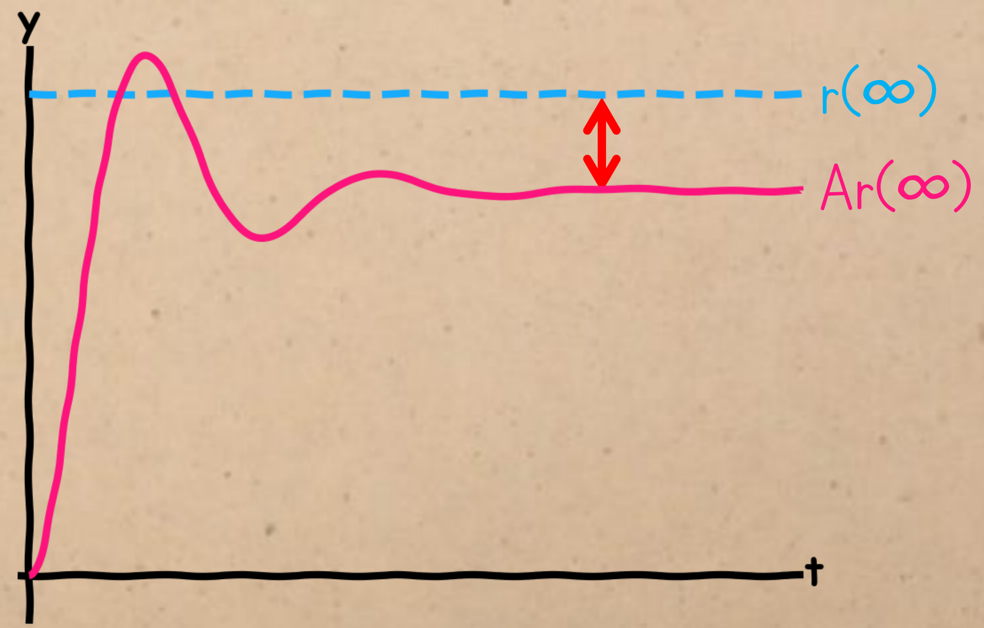
$$G_{CL}(s) = \frac{1\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2}$$

To simplify the derivation, let's assume A = 1

DAMPING vs. PHASE MARGIN

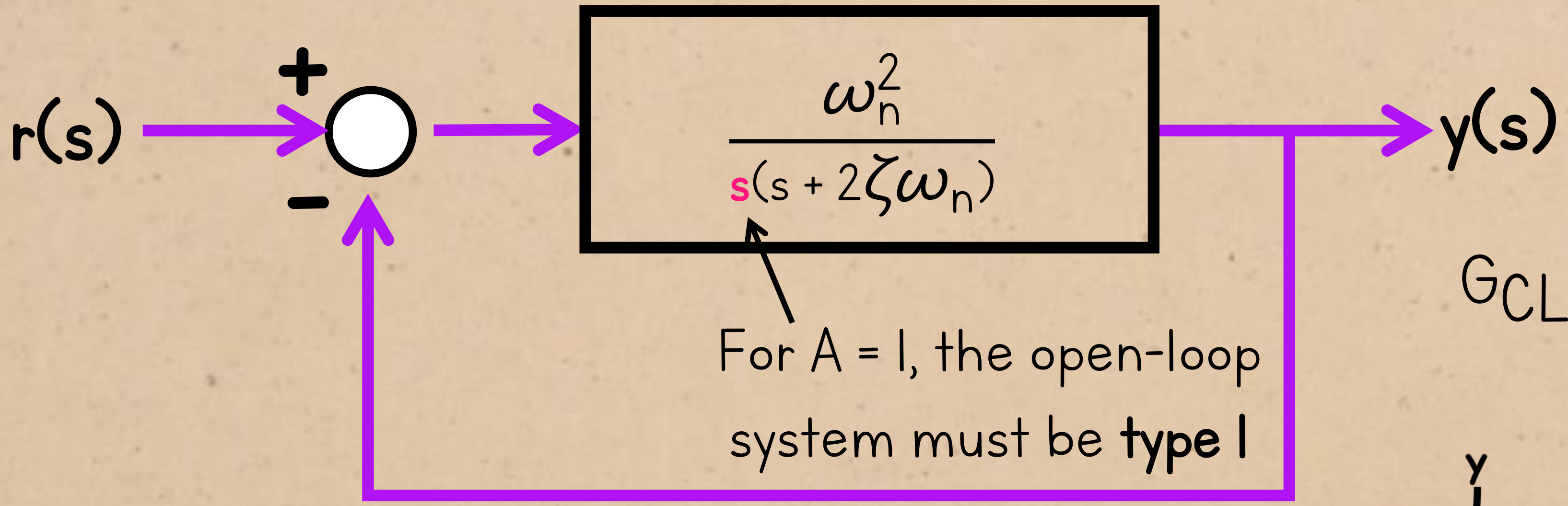


$$G_{CL}(s) = \frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2}$$



We can assume that A = 1, because we need the closed-loop gain to be as close to 1 as possible for accurate setpoint tracking.

DAMPING vs. PHASE MARGIN



$$G_{CL}(s) = \frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2}$$

Notice that we get an integrating in open-loop when we assume the gain is I. This makes sense, given what we know about integrators and accuracy.

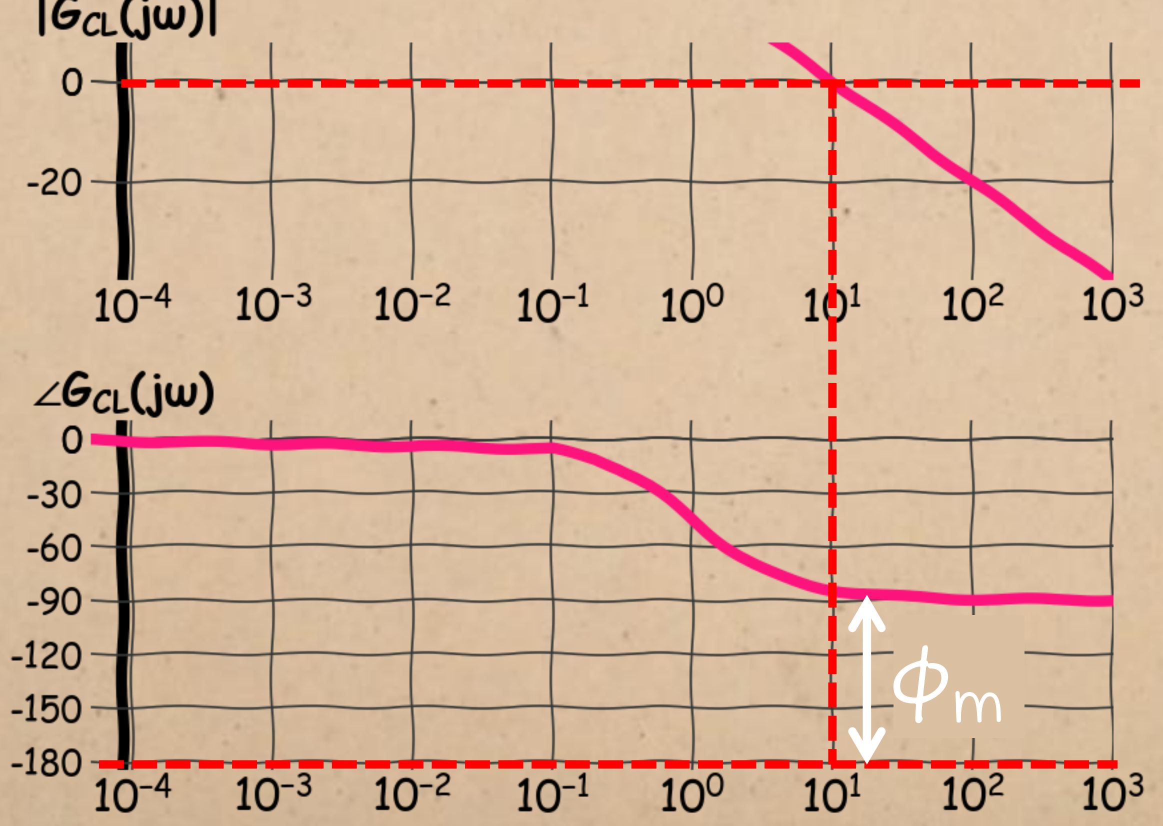
DAMPING vs. PHASE MARGIN

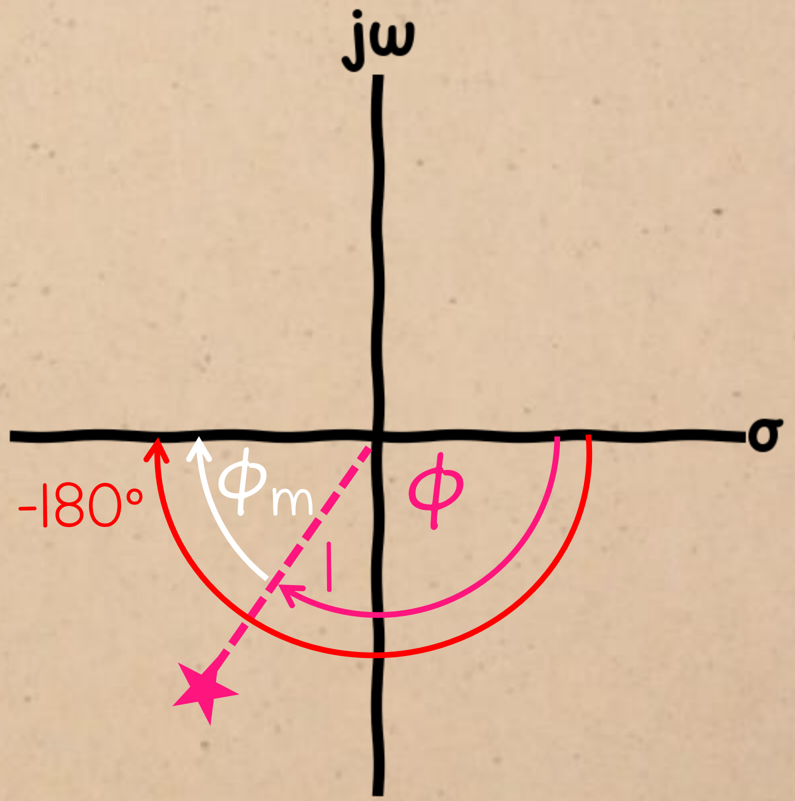
$$\frac{\omega_n^2}{(j\omega)^2 + 2\zeta\omega_n(j\omega)}$$

We need to find the phase margin from the open-loop frequency response.

DAMPING vs. PHASE MARGIN

$$\frac{\omega_n^2}{-\omega^2 + j2\zeta\omega_n\omega}$$

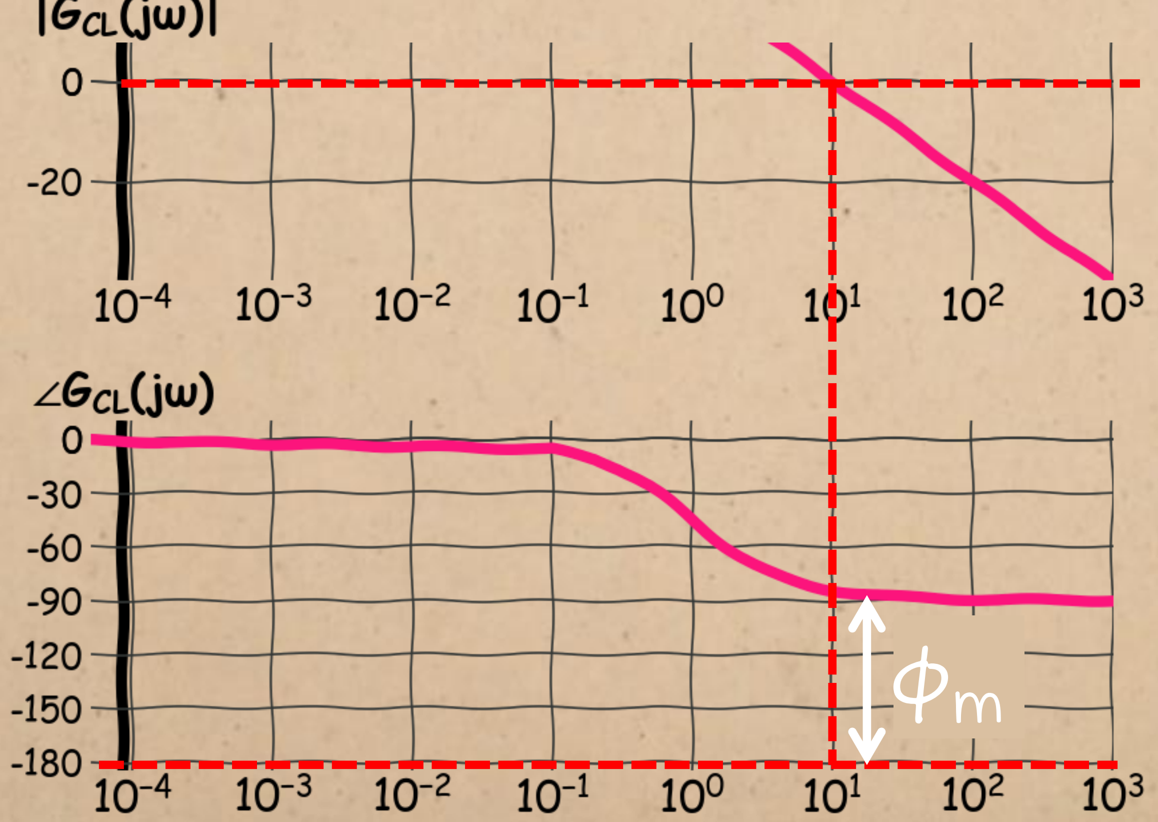


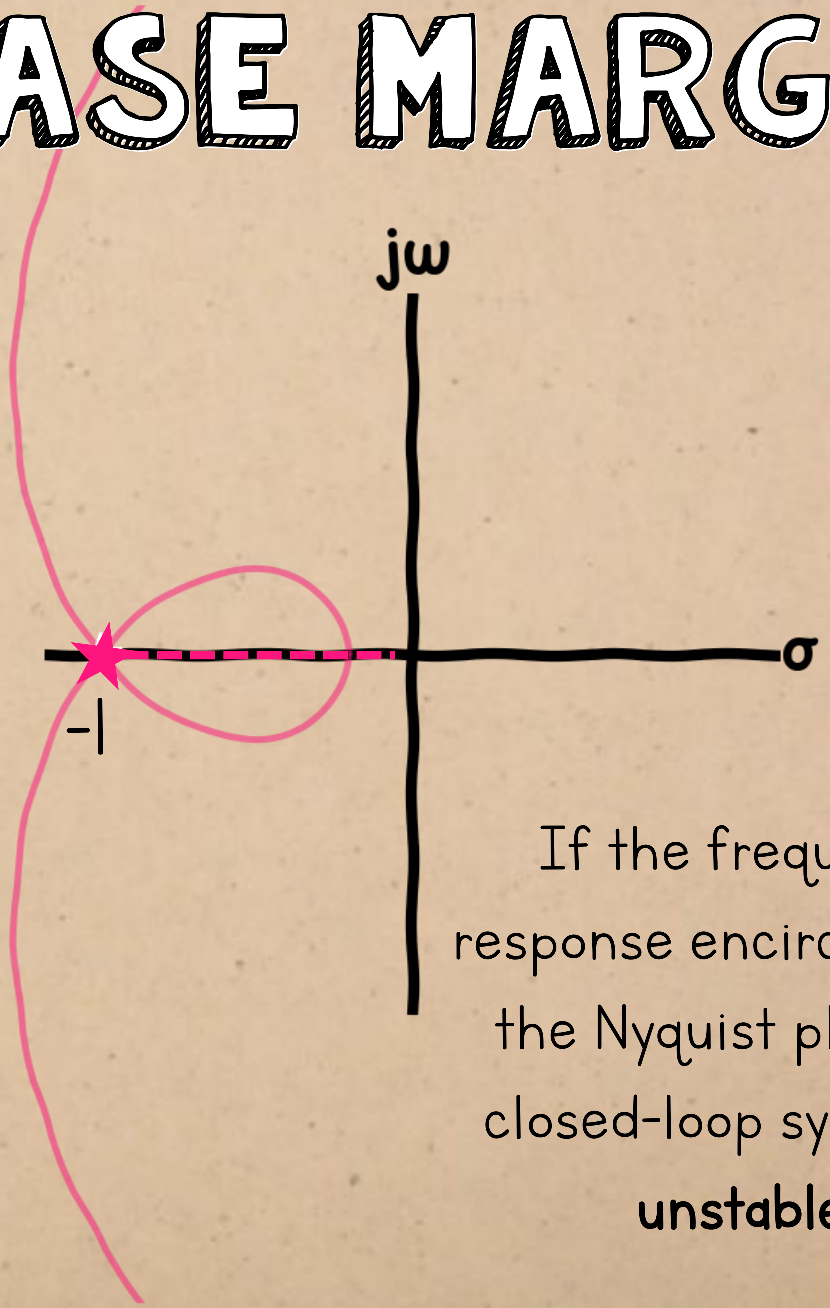


The phase margin is the difference between the system's phase response and -180° when its magnitude = 1

DAMPING vs. PHASE MARGIN

$$\frac{\omega_n^2}{-\omega^2 + j2\zeta\omega_n\omega}$$

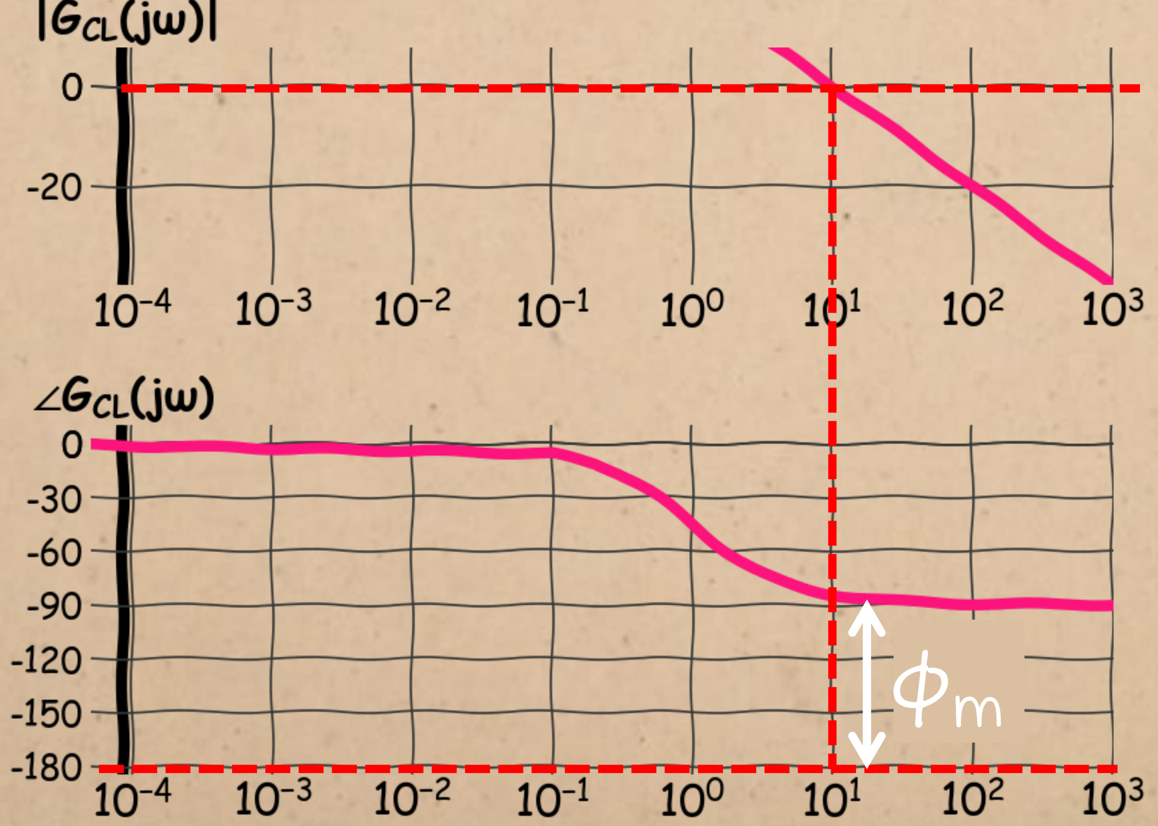


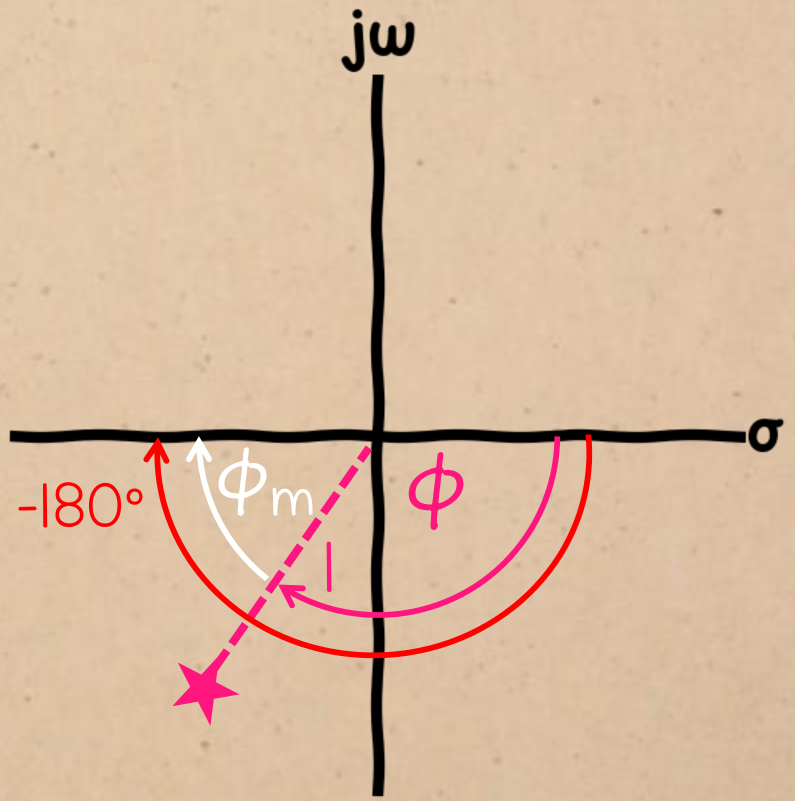


If the frequency response encircles -1 on the Nyquist plot, the closed-loop system is **unstable.**

DAMPING vs. PHASE MARGIN

$$\frac{\omega_n^2}{-\omega^2 + j2\zeta\omega_n\omega}$$

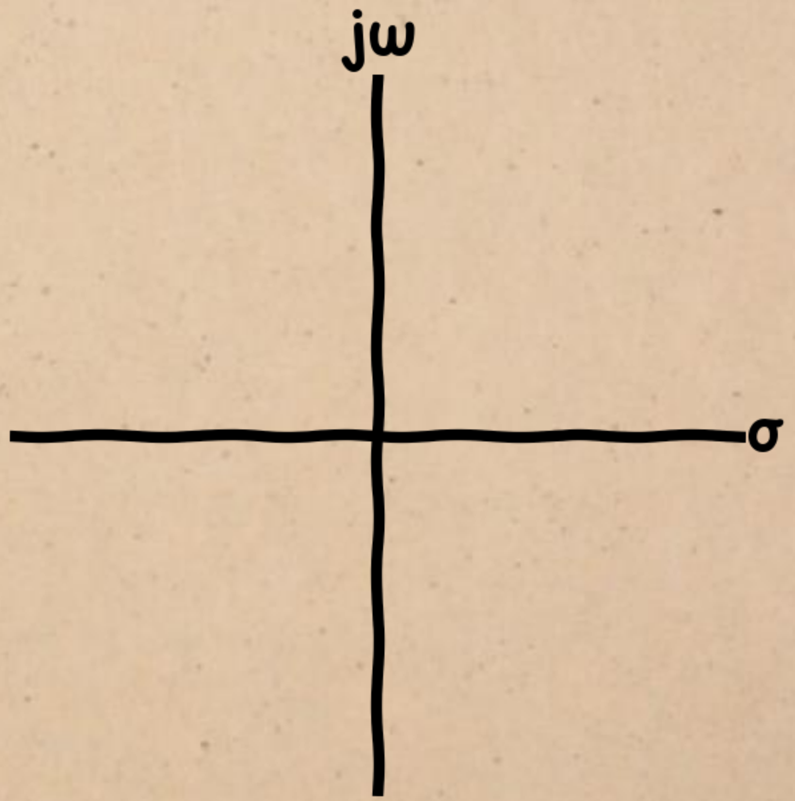




∴ the phase margin shows how much the phase can change before the system loses stability in feedback.

DAMPING vs. PHASE MARGIN

$$\frac{\omega_n^2}{-\omega^2 + j2\zeta\omega_n\omega}$$



How does the damping ratio affect the phase response?

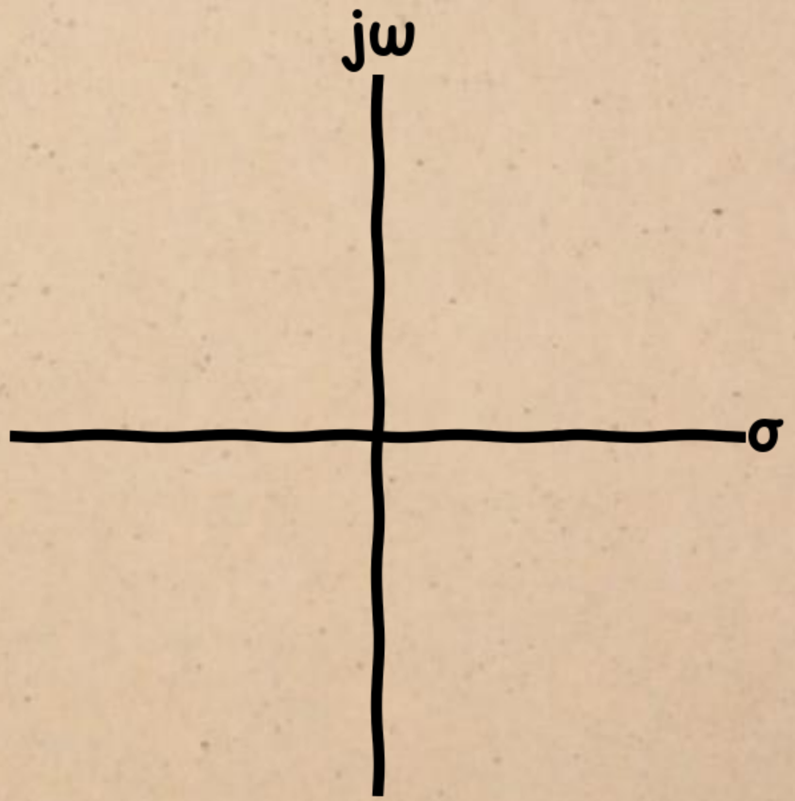
(and ∴ the phase margin?)

DAMPING vs. PHASE MARGIN

$$\frac{\omega_n^2}{-\omega^2 + j2\zeta\omega_n\omega}$$

$$\angle G(j\omega) = \angle\omega_n^2 - \angle(-\omega^2 + j2\zeta\omega_n\omega)$$

The phase is the difference between the numerator and denominator angles.

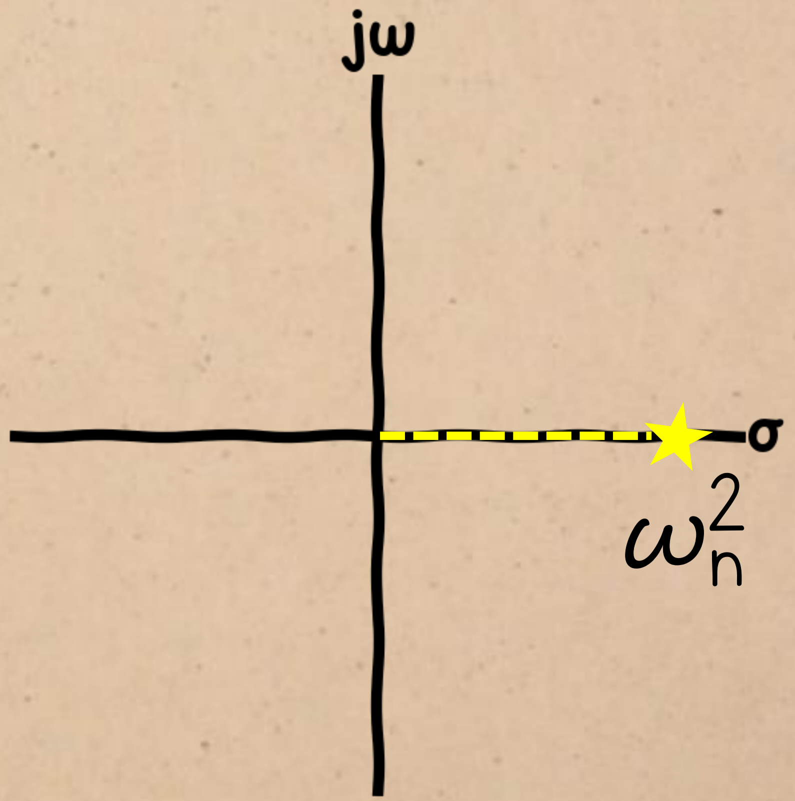


DAMPING vs. PHASE MARGIN

$$\frac{\omega_n^2}{-\omega^2 + j2\zeta\omega_n\omega}$$

$$\angle G(j\omega) = \underbrace{\cancel{\angle\omega_n^2}}_{=0} - \angle(-\omega^2 + j2\zeta\omega_n\omega)$$

The numerator is a positive real number, so its angle is zero.

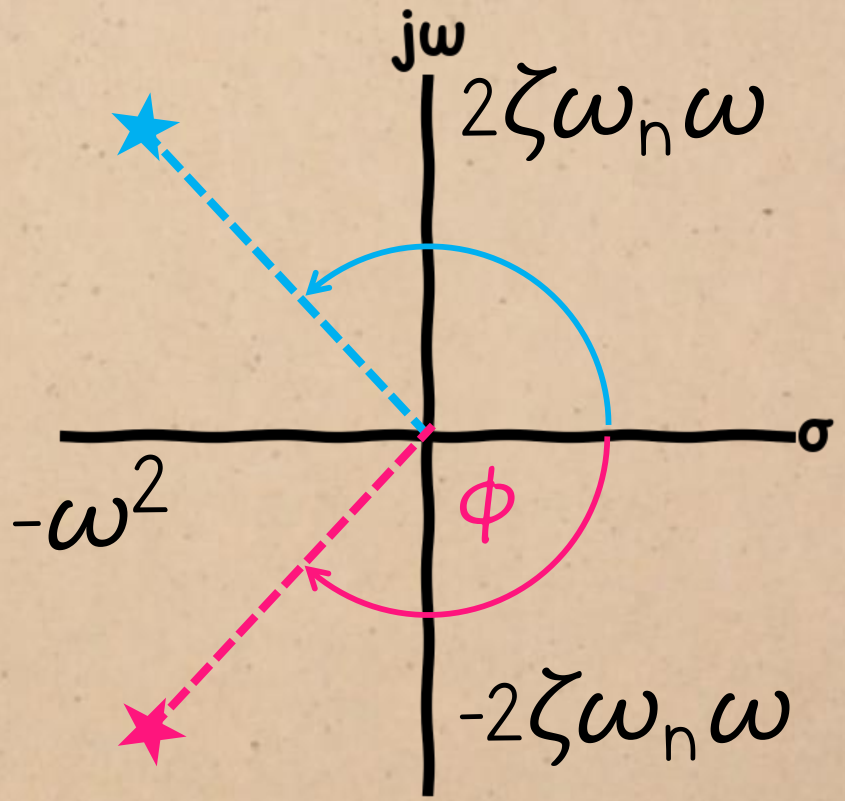


DAMPING vs. PHASE MARGIN

$$\frac{\omega_n^2}{-\omega^2 + j2\zeta\omega_n\omega}$$

$$\angle G(j\omega) = -\angle(-\omega^2 + j2\zeta\omega_n\omega)$$

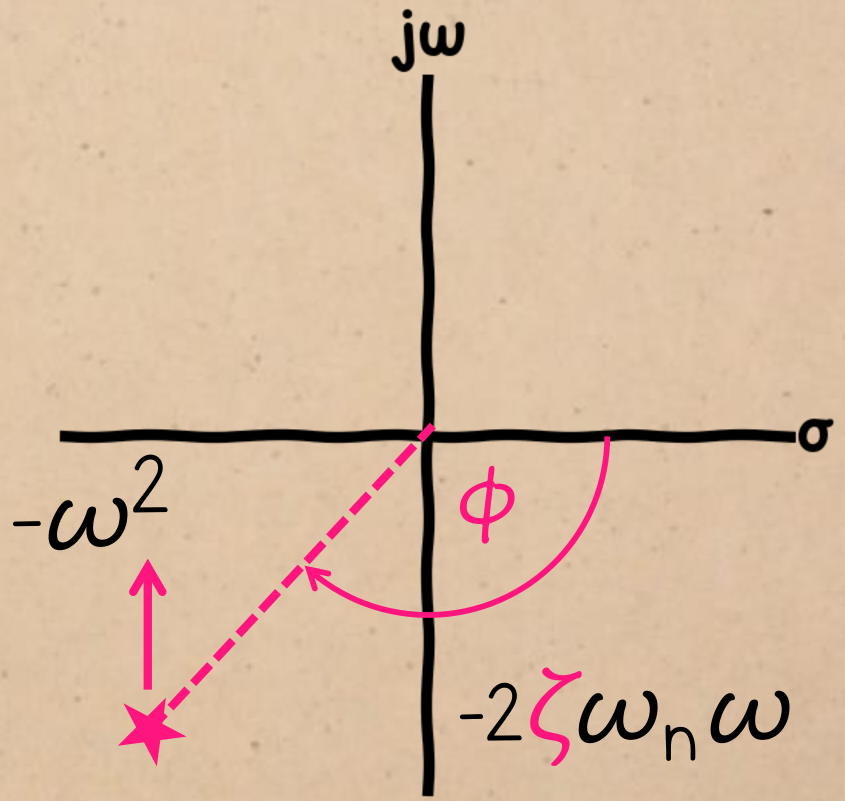
$\therefore$ the phase is just the negative of the denominator angle.



DAMPING vs. PHASE MARGIN

$$\frac{\omega_n^2}{-\omega^2 + j2\zeta\omega_n\omega}$$

$$\angle G(j\omega) = -\angle(-\omega^2 + j2\zeta\omega_n\omega)$$

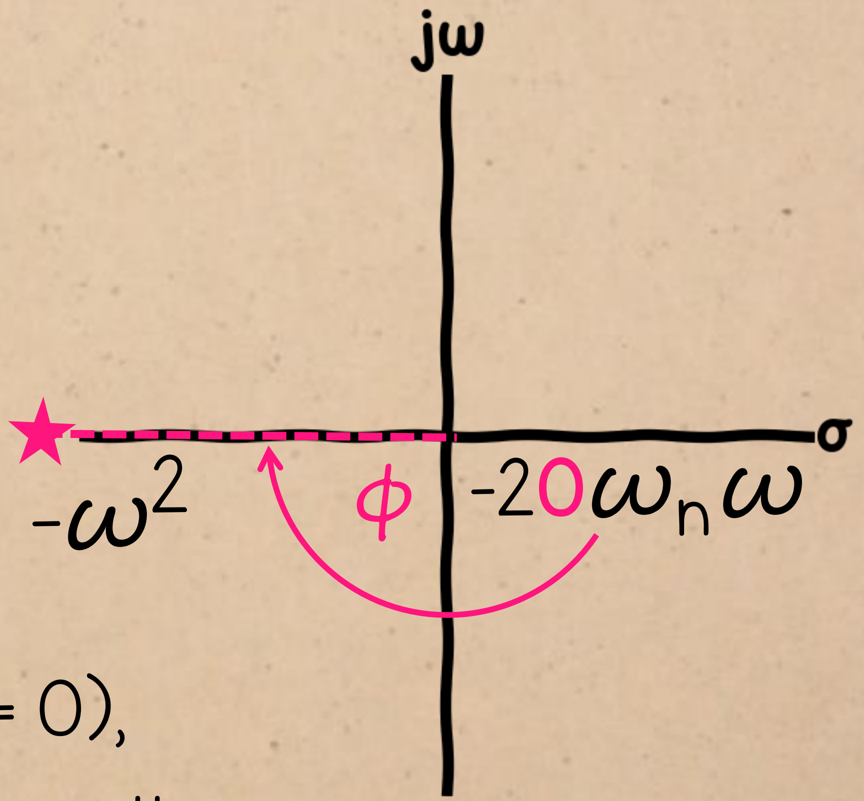


Decreasing the damping ratio will move the point **closer** to the real axis.

DAMPING vs. PHASE MARGIN

$$\frac{\omega_n^2}{-\omega^2 + j2\zeta\omega_n\omega}$$

$$\angle G(j\omega) = -\angle(-\omega^2 + j2\zeta\omega_n\omega)$$

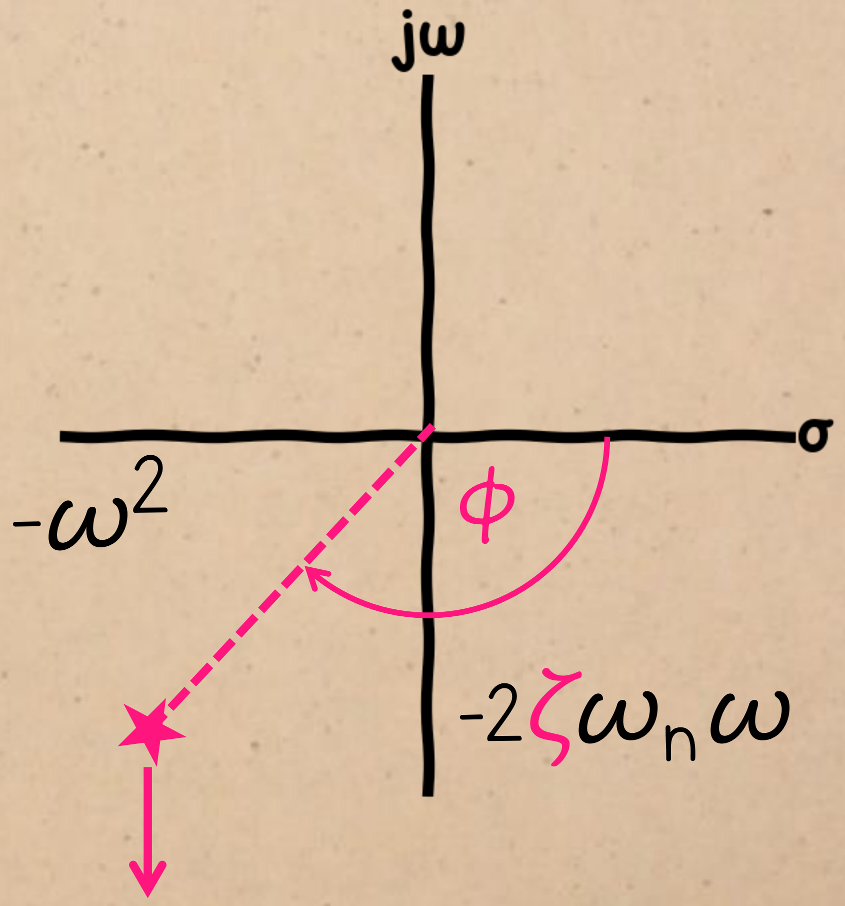


If system is **undamped** (damping ratio = 0), the phase will be -180° for all ω, so there will be **zero phase margin.**

DAMPING vs. PHASE MARGIN

$$\frac{\omega_n^2}{-\omega^2 + j2\zeta\omega_n\omega}$$

$$\angle G(j\omega) = -\angle(-\omega^2 + j2\zeta\omega_n\omega)$$



Increasing the damping increases the angle between this arbitrary point and the real axis.

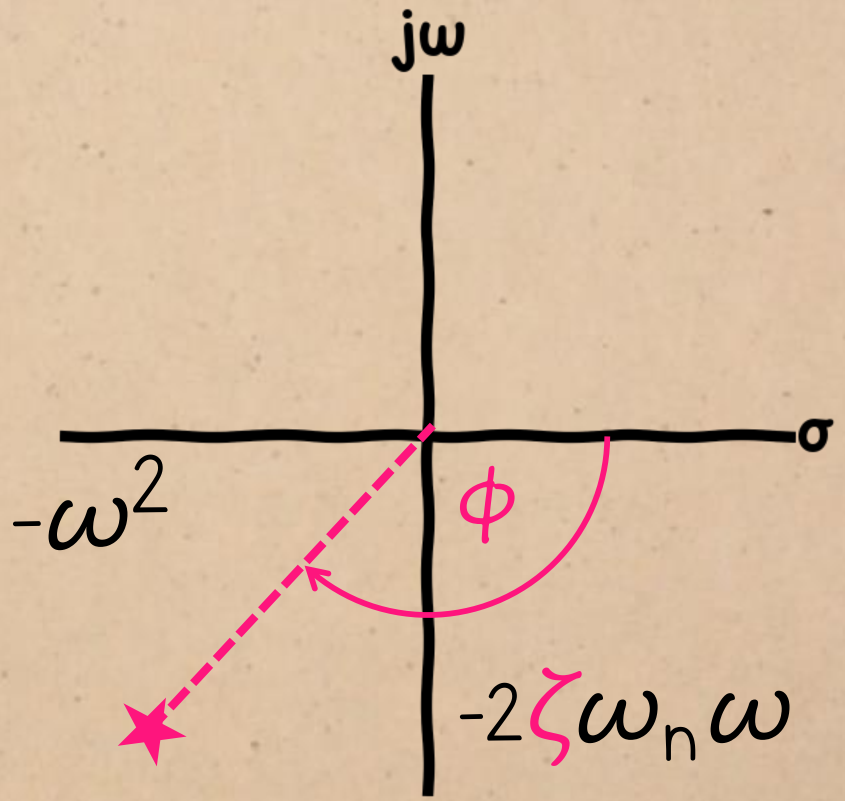
This suggests that

more damping = larger phase margin

DAMPING vs. PHASE MARGIN

$$\frac{\omega_n^2}{-\omega^2 + j2\zeta\omega_n\omega}$$

$$\angle G(j\omega) = -\angle(-\omega^2 + j2\zeta\omega_n\omega)$$



more damping = larger phase margin

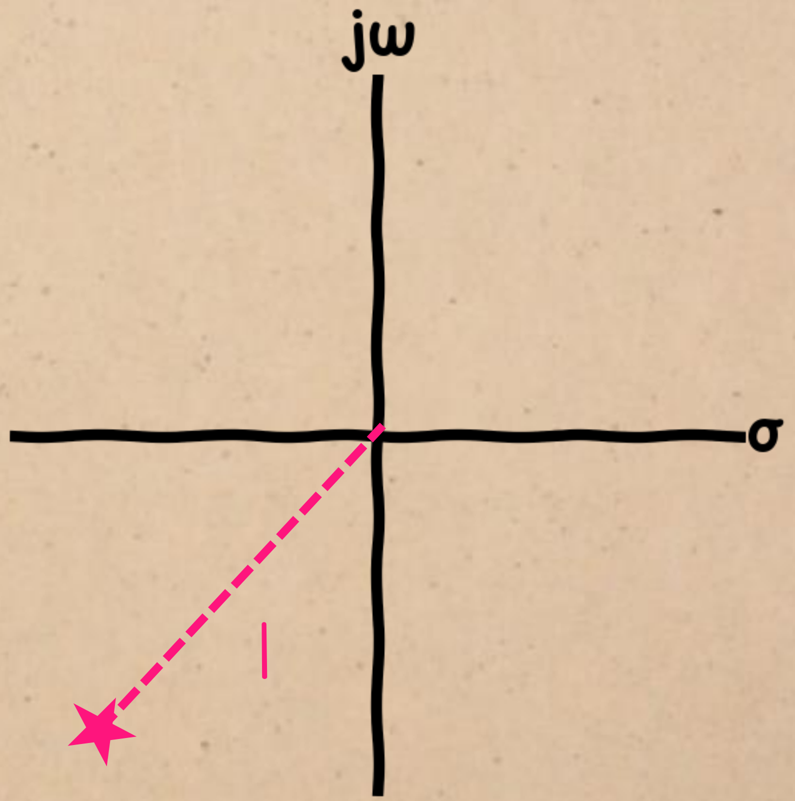
We can confirm this by deriving an expression for the phase margin.

DAMPING vs. PHASE MARGIN

$$|G(j\omega)| = \left| \frac{\omega_n^2}{-\omega_c^2 + j2\zeta\omega_n\omega_c} \right| = 1$$

First, we need to find the gain crossover frequency ω_c.

(The frequency where the magnitude = 1)

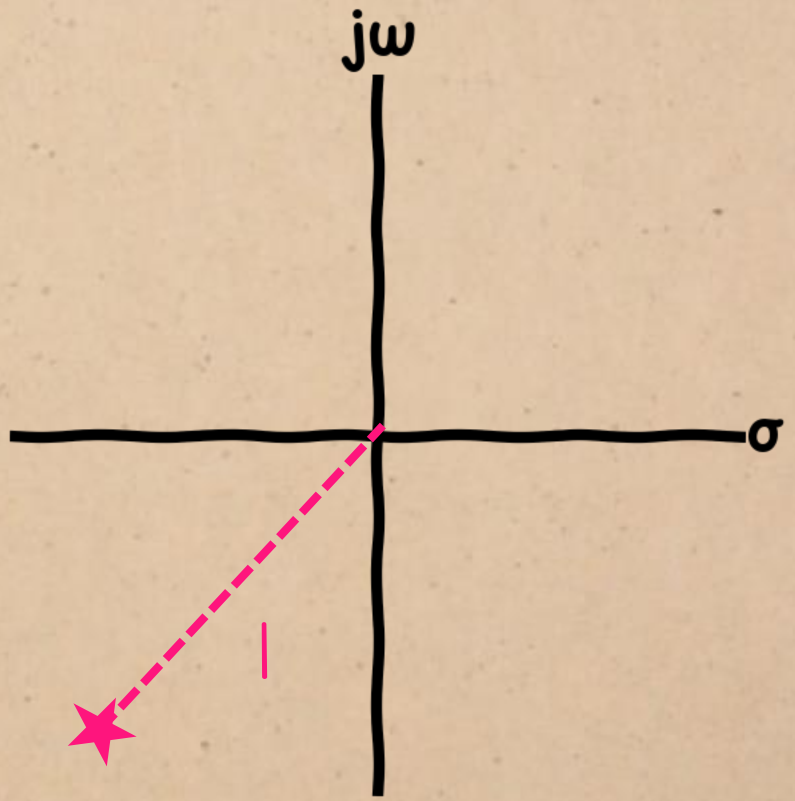


DAMPING vs. PHASE MARGIN

$$|G(j\omega)| = \frac{|\omega_n^2|}{|-\omega_c{}^2 + j2\zeta\omega_n\omega_c|} = 1$$

$$\therefore\ |\omega_n^2| = |-\omega_c{}^2 + j2\zeta\omega_n\omega_c|$$

For the gain to be 1, the numerator and denominator magnitudes must be equal.

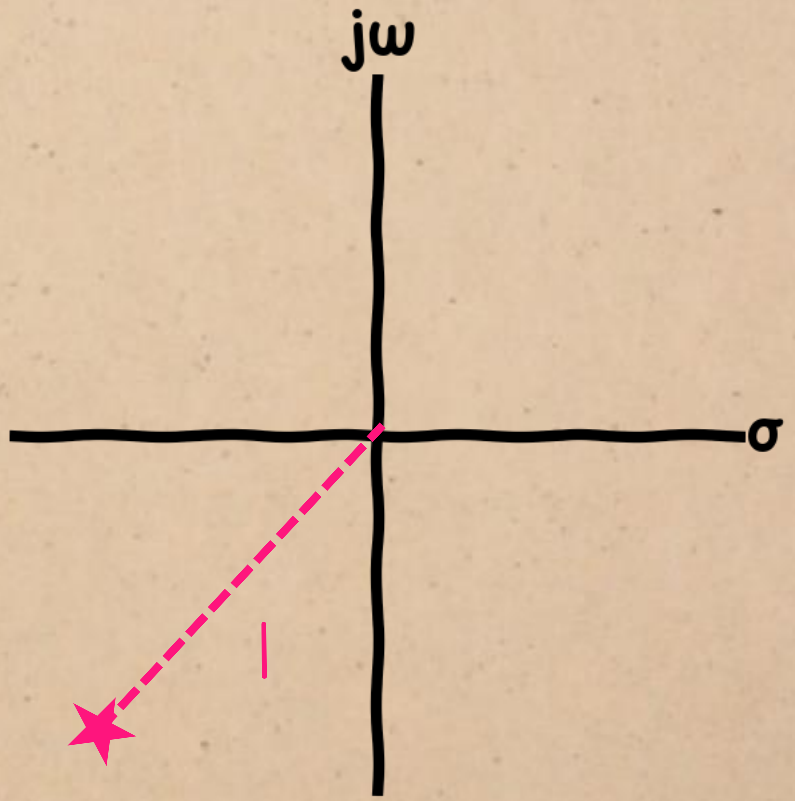


DAMPING vs. PHASE MARGIN

$$|G(j\omega)| = \frac{|\omega_n^2|}{|-\omega_c{}^2 + j2\zeta\omega_n\omega_c|} = 1$$

$$\therefore \omega_n^2 = \sqrt{(\omega_c{}^2)^2 + (2\zeta\omega_n\omega_c)^2}$$

For the gain to be 1, the numerator and denominator magnitudes must be equal.

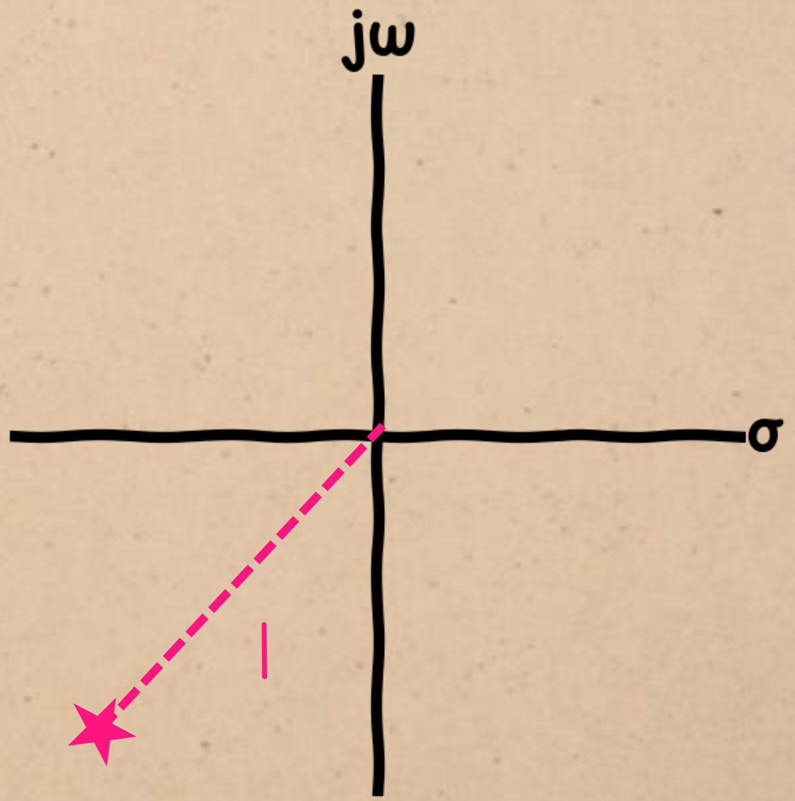


DAMPING vs. PHASE MARGIN

$$|G(j\omega)| = \frac{|\omega_n^2|}{|-\omega_c^{\,2} + j2\zeta\omega_n\omega_c|} = 1$$

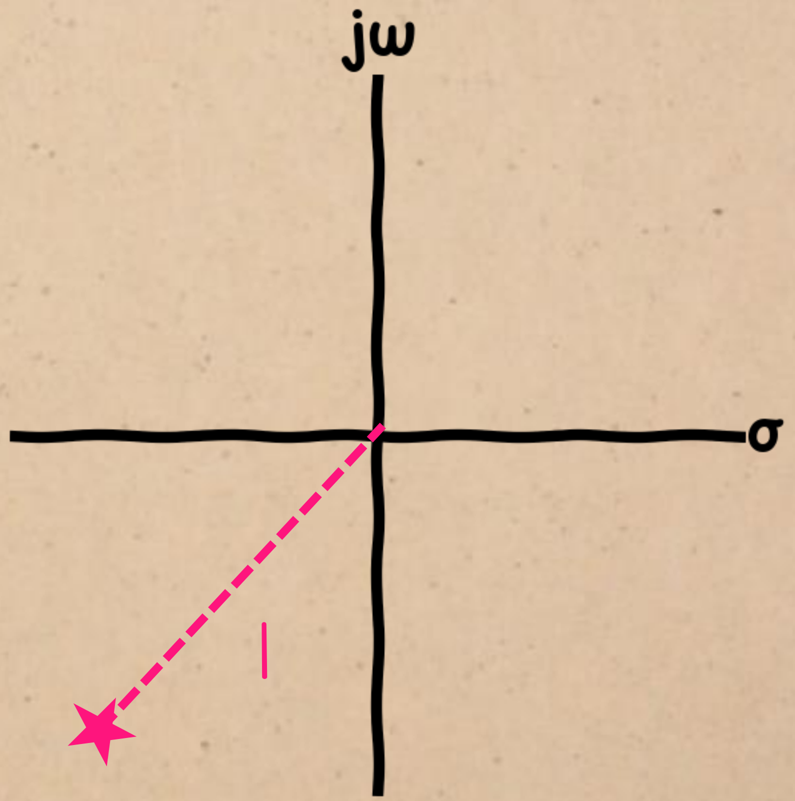
square both sides:

$$\therefore \omega_n^4 = (\omega_c^{\,2})^2 + (2\zeta\omega_n\omega_c)^2$$



DAMPING vs. PHASE MARGIN

$$|G(j\omega)| = \frac{|\omega_n^2|}{|-\omega_c^{\,2} + j2\zeta\omega_n\omega_c|} = 1$$



expand the brackets

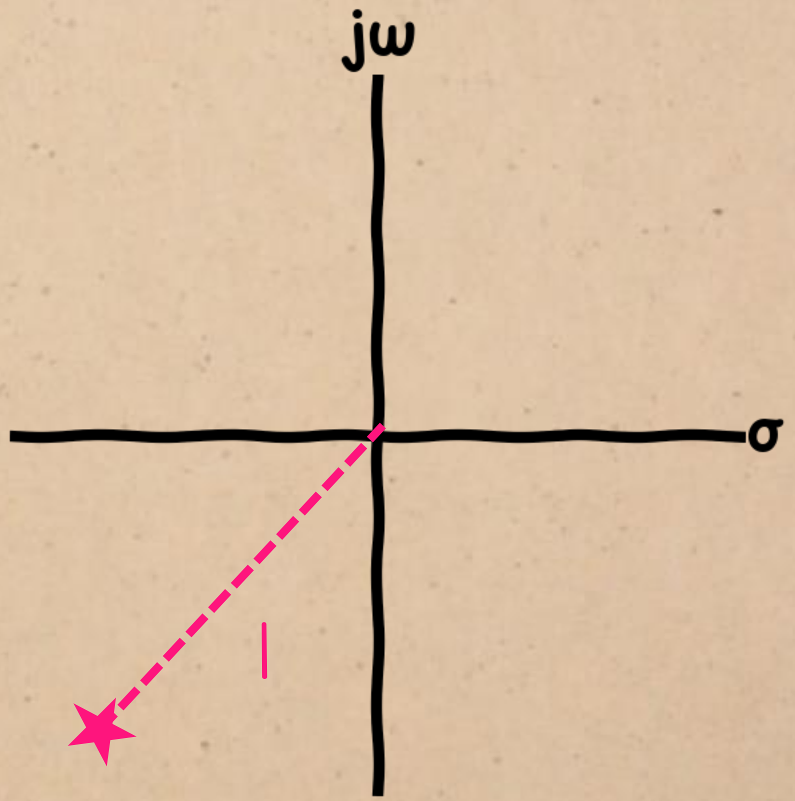
$$\therefore\ \omega_n^4 = \omega_c^{\,4} + 4\zeta^2\omega_n^{\,2}\omega_c^{\,2}$$

DAMPING vs. PHASE MARGIN

$$|G(j\omega)| = \frac{|\omega_n^2|}{|-\omega_c^{\,2} + j2\zeta\omega_n\omega_c|} = 1$$

move everything to one side:

$$\therefore\ \omega_n^4 - \omega_c^{\,4} - 4\zeta^2\omega_n^{\,2}\omega_c^{\,2} = 0$$

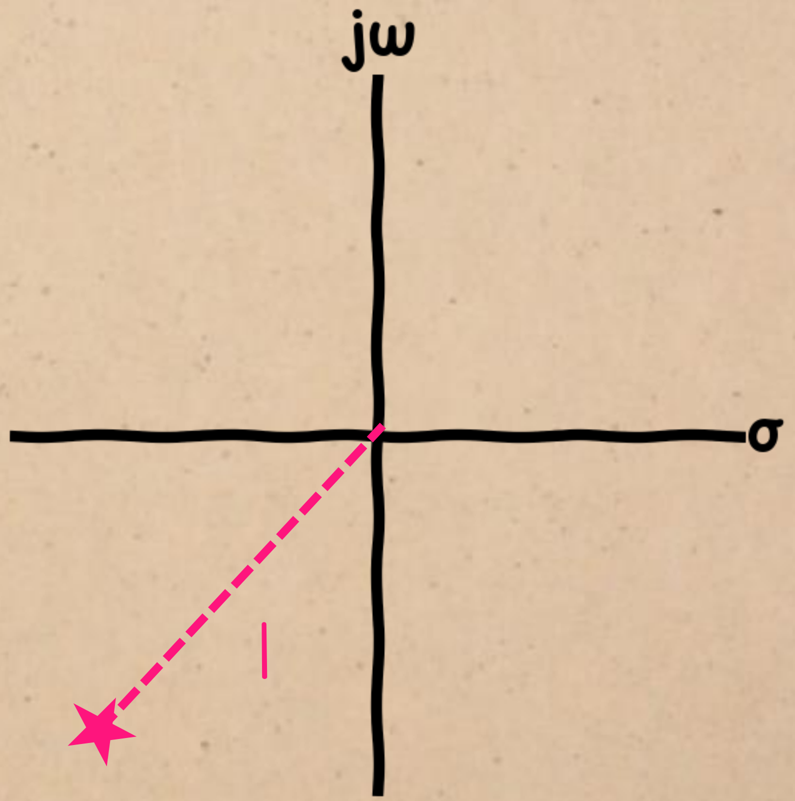


DAMPING vs. PHASE MARGIN

$$|G(j\omega)| = \frac{|\omega_n^2|}{|-\omega_c^{\,2} + j2\zeta\omega_n\omega_c|} = 1$$

divide by ω_n^4:

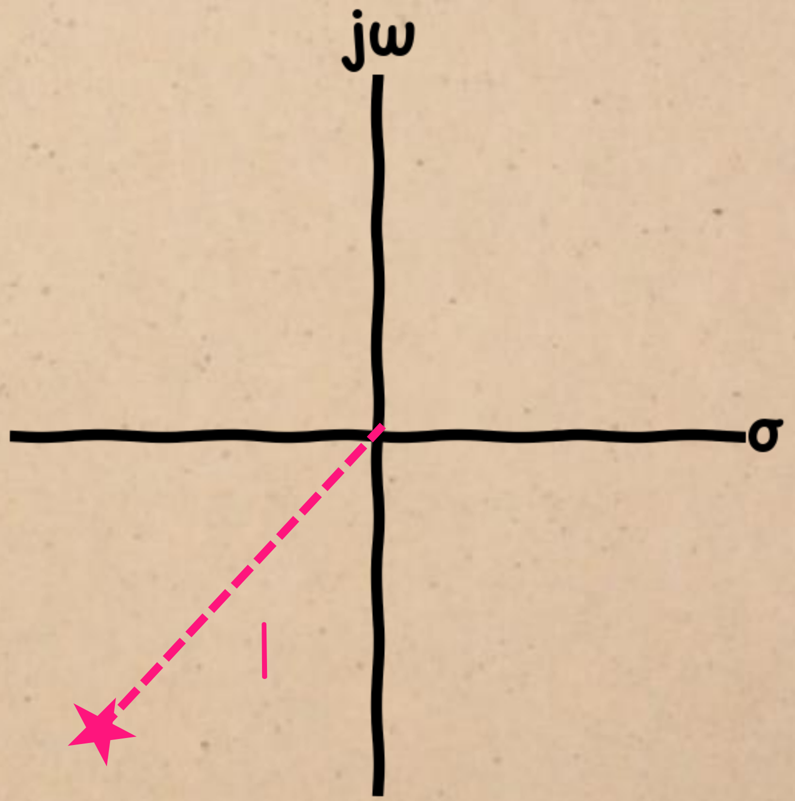
$$\therefore 1 - \frac{\omega_c^{\,4}}{\omega_n^{\,4}} - 4\zeta^2\frac{\omega_c^{\,2}}{\omega_n^{\,2}} = 0$$



DAMPING vs. PHASE MARGIN

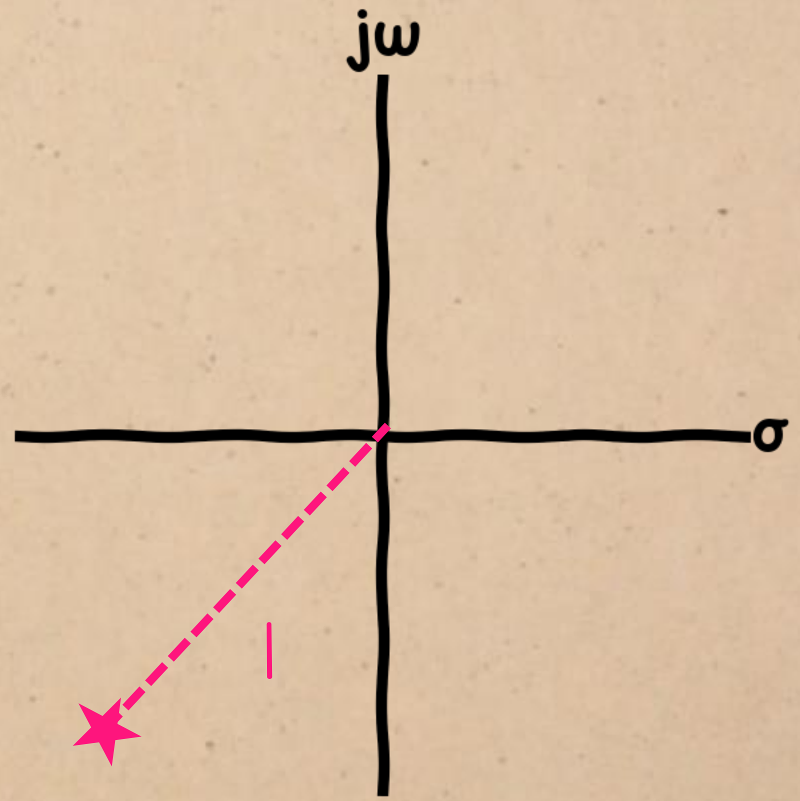
$$|G(j\omega)| = \frac{|\omega_n^2|}{|-\omega_c^{\,2} + j2\zeta\omega_n\omega_c|} = 1$$

$$\therefore \left(\frac{\omega_c^{\,2}}{\omega_n^{\,2}}\right)^4 + 4\zeta^2\frac{\omega_c^{\,2}}{\omega_n^{\,2}} - 1 = 0$$



DAMPING vs. PHASE MARGIN

$$|G(j\omega)| = \frac{|\omega_n^2|}{|-\omega_c^2 + j2\zeta\omega_n\omega_c|} = 1$$



$$\therefore \left(\frac{\omega_c^2}{\omega_n^2}\right)^4 + 4\zeta^2 \frac{\omega_c^2}{\omega_n^2} - 1 = 0$$

We have a quadratic equation in $\frac{\omega_c^2}{\omega_n^2}$

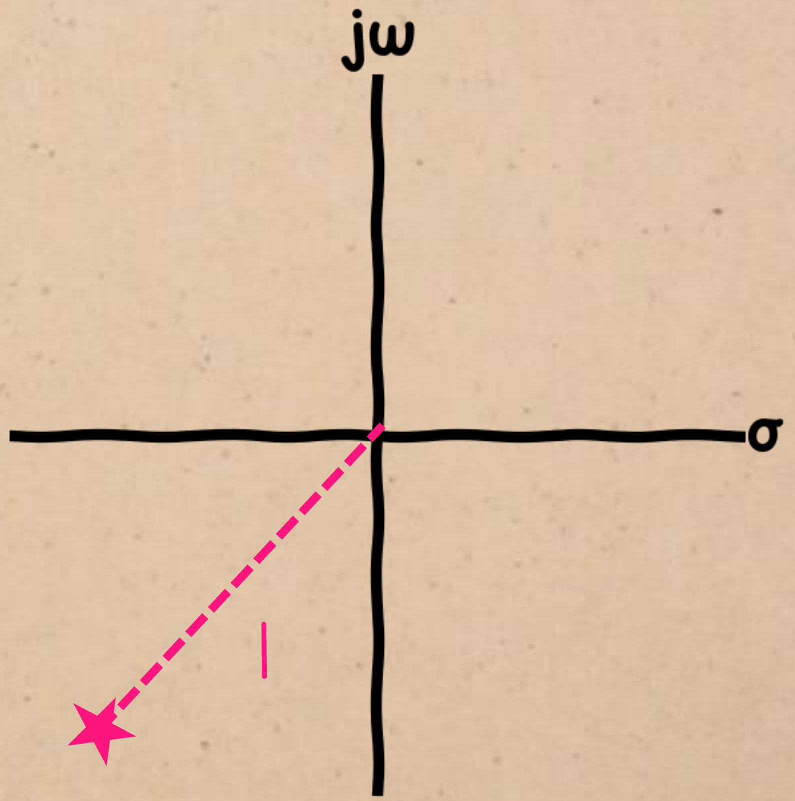
DAMPING vs. PHASE MARGIN

$$|G(j\omega)| = \frac{|\omega_n^2|}{|-\omega_c^{\,2} + j2\zeta\omega_n\omega_c|} = 1$$

$$\therefore \left(\frac{\omega_c^{\,2}}{\omega_n^{\,2}}\right)^4 + 4\zeta^2\frac{\omega_c^{\,2}}{\omega_n^{\,2}} - 1 = 0$$

From the quadratic formula,

$$\frac{\omega_c^{\,2}}{\omega_n^{\,2}} = -2\zeta^2 \pm \sqrt{4\zeta^4 + 1}$$



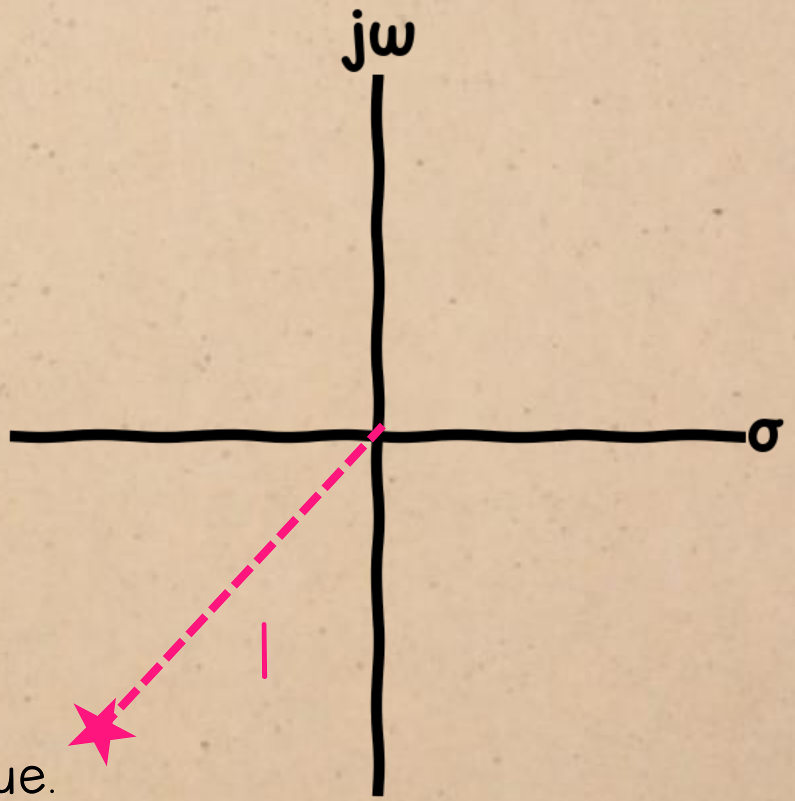
DAMPING vs. PHASE MARGIN

$$|G(j\omega)| = \frac{|\omega_n^2|}{|-\omega_c^{\,2} + j2\zeta\omega_n\omega_c|} = 1$$

$$\therefore \left(\frac{\omega_c^{\,2}}{\omega_n^{\,2}}\right)^4 + 4\zeta^2\frac{\omega_c^{\,2}}{\omega_n^{\,2}} - 1 = 0$$

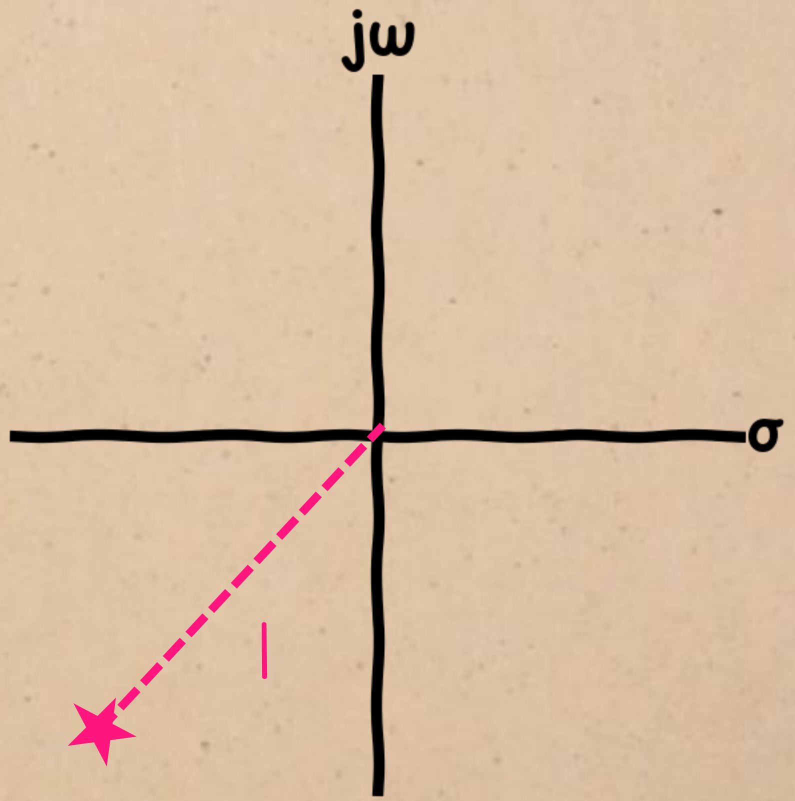
We'll take the + solution, because we need a **positive** value.

$$\frac{\omega_c^{\,2}}{\omega_n^{\,2}} = -2\zeta^2 + \sqrt{4\zeta^4 + 1}$$



DAMPING vs. PHASE MARGIN

$$|G(j\omega)| = \frac{|\omega_n^2|}{|-\omega_c^2 + j2\zeta\omega_n\omega_c|} = 1$$

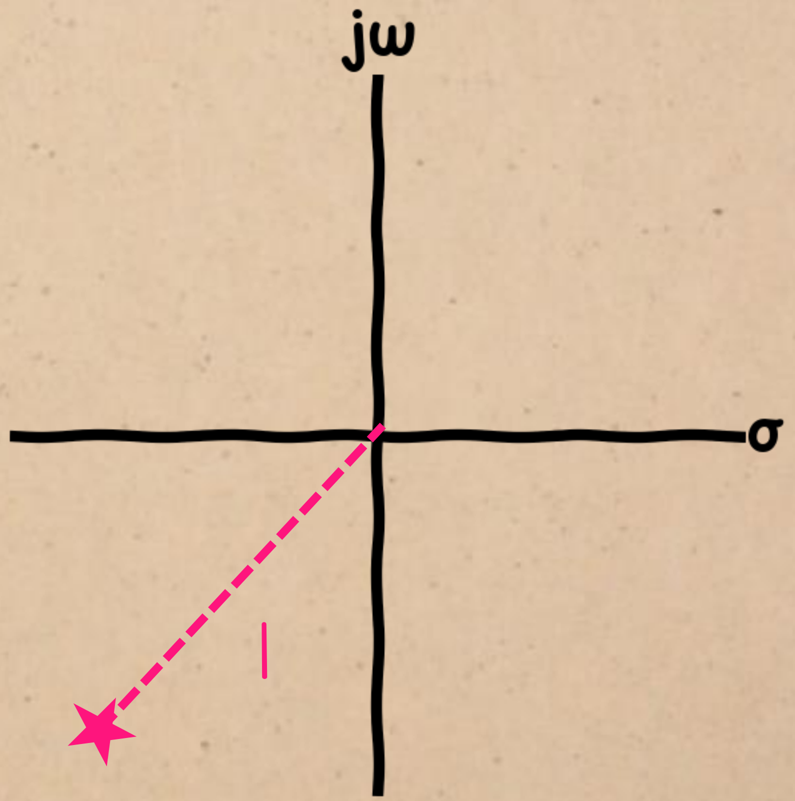


multiply out:

$$\omega_c^2 = \omega_n^2\left(-2\zeta^2 + \sqrt{4\zeta^4 + 1}\right)$$

DAMPING vs. PHASE MARGIN

$$|G(j\omega)| = \frac{|\omega_n^2|}{|-\omega_c^{\,2} + j2\zeta\omega_n\omega_c|} = 1$$

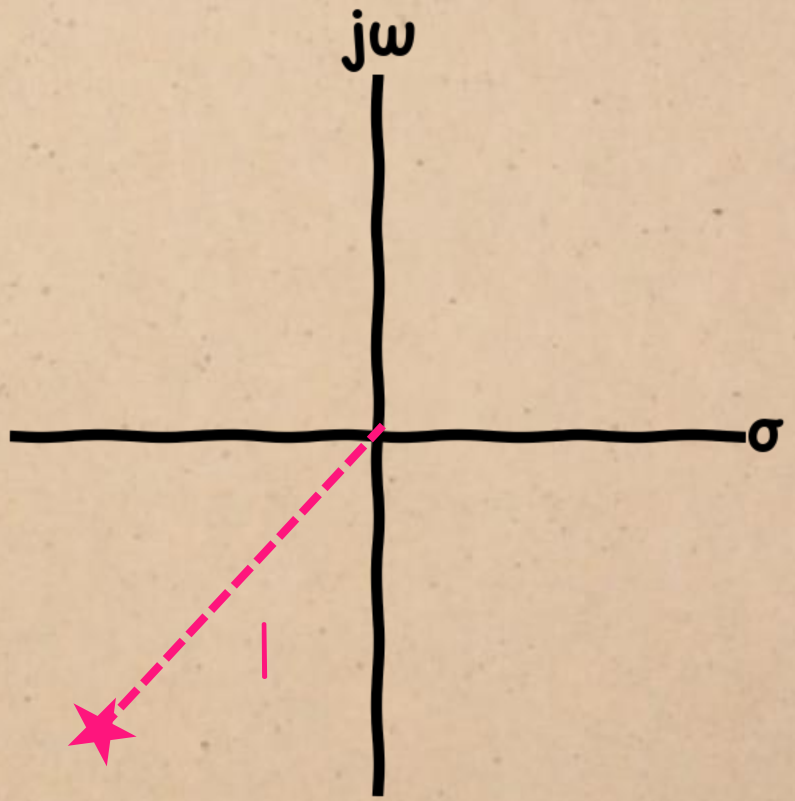


take the square root:

$$\omega_c = \omega_n\sqrt{-2\zeta^2 + \sqrt{4\zeta^4 + 1}}$$

DAMPING vs. PHASE MARGIN

$$|G(j\omega)| = \frac{|\omega_n^2|}{|-\omega_c^{\,2} + j2\zeta\omega_n\omega_c|} = 1$$

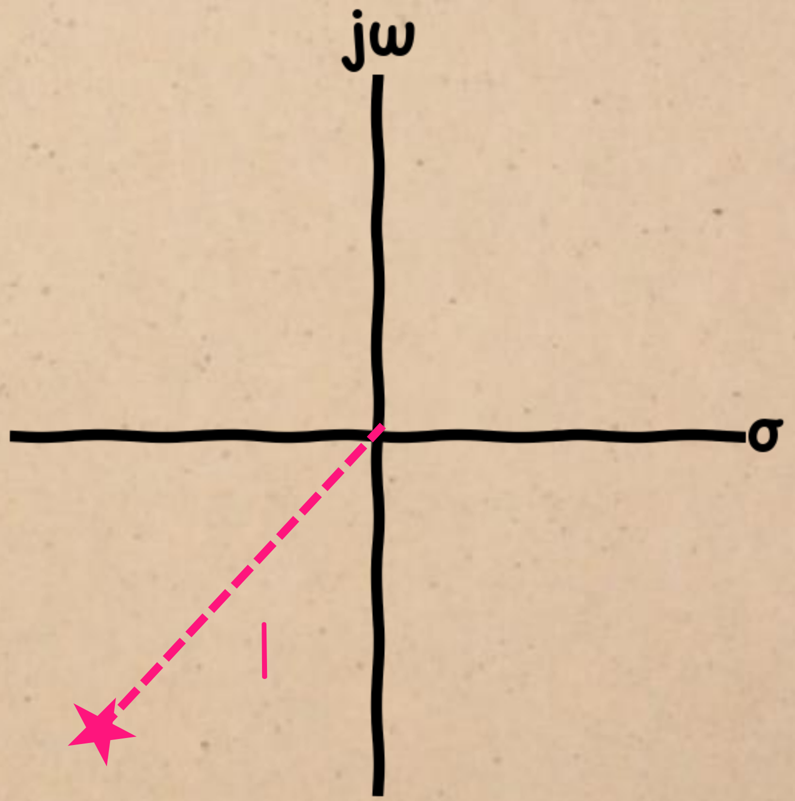


This is the gain crossover frequency:

$$\omega_c = \omega_n\sqrt{-2\zeta^2 + \sqrt{4\zeta^4 + 1}}$$

DAMPING vs. PHASE MARGIN

$$|G(j\omega)| = \frac{|\omega_n^2|}{|-\omega_c^{\,2} + j2\zeta\omega_n\omega_c|} = 1$$



substitute:

$$\omega_c = \omega_n\lambda(\zeta)$$

where

$$\lambda(\zeta) = \sqrt{-2\zeta^2 + \sqrt{4\zeta^4 + 1}}$$

To make this expression easier to work with, let's replace that big square root with a function $\lambda(\zeta)$

DAMPING vs. PHASE MARGIN

Now we can find the phase at this frequency:

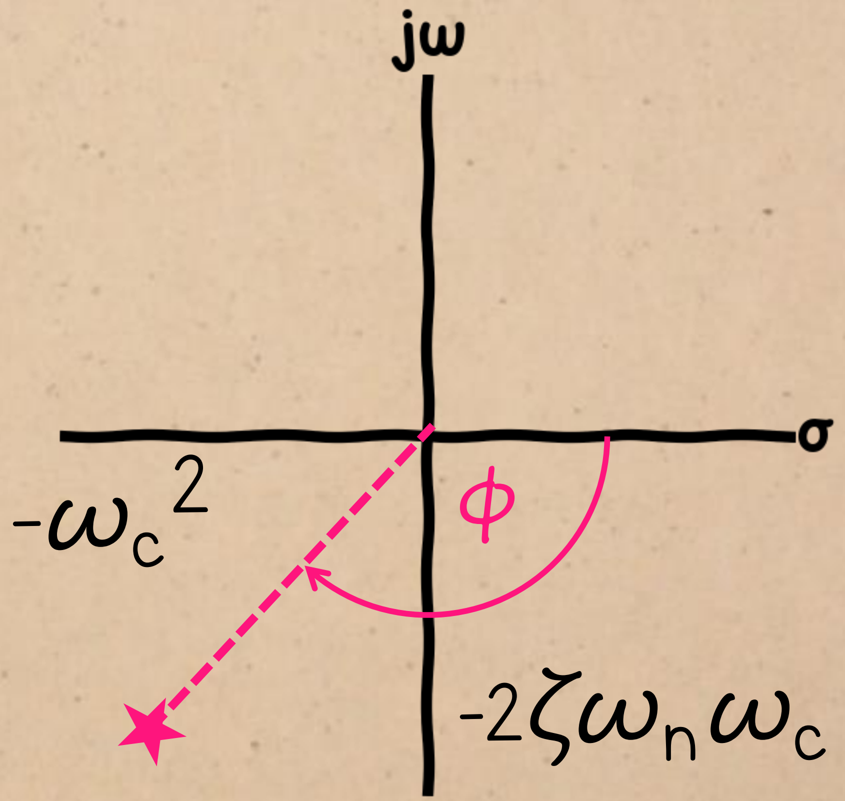
$$\frac{\omega_n^2}{-\omega_c^{\,2} + j2\zeta\omega_n\omega_c}$$

$$\angle G(j\omega_c) = -\angle\left(-\omega_c^{\,2} + j2\zeta\omega_n\omega_c\right)$$

$$= -\angle\left(-(\omega_n\lambda)^2 + j2\zeta\omega_n(\omega_n\lambda)\right)$$

where

$$\lambda(\zeta) = \sqrt{-2\zeta^2 + \sqrt{4\zeta^4 + 1}}$$



DAMPING vs. PHASE MARGIN

Now we can find the phase at this frequency:

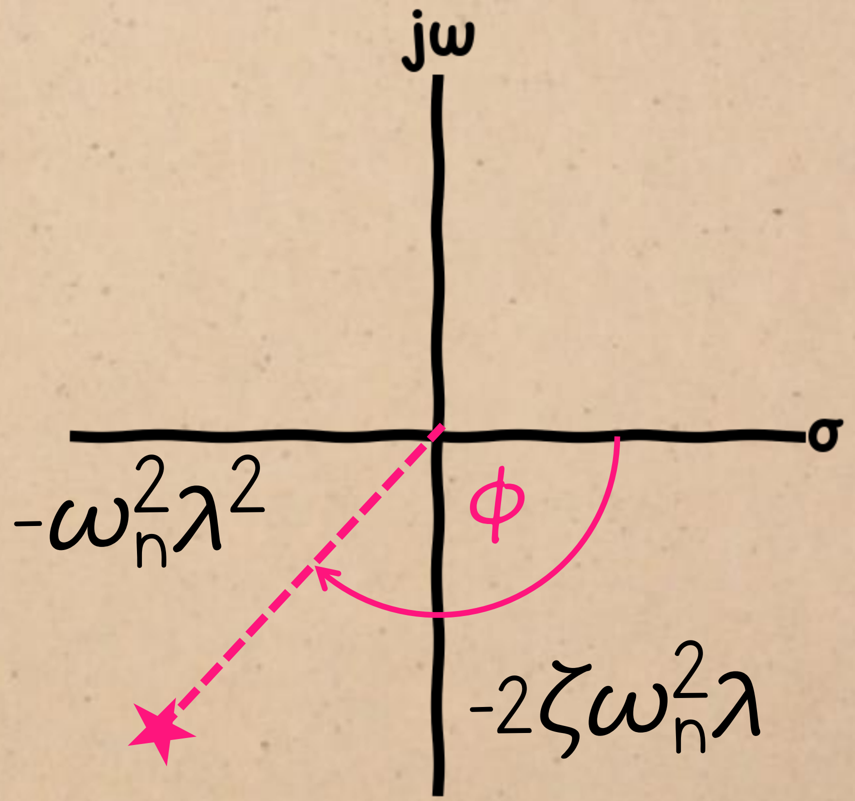
$$\frac{\omega_n^2}{-\omega_c^{\,2} + j2\zeta\omega_n\omega_c}$$

$$\angle G(j\omega_c) = -\angle(-\omega_c^{\,2} + j2\zeta\omega_n\omega_c)$$

$$= -\angle(-\omega_n^2\lambda^2 + j2\zeta\omega_n^2\lambda)$$

where

$$\lambda(\zeta) = \sqrt{-2\zeta^2 + \sqrt{4\zeta^4 + 1}}$$

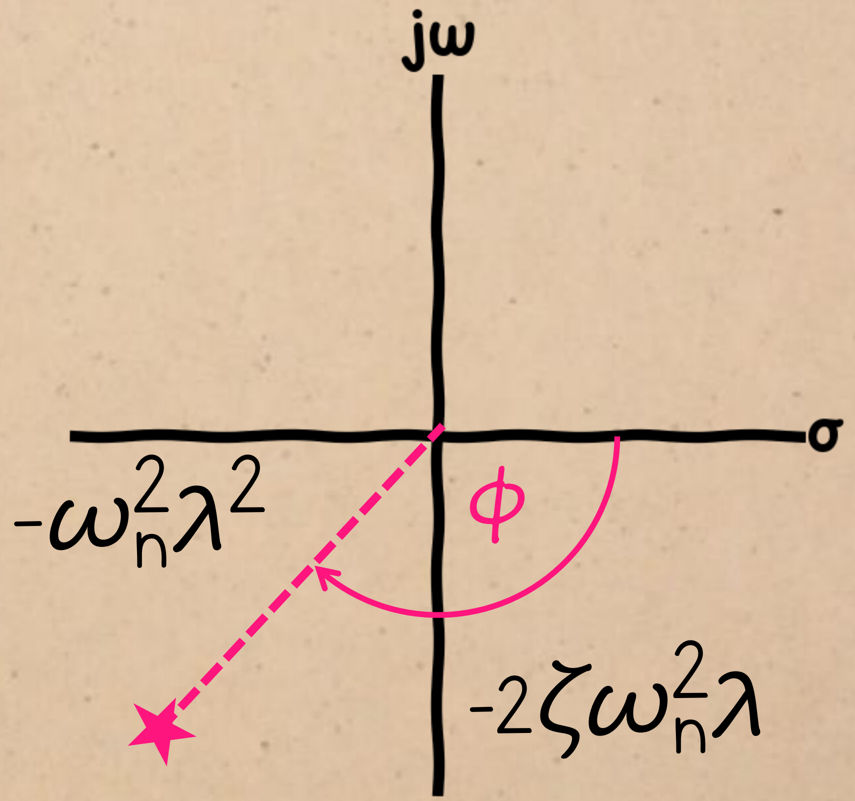


DAMPING vs. PHASE MARGIN

Now we can find the phase at this frequency:

$$\frac{\omega_n^2}{-\omega_c{}^2 + j2\zeta\omega_n\omega_c}$$

$$\angle G(j\omega_c) = \arctan\left(\frac{-2\zeta\omega_n^2\lambda}{-\omega_n^2\lambda^2}\right)$$



where

$$\lambda(\zeta) = \sqrt{-2\zeta^2 + \sqrt{4\zeta^4 + 1}}$$

DAMPING vs. PHASE MARGIN

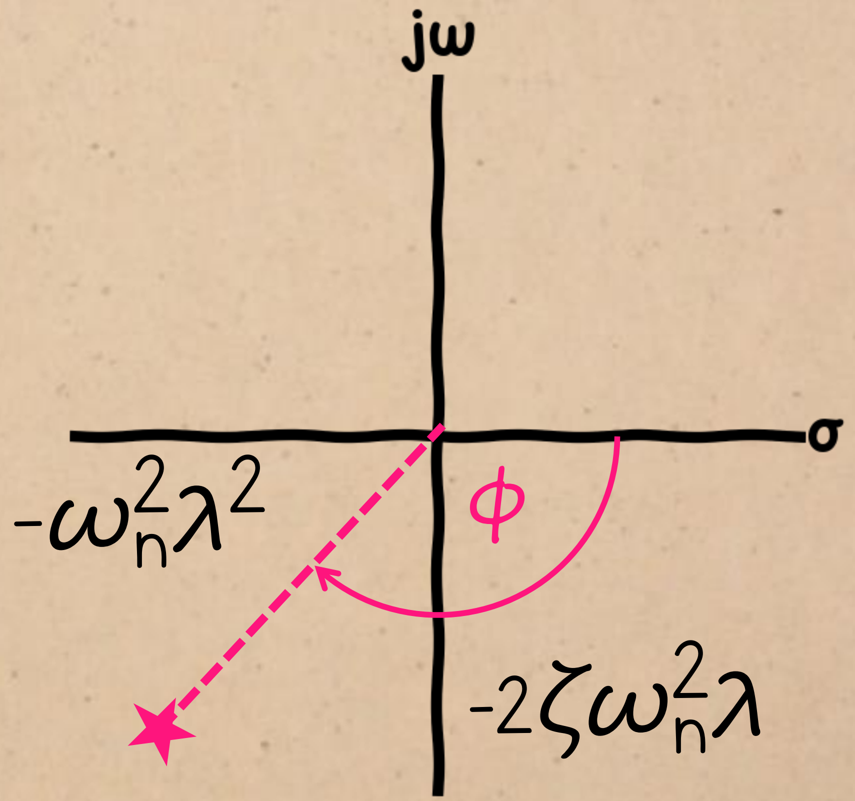
$$\frac{\omega_n^2}{-\omega_c^{\,2} + j2\zeta\omega_n\omega_c}$$

$$\angle G(j\omega_c) = \arctan\left(\frac{-2\zeta\cancel{\omega_n^2}\lambda}{-\cancel{\omega_n^2}\lambda^2}\right)$$

The ω_n's cancel, so the phase is only a function of the damping ratio.

where

$$\lambda(\zeta) = \sqrt{-2\zeta^2 + \sqrt{4\zeta^4 + 1}}$$



DAMPING vs. PHASE MARGIN

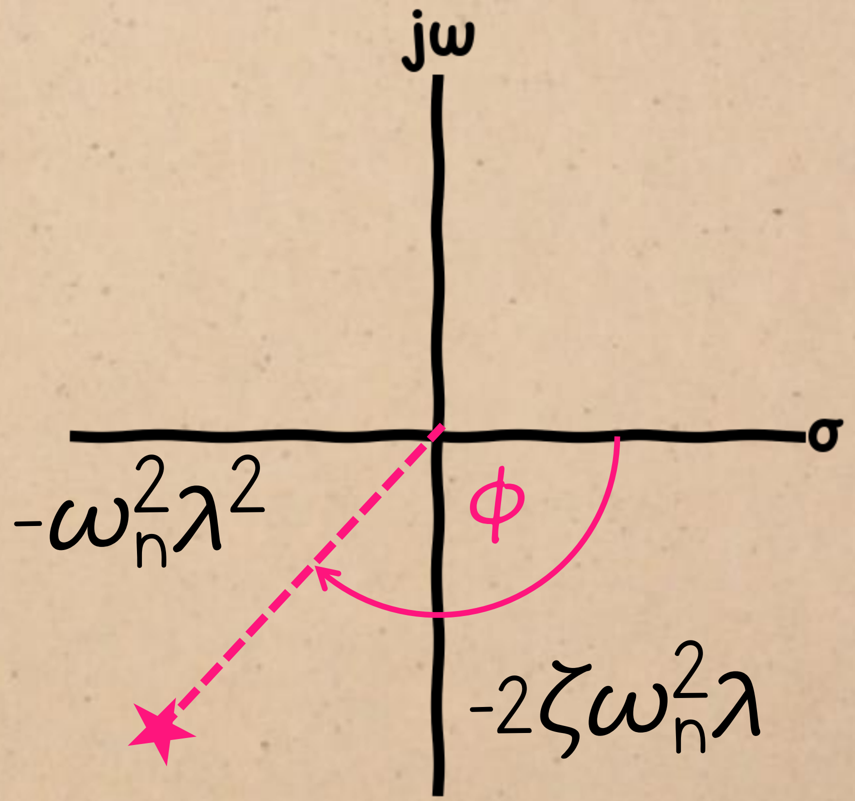
$$\frac{\omega_n^2}{-\omega_c^{\;2} + j2\zeta\omega_n\omega_c}$$

$$\angle G(j\omega_c) = \arctan\left(\frac{-2\zeta}{-\lambda}\right)$$

The ω_n's cancel, so the phase is only a function of the damping ratio.

where

$$\lambda(\zeta) = \sqrt{-2\zeta^2 + \sqrt{4\zeta^4 + 1}}$$



DAMPING vs. PHASE MARGIN

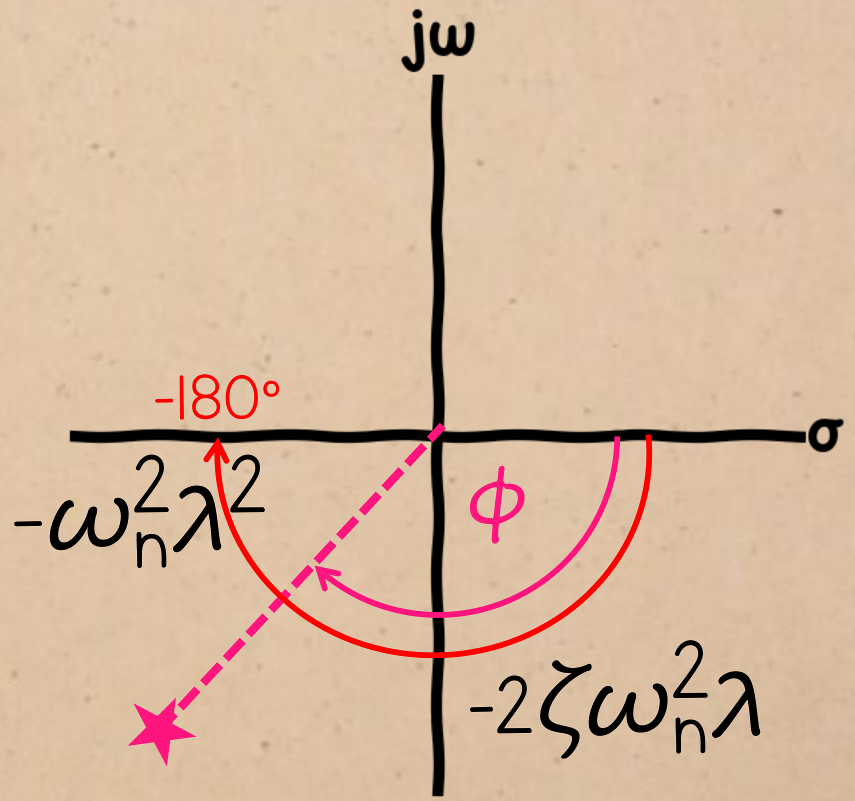
$$\frac{\omega_n^2}{-\omega_c^2 + j2\zeta\omega_n\omega_c}$$

$$\angle G(j\omega_c) = \arctan\left(\frac{2\zeta}{\lambda}\right)$$

$$\therefore \phi_m = \arctan\left(\frac{2\zeta}{\lambda}\right) - (-180°)$$

where

$$\lambda(\zeta) = \sqrt{-2\zeta^2 + \sqrt{4\zeta^4 + 1}}$$

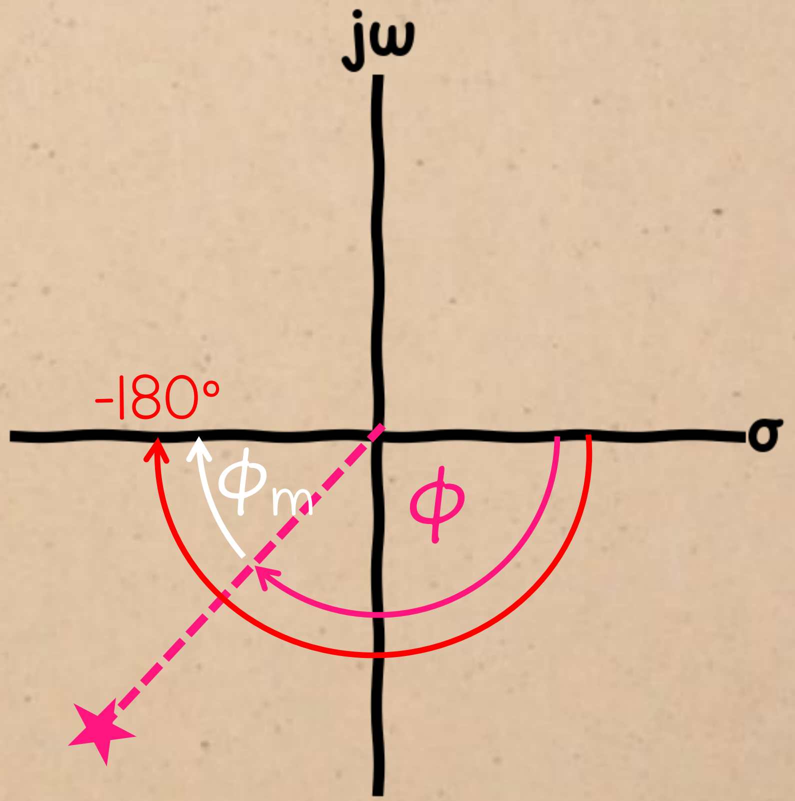


DAMPING vs. PHASE MARGIN

$$\frac{\omega_n^2}{-\omega_c{}^2 + j2\zeta\omega_n\omega_c}$$

$$\angle G(j\omega_c) = \arctan\left(\frac{2\zeta}{\lambda}\right)$$

$$\therefore \phi_m = \arctan\left(\frac{2\zeta}{\sqrt{-2\zeta^2 + \sqrt{4\zeta^4 + 1}}}\right) + 180°$$

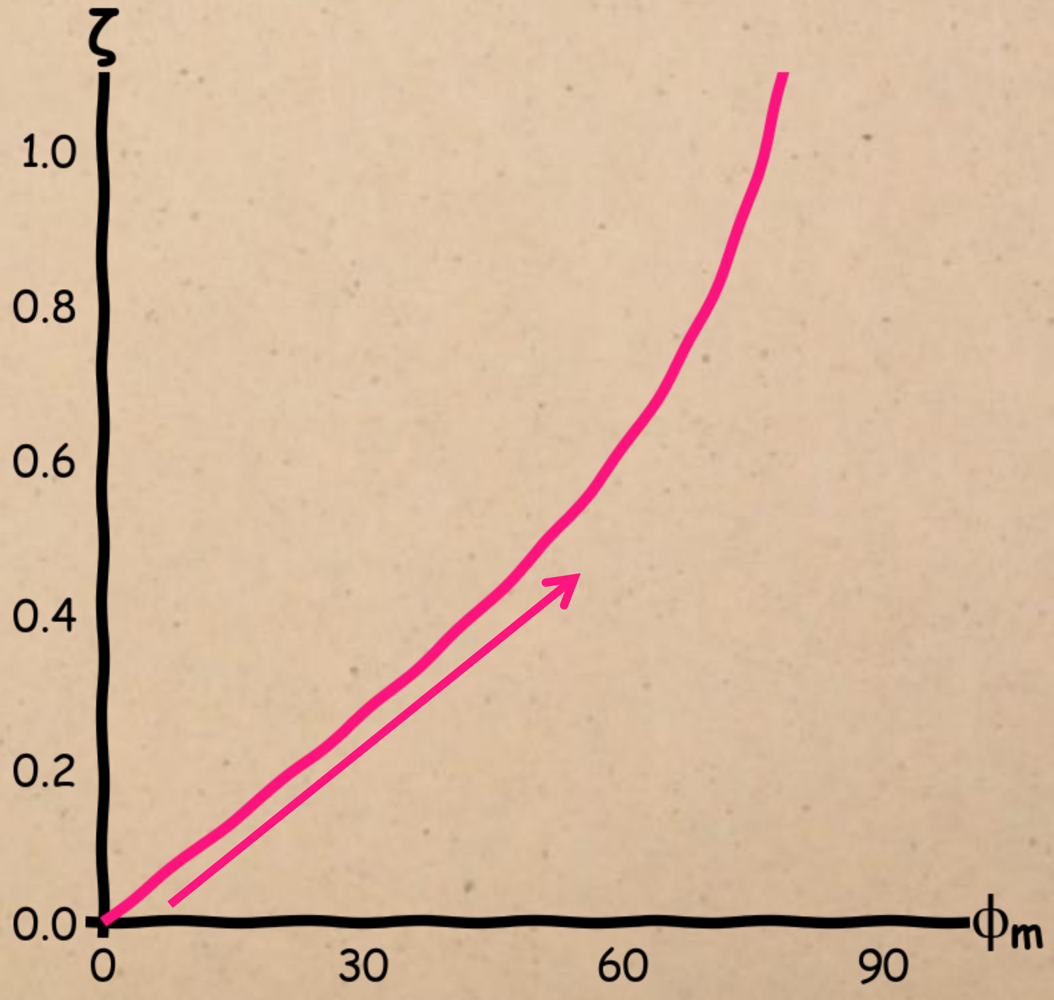


This equation isn't nice, but you don't need to remember it: we're just going to use it to plot the relationship between the damping ratio and phase margin so we can see how close the rule of thumb is.

DAMPING vs. PHASE MARGIN

This relationship is approximately linear for smaller phase margins.

$$\therefore \phi_m = \arctan\left(\frac{2\zeta}{\sqrt{-2\zeta^2 + \sqrt{4\zeta^4 + 1}}}\right) + 180°$$

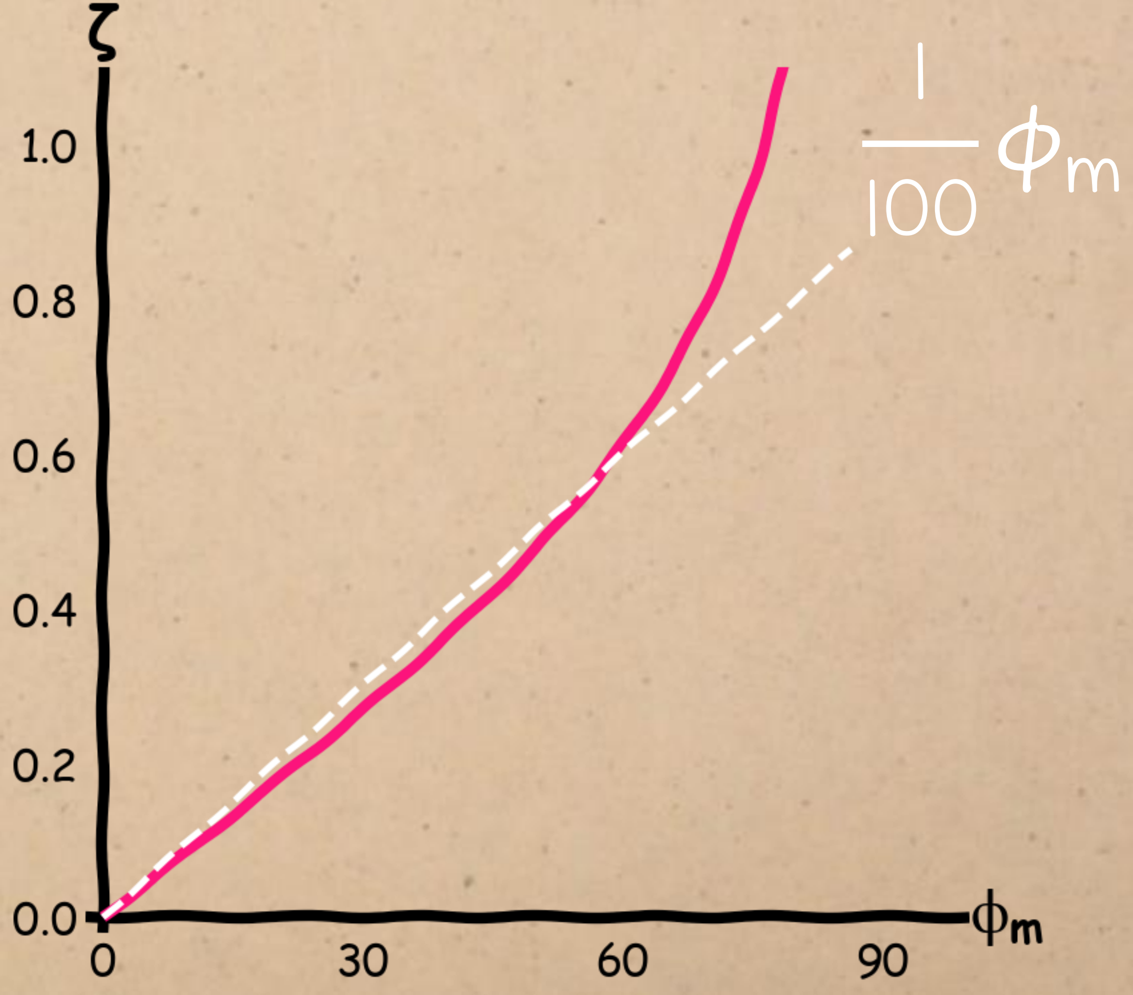


DAMPING vs. PHASE MARGIN

We can approximate the linear region like this:

$$\therefore \phi_m = \arctan\left(\frac{2\zeta}{\sqrt{-2\zeta^2 + \sqrt{4\zeta^4 + 1}}}\right) + 180°$$

$$\approx 100\zeta$$

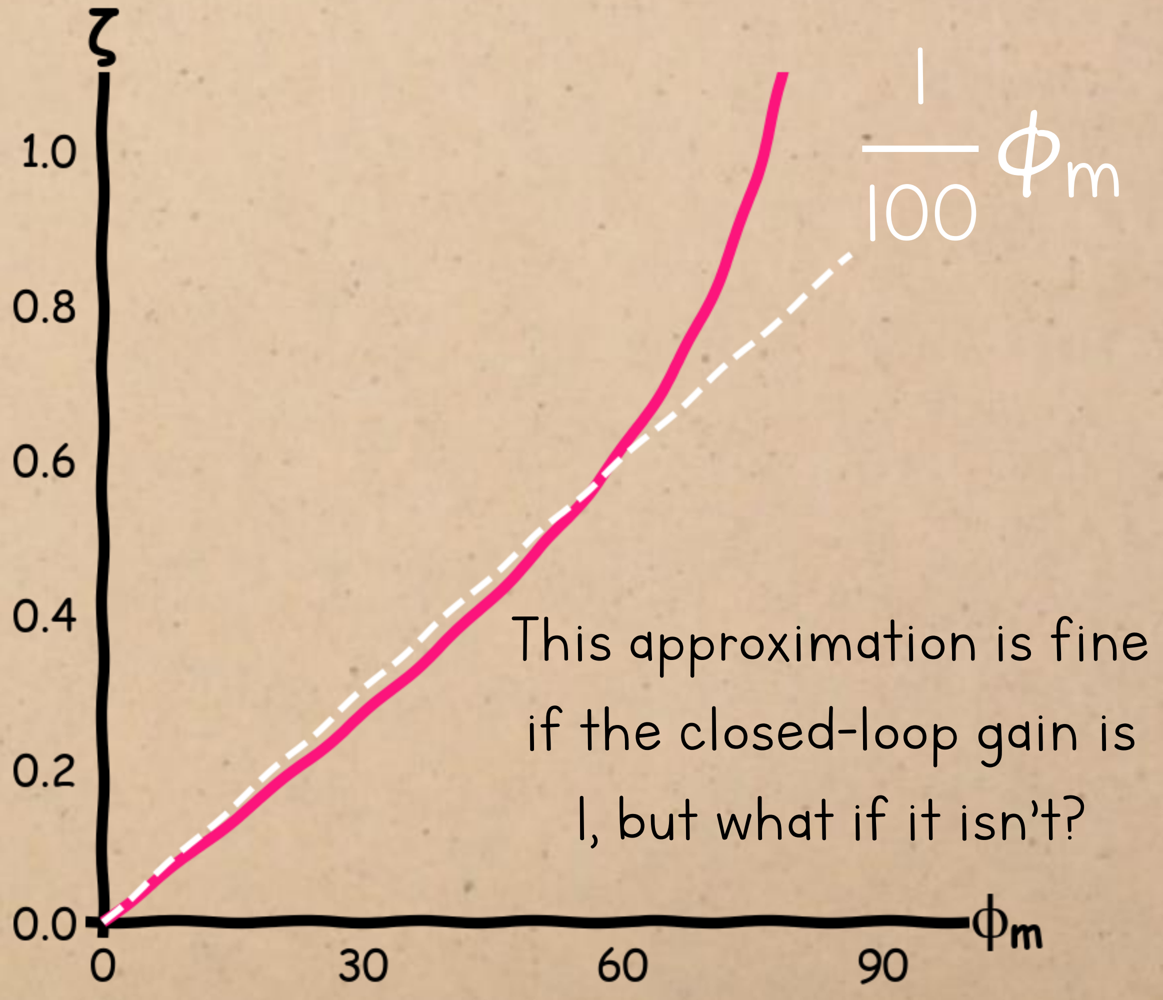


DAMPING vs. PHASE MARGIN

We can approximate the linear region like this:

$$\therefore \phi_m = \arctan\left(\frac{2\zeta}{\sqrt{-2\zeta^2 + \sqrt{4\zeta^4 + 1}}}\right) + 180^\circ$$

$$\approx 100\zeta$$

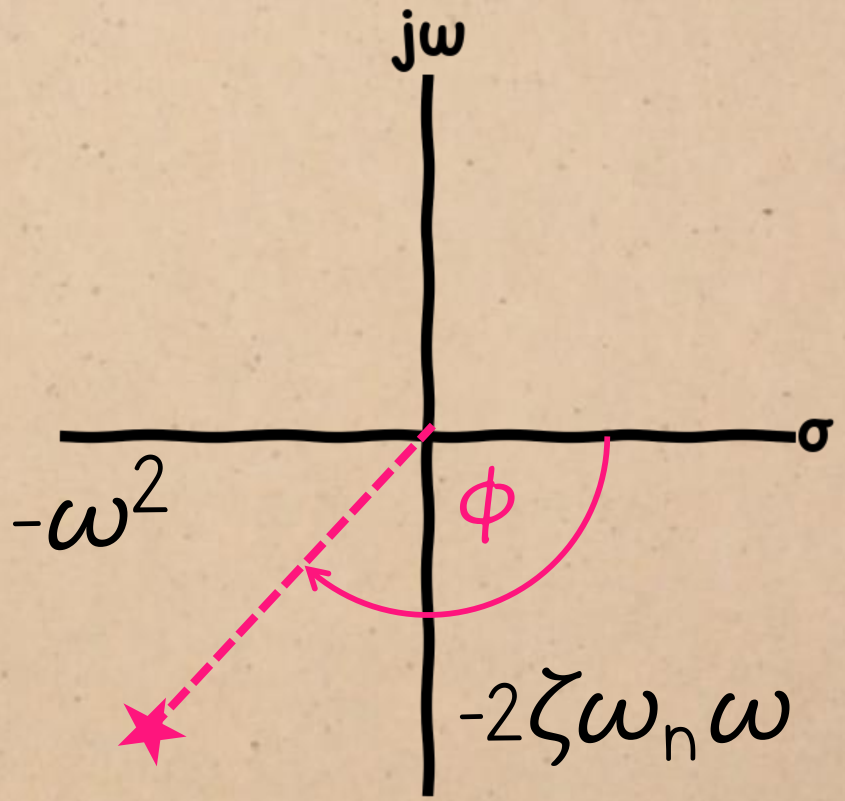


DAMPING vs. PHASE MARGIN

$$\frac{\omega_n^2}{-\omega^2 + j2\zeta\omega_n\omega}$$

$$\angle G(j\omega) = -\angle(-\omega^2 + j2\zeta\omega_n\omega)$$

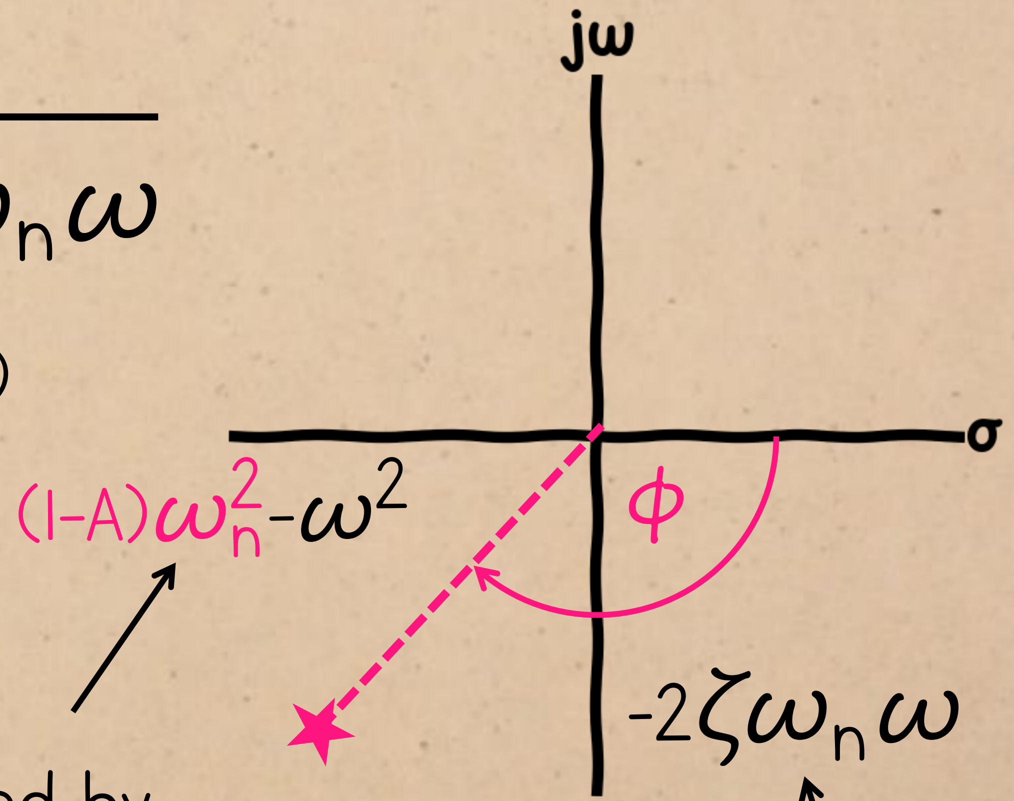
Let's see how including the gain affects the expression for the phase at an arbitrary frequency.



DAMPING vs. PHASE MARGIN

$$\frac{A\omega_n^2}{(1-A)\omega_n^2-\omega^2+j2\zeta\omega_n\omega}$$

$$\angle G(j\omega) = -\angle((1-A)\omega_n^2-\omega^2+j2\zeta\omega_n\omega)$$

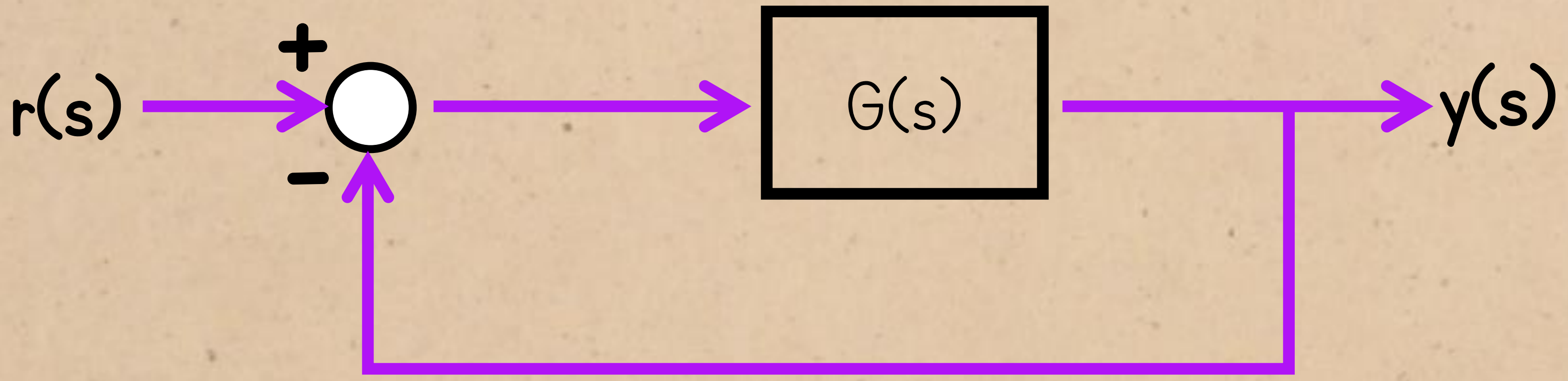


real part shifted by

$(1\text{-}A)\omega_n^2$

same imaginary part

DAMPING vs. PHASE MARGIN

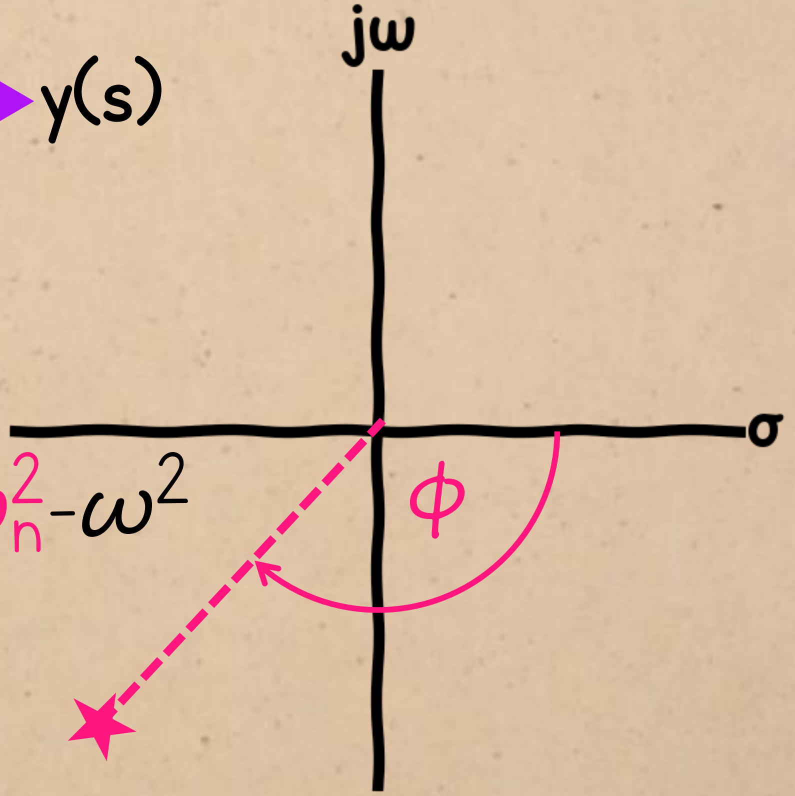


$$G_{CL}(s) = \frac{G(s)}{1 + G(s)} \quad \therefore\ A = \frac{|G(s)|}{1 + |G(s)|}$$

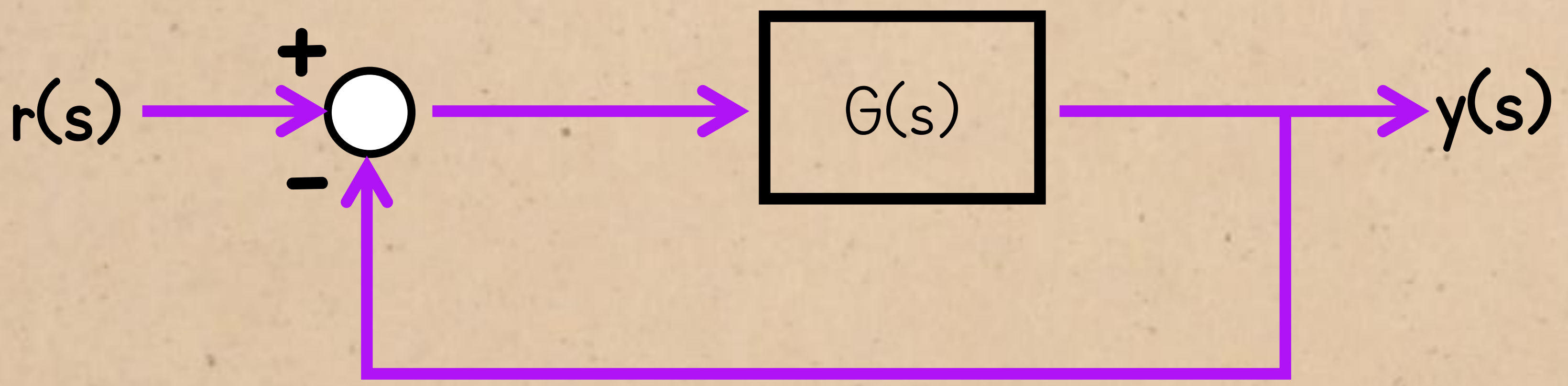
So A < 1 for any positive, finite open-loop gain.

$(1-A)\omega_n^2 - \omega^2$

$\therefore$ positive



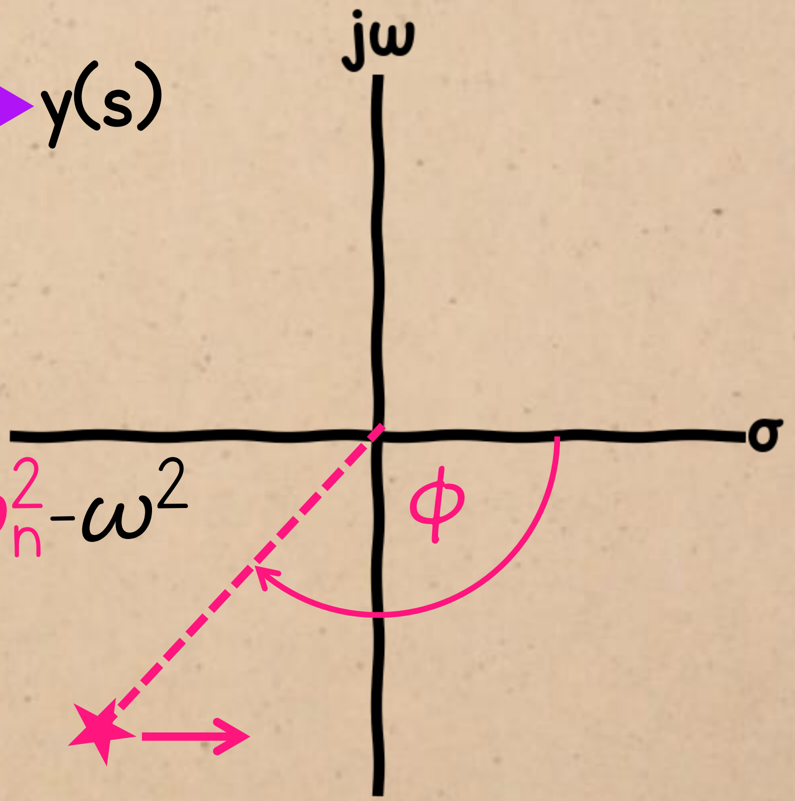
DAMPING vs. PHASE MARGIN



$$G_{CL}(s) = \frac{G(s)}{1 + G(s)} \quad \therefore A = \frac{|G(s)|}{1 + |G(s)|}$$

$(1-A)\omega_n^2 - \omega^2$

$\therefore$ positive

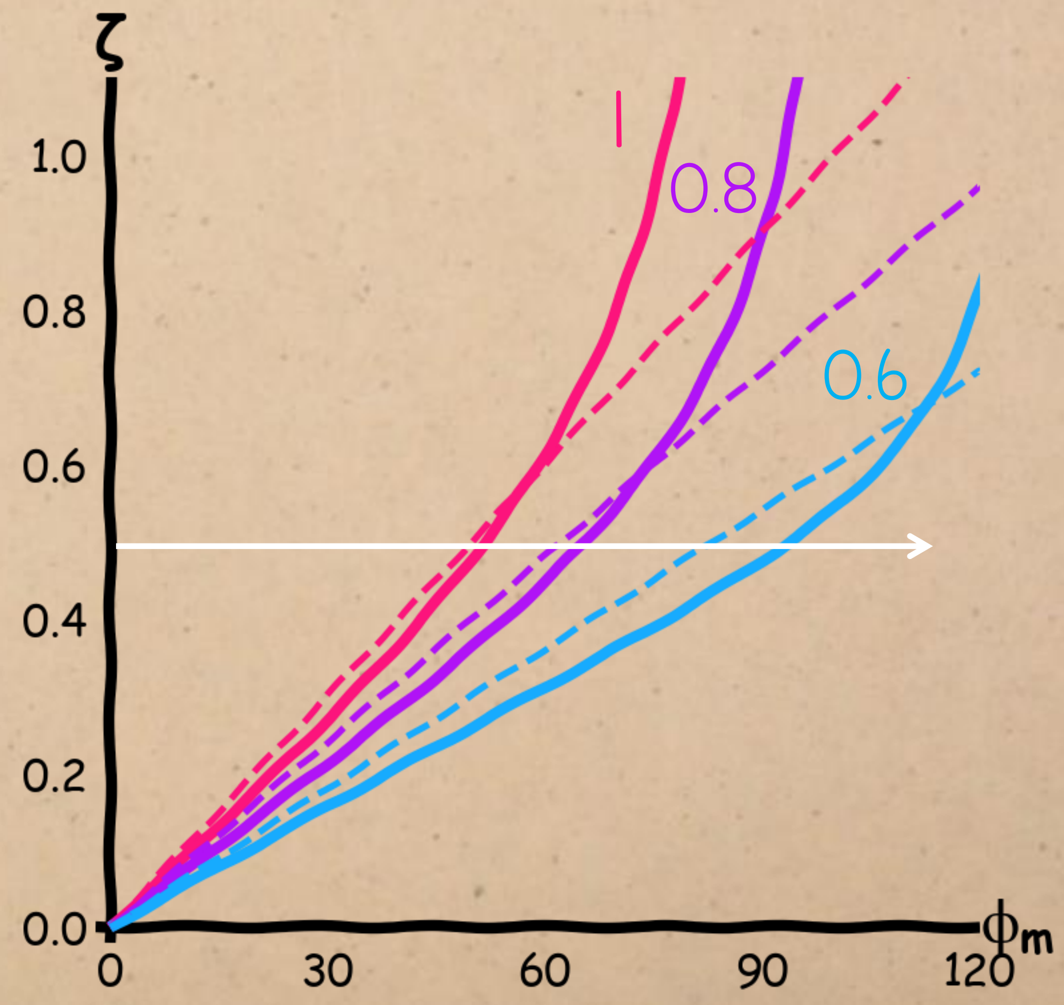


If the gain isn't 1, the point will shift right, which **increases** the angle to the real axis.

So the phase margin will be **larger** if the closed-loop gain is < 1

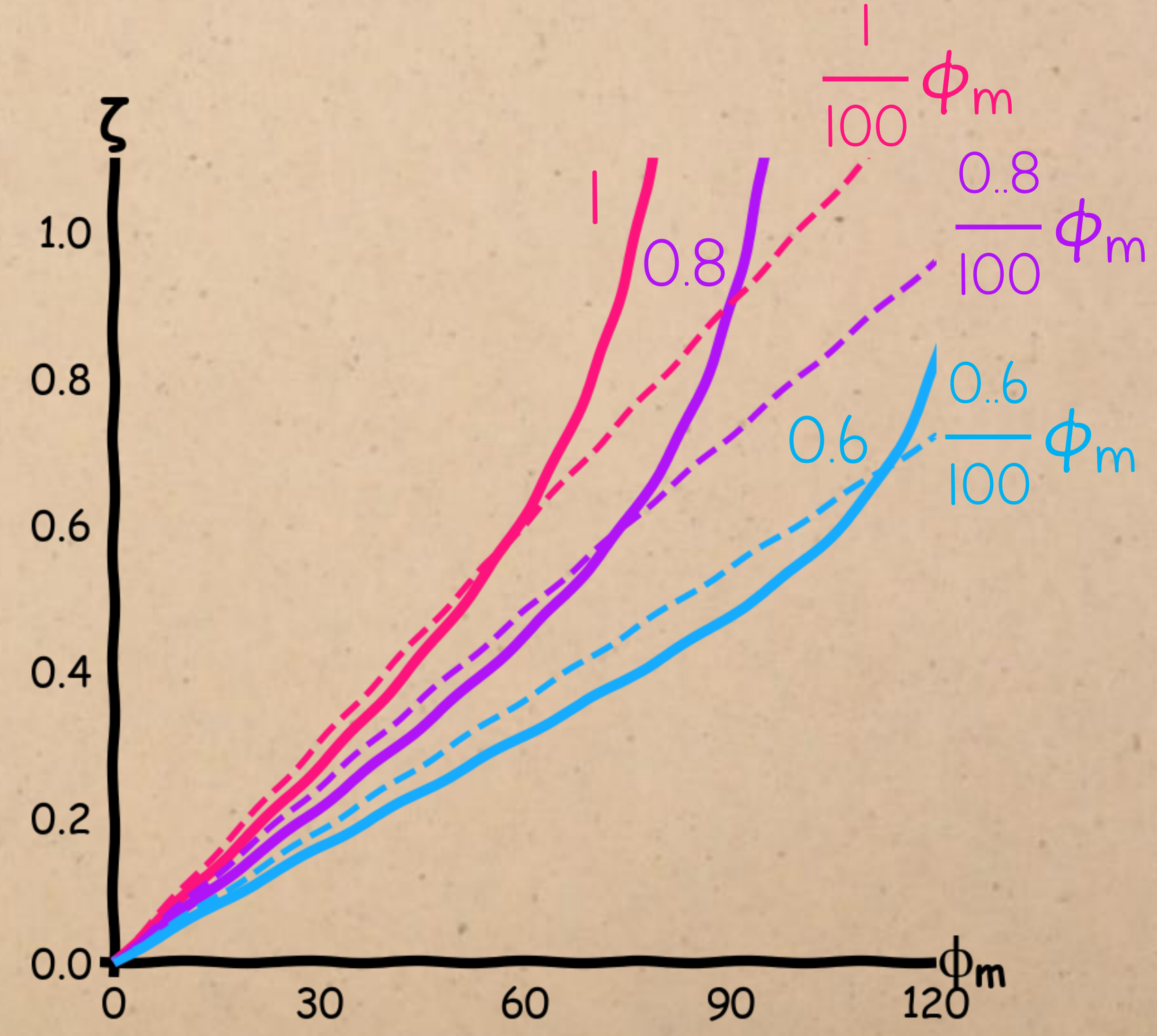
DAMPING vs. PHASE MARGIN

This means that, for a given ratio, the phase margin will tend to be larger if the gain is < 1.



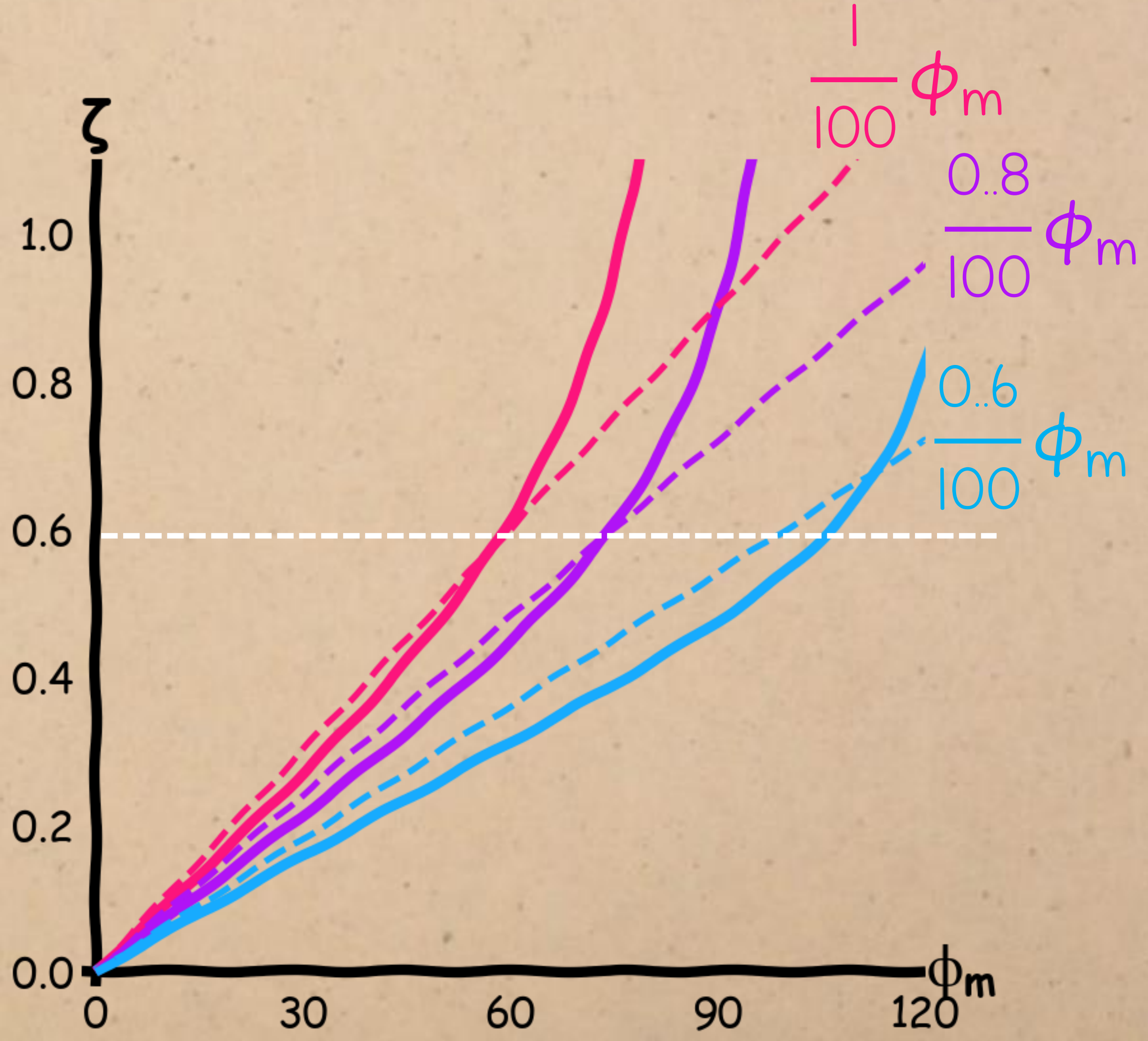
DAMPING vs. PHASE MARGIN

We can still get a decent linear approximation if we multiply the phase margin by the gain.



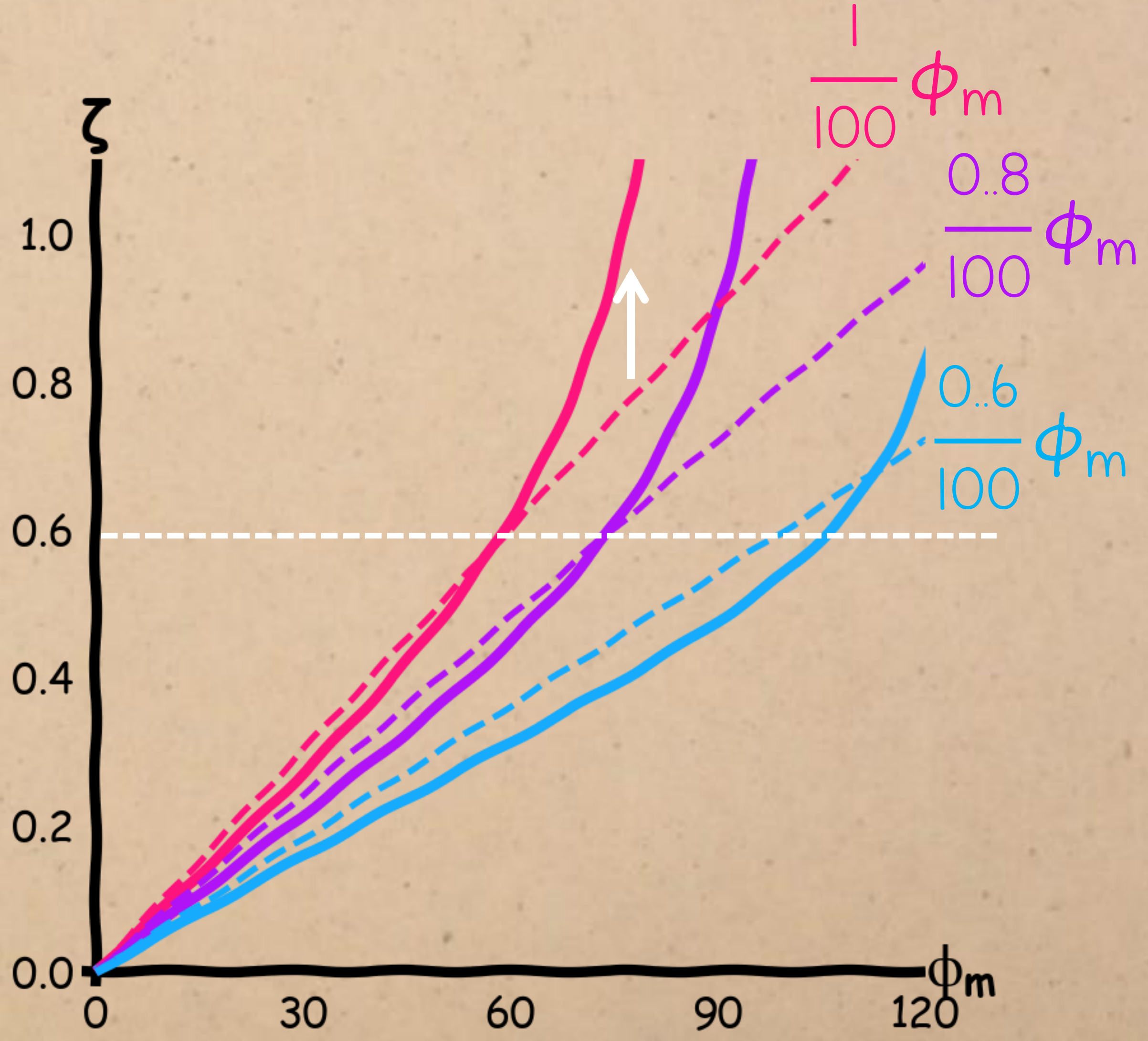
DAMPING vs. PHASE MARGIN

The linear approximation holds up to a damping ratio of around 0.6



DAMPING vs. PHASE MARGIN

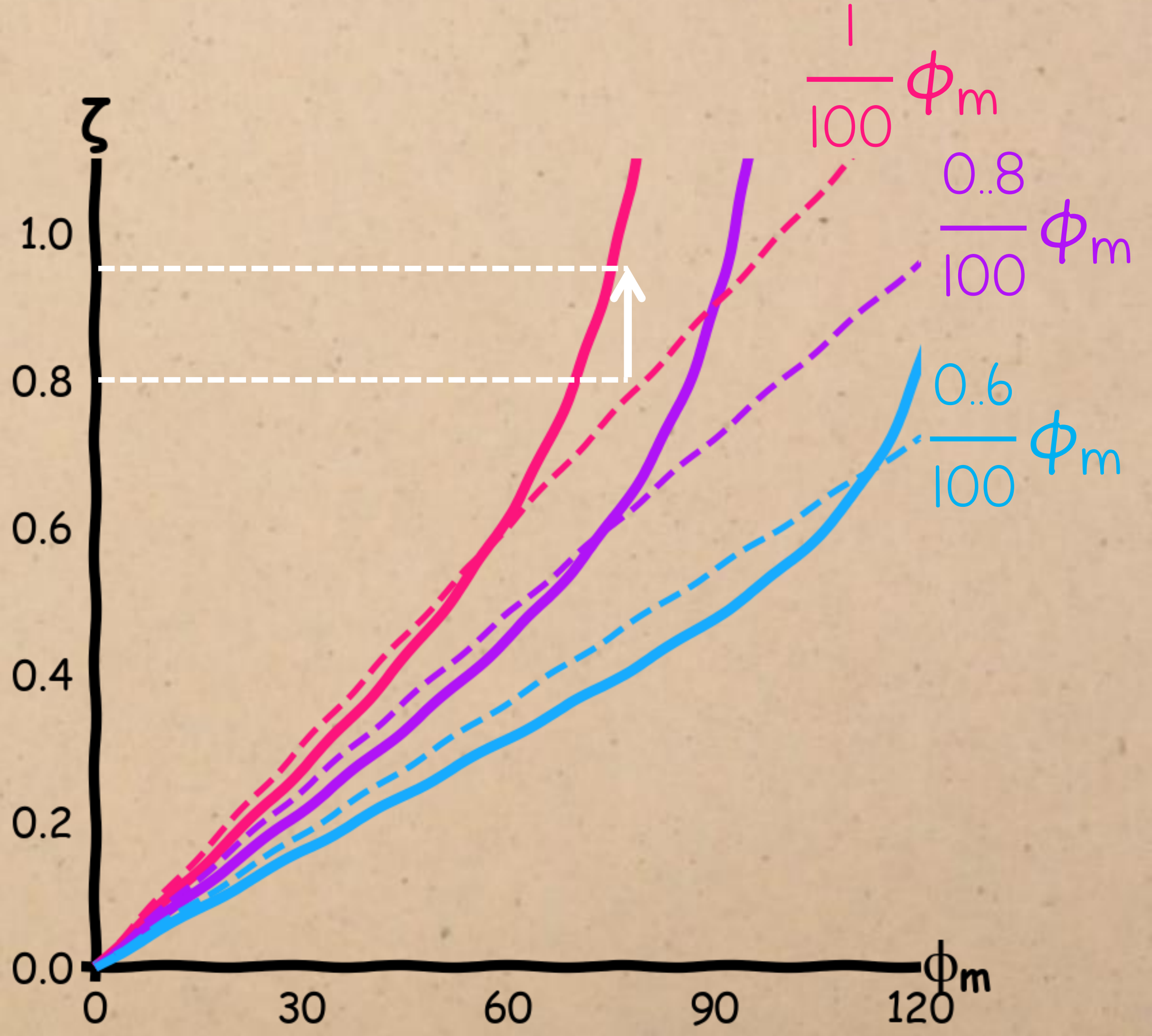
but it will tend to underestimate larger damping ratios.



DAMPING vs. PHASE MARGIN

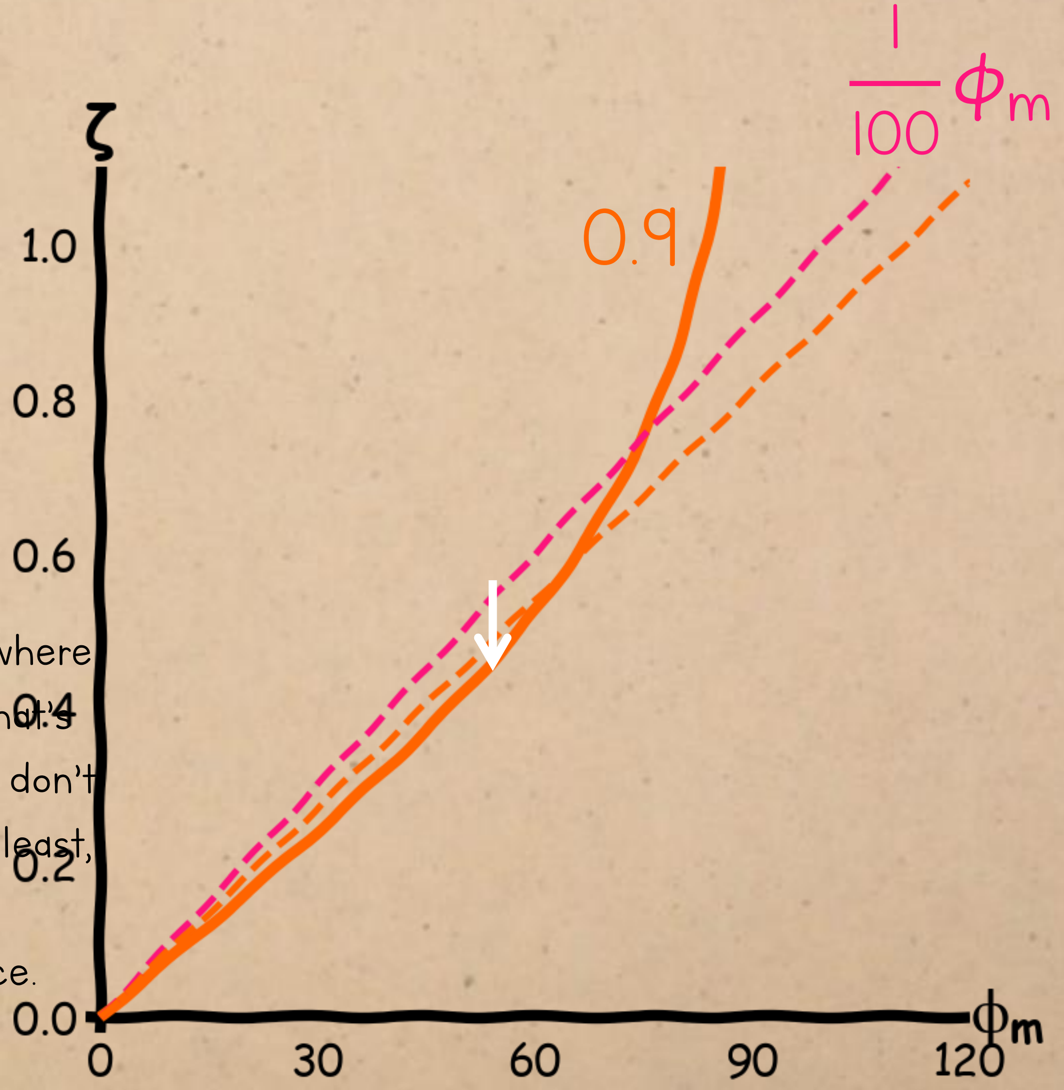
For example, if the rule returns a damping ratio of 0.8, the real value could be more than 0.9.

(which is fine, because more damping is usually a good thing) 👍



DAMPING vs. PHASE MARGIN

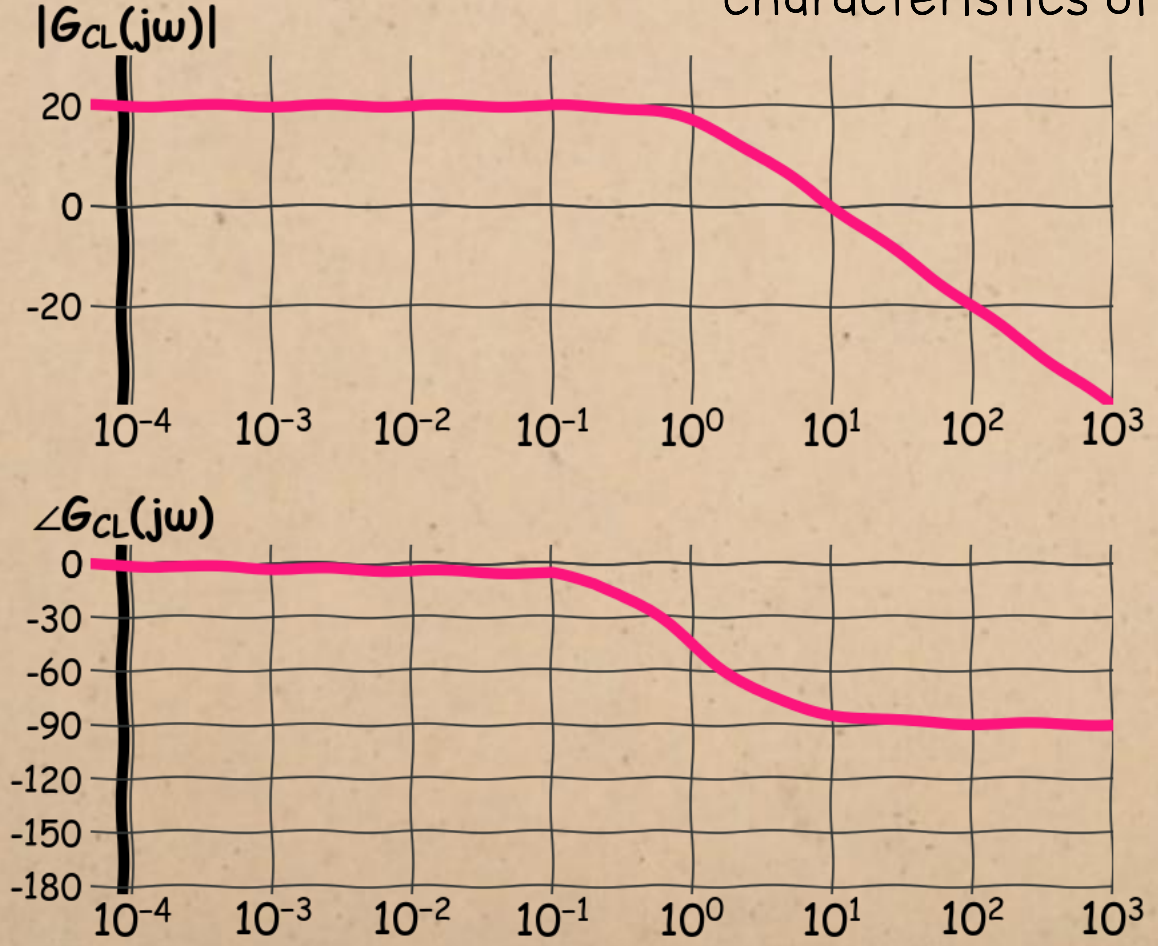
If you assume the closed-loop gain is 1 when it is actually smaller, you will tend to overestimate the damping ratio.



This is unlikely to be a problem in real-life design situations, where the difference will probably be negligibly small at any gain that's close enough to 1 to satisfy the accuracy specs. May textbooks don't even include the gain in this analysis. But I thought it was, at least, worth talking about so you can either keep the possible underestimation in mind, or ignore the gain with confidence.

SUMMARY:

We can relate each of the time-domain parameters to characteristics of the frequency response:



1. **Accuracy (steady-state error)** depends on the **DC gain**
2. **Speed (time constant)** depends on the **bandwidth**
3. **Damping (damping ratio)** depends on the **phase margin**